Proceedings of Machine Learning Research vol vvv:1–15, 2027

One Budget for Every Gap: Certifying Reduced-Order, Quantized, Sampled-Data Neural Barrier Controllers on the Full Plant

Anu Karthika IAMANUKARTHIKA@GMAIL.COM
Independent Researcher, Chennai, India

Abstract

A neural safety filter for a high-dimensional plant is verified on a reduced model, in exact arithmetic and continuous time, but runs as a quantized network and a look-up table, sampled and latched, on the full plant under a bounded disturbance. We charge each gap to one term of a single control-authority budget on a reduced control barrier function. The reduced-order tail, the table quantization, the verified deviation of the quantized network, the latch lag, the sample-and-hold error and the adversary each consume part of the certified authority δ_{cert}. When their sum does not exceed it, and under standing hypotheses on the plant, the reduction and the initial state, the full-order plant is safe for all time. For underdamped second-order port-Hamiltonian plants with uniform modal damping and $m \leq K$ actuators, δ_{cert} has a closed-form lower bound. A coupled twin-network CROWN pass bounds the deviation of the quantized network from its teacher, and both this bound and a closed-form companion are of order 2^{-b_B}. On a 320-state reaction–diffusion–advection lattice at Péclet 20, open-loop unsafe on every seed, the budget closes at utilisation 0.931 and no certified trajectory leaves the safe set. Without the tail or quantization term, the certificate accepts a controller the plant then violates, and the single-rate alternative is refused at the deployed period. The certified plant filters its advection spectrally, and a sweep over the filter weight maps where the L^2-based tail bound is non-vacuous; unfiltered advection is refused. Every constant is computed in double precision from the operators, and the code reproduces every number.

Keywords: control barrier functions; sampled-data control; reduced-order models; neural network verification; quantized neural networks; port-Hamiltonian systems; safety certificates

1. Introduction

A neural safety filter is rarely verified on the system it runs on. It is designed on a reduced model because the plant is too large. It is specified as a floating-point teacher, although the deployed student is a b_B-bit network paired with a b_A-bit look-up table. It is analysed in continuous time because that is where barrier conditions are stated, while the hardware samples the state at period T_s and latches two commands after two different delays. It is certified against a nominal model, while the plant is driven by an unknown bounded disturbance. Each of these gaps has its own remedy; what is missing is a statement of what survives all of them at once, on the plant, for all time.

We give one: each of the six gaps is charged to one term of one inequality. When the inequality closes, under standing hypotheses on the plant, the reduction and the initial state, the full-order plant is safe for all time; when the tail or the quantization term is dropped, the certificate accepts a controller the plant then violates. The single-rate alternative, refused at the deployed period, leaves the safe set at a longer one, and so does the plant under an adversary beyond the certified level. Every gap consumes part of the certified control authority δ_{cert} of a reduced barrier, the amount by which the teacher holds the barrier rate above its prescribed decay on a shell near the boundary.

© 2027 A. Karthika.

The terms are not safety factors stacked on a nominal design; each bounds one summand of the exponentially weighted integral of the reduced barrier rate over one sampling interval along the full-order flow.

On a 320-state telegraph reaction–diffusion–advection lattice at Péclet 20, open-loop unsafe on all 16 seeds, the deployed configuration consumes 0.931 of its certified authority and no certified trajectory leaves the safe set. The certified plant filters its advection spectrally, unfiltered advection is refused, and a sweep over the filter weight locates where the certificate stops accepting. The contributions are four. (i) The budget (7), named the Certified Control Authority Condition, and Theorem 1: full-order safety in continuous time, for all time and under a bounded adversary, of a reduced-order, quantized, multi-rate sampled-data neural controller (Sections 3 and 4). (ii) A closed-form lower bound on δ_{cert} for class-(H) reduced models with uniform modal damping, $\gamma < 2\omega_{\min}$ and $m \leq K$ actuators, within one percent of brute force in both regimes (Lemma 1, Table 1). (iii) A bound on the deviation of a quantized GroupSort-2 student from its isometric teacher, obtained by a twin-network CROWN pass, an instance of differential verification, with a closed-form companion; both are of order 2^{-b_B} (Proposition 1, Section 5). (iv) The certificate on a 320-state lattice in two regimes, with ablations, refusals and the frontier sweep (Section 5, Figure 1). Section 2 fixes the setting, Section 3 derives the six terms, Section 4 states the theorem, Section 5 certifies the lattice and Section 6 states what the certificate does not cover.

Sampled-data barriers. Control barrier functions certify forward invariance for continuous-time control-affine systems (Ames et al., 2017, 2019). Sampled-data versions tighten the condition to hold between samples, under input delay (Singletary et al., 2020), for a fixed time step (Breeden et al., 2022), through approximate discrete-time models (Taylor et al., 2022), or by a zero-order condition free of derivatives (Tan et al., 2025). The closest work is Niu et al. (2024), which bounds the gap between digital and true barrier constraints under sampled, quantized state measurements and finite-precision coefficients on the full-order plant; here the quantized object is a learned controller and a reduced-order tail term is added. These works charge neither the tail term nor the verification slack.

Reduced-order and model-error barriers. Robust and input-to-state safe barriers add a margin for an unstructured disturbance (Jankovic, 2018; Kolathaya and Ames, 2019). Reduced-order designs filter on a low-dimensional model and carry a tracking error to the full system (Singletary et al., 2022; Molnar et al., 2022; Cohen et al., 2024). The closest reduced-order work is Cohen et al. (2025), which relates the two models by projection maps, bounds their discrepancy by an assumed supremum and proves invariance of a safe set inflated by that bound. Here the residual is computed from the Petrov–Galerkin truncation of a dissipative port-Hamiltonian operator and enters as one budget term; these works charge no quantization, slack, lag or hold term.

Verified and quantized networks. Bound propagation gives sound linear bounds on a network's output over an input box (Zhang et al., 2018; Xu et al., 2020; Wang et al., 2021), GroupSort networks carry a provable Lipschitz constant (Anil et al., 2019), and semidefinite programs bound that of a general network (Fazlyab et al., 2019), though Pauli et al. (2022) showed the coupled-neuron variant invalid. Differential verification bounds the gap between a network and its compressed or quantized copy (Paulsen et al., 2020; Zhang et al., 2023b). Neural barriers and Lyapunov controllers are verified, exactly or by CROWN-type bounds, on a fixed model with real-valued weights (Zhang et al., 2023a; Yang et al., 2024; Dawson et al., 2023); none charges the tail, lag or hold term.

Structure-preserving reduction. Port-Hamiltonian and GENERIC models encode storage and dissipation in their structure (van der Schaft and Jeltsema, 2014; Grmela and Öttinger, 1997). Structure-preserving reduction by interpolation (Gugercin et al., 2012), POD/DEIM (Chaturantabut et al., 2016) or discretization of boundary-controlled PDEs (Toledo-Zucco et al., 2024) keeps the reduced model passive; matrix measures and non-Euclidean contraction bound error growth (Söderlind, 2006; Davydov et al., 2022). Califano (2023) pairs barrier filters with port-Hamiltonian structure on the full-order model.

Stochastic and sector-bounded analyses. Stochastic barrier certificates bound the probability of reaching the unsafe set (Prajna et al., 2007; Kushner, 1967; Santoyo et al., 2021; Jagtap et al., 2021; Cosner et al., 2024), the probability-one guarantee of the zeroing barriers of Clark (2021) was shown not to hold by So et al. (2023), and classical quantized feedback treats quantization as a stability problem (Delchamps, 1990; Elia and Mitter, 2001; Fu and Xie, 2005); none of them charges the tail term, the verification slack or the lag term.

2. Setting

Plant and operating domain. The plant is $\dot{x} = F(x) + Gu + Dw$ on $\mathbb{R}^N$, with state in a compact operational design domain $\mathcal{X}_{\mathrm{ODD}}$ containing the origin in its interior, input $u \in U_{\mathrm{adm}} := \{\|u\|_2 \le u_{\max}\}$, $u \in \mathbb{R}^m$, and measurable disturbance $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}}) := \{\|w\|_2 \le \varepsilon_{\mathrm{adv}}\}$; solutions are Carathéodory. Let ρ_{ODD} be the radius of the largest origin-centred ball in $\mathcal{X}_{\mathrm{ODD}}$.

Assumption 1 (Plant) *$F(x) = A_{\mathrm{full}}x + F_{\mathrm{nl}}(x)$ with $A_{\mathrm{full}} := (J_{\mathrm{full}} - R_{\mathrm{full}})P_{\mathrm{full}}$, $J_{\mathrm{full}} = -J_{\mathrm{full}}^\top$, $R_{\mathrm{full}} = R_{\mathrm{full}}^\top \succeq 0$, $P_{\mathrm{full}} \succ 0$, $F_{\mathrm{nl}}(0) = 0$, F_{nl} locally Lipschitz on $\mathbb{R}^N$ and $L_{F_{\mathrm{nl}}}$-Lipschitz on $\mathcal{X}_{\mathrm{ODD}}$.*

Two classes are treated. In class (D), $R_{\mathrm{full}} \succ 0$ and $P_{\mathrm{full},\epsilon} := P_{\mathrm{full}}$. In class (H), the plant is an underdamped second-order Hamiltonian system, $x = (q, p)$, J_{full} canonical, $P_{\mathrm{full}} = \mathrm{blkdiag}(\Omega_{\mathrm{full}}^2, I)$, $R_{\mathrm{full}} = \mathrm{blkdiag}(0, R_p)$, and for a damping level $0 < \gamma < 2\omega_{\min}$ (with $\omega_{\min}$ the smallest full-order natural frequency) the Lyapunov metric is $P_{\mathrm{full},\epsilon} := \left[\begin{smallmatrix} \Omega_{\mathrm{full}}^2 & \epsilon I \\ \epsilon I & I \end{smallmatrix}\right] \succ 0$ with $\epsilon := \gamma/2$. The barrier and the safe set are $h(x) := E_{\mathrm{safe}} - x^\top P_{\mathrm{full},\epsilon} x$ and $\mathcal{C} := \{x \in \mathcal{X}_{\mathrm{ODD}} : h(x) \ge 0\}$; under the containment condition (H0) of Theorem 1, $\mathcal{C}$ is the full ellipsoid and lies in the interior of $\mathcal{X}_{\mathrm{ODD}}$ (Lemma S1). The residual F_{nl} may carry a linear part $A_{\mathrm{adv}}x$, its linearisation at the origin; in the benchmark it is the advection, which the certified plant carries through a spectral filter (Section 5).

Reduction and tail. For a full-rank basis $\Phi \in \mathbb{R}^{N\times K}$ let $P_0 := \Phi^\top P_{\mathrm{full}} \Phi$, $z(x) := P_0^{-1}\Phi^\top P_{\mathrm{full}} x$, $P_\perp := I - \Phi P_0^{-1}\Phi^\top P_{\mathrm{full}}$, $x_\perp := P_\perp x$ and $\mathcal{V}_\perp := \mathrm{Im}(P_\perp)$, so that $x = \Phi z + x_\perp$. In class (H), $\Phi = \mathrm{blkdiag}(\Phi_q, \Phi_q)$ with Φ_q orthonormal eigenvectors of Ω_{full}^2, which defines $\Omega_r^2 := \Phi_q^\top \Omega_{\mathrm{full}}^2 \Phi_q$ and $R_{p,\perp} := (I - \Phi_q\Phi_q^\top) R_p (I - \Phi_q\Phi_q^\top)$. The reduced coordinates obey exactly

$$\dot{z} = f_r(z) + g_r u + d_r w + \Delta_\perp(z, x_\perp), \qquad g_r := P_0^{-1}\Phi^\top P_{\mathrm{full}} G, \quad d_r := P_0^{-1}\Phi^\top P_{\mathrm{full}} D, \tag{1}$$

with $f_r(z) = A_{r,0}z + F_{\mathrm{nl},r}(z)$ the Petrov–Galerkin reduction of F and $\Delta_\perp$ the tail coupling.

Assumption 2 (Tail) *(H2a) $\Phi^\top P_{\mathrm{full},\epsilon} P_\perp = 0$. (H2b) In class (H), the tail-restricted damping satisfies $v^\top R_{p,\perp} v > \epsilon \|v\|_2^2$ for nonzero tail vectors v. (H2c) There is $\lambda_{K+1} > 0$ with $-x_\perp^\top (P_{\mathrm{full},\epsilon} A_{\mathrm{full}} + A_{\mathrm{full}}^\top P_{\mathrm{full},\epsilon}) x_\perp \ge 2\lambda_{K+1} \|x_\perp\|_{P_{\mathrm{full},\epsilon}}^2$ on $\mathcal{V}_\perp$, and the tail excitation*

$$M_\perp := \sup_{\mathcal{X}_{\mathrm{ODD}} \times U_{\mathrm{adm}} \times \mathcal{B}(\varepsilon_{\mathrm{adv}})} \|P_\perp [A_{\mathrm{full}} \Phi z(x) + F_{\mathrm{nl}}(x) + Gu + Dw]\|_{P_{\mathrm{full},\epsilon}} < \infty.$$

The tail radius is $\varepsilon_{\mathrm{basis},P} := M_\perp/\lambda_{K+1}$, and the tail coupling constant $L_{\perp,P}$ is the smallest constant with $\|\Delta_\perp(z, x_\perp)\|_2 \le L_{\perp,P}\|x_\perp\|_{P_{\mathrm{full},\epsilon}}$ for all $x = \Phi z + x_\perp \in \mathcal{X}_{\mathrm{ODD}}$ with $\Phi z \in \mathcal{X}_{\mathrm{ODD}}$. Both are computed from the operators. Under (H2a), $h(\Phi z + x_\perp) = h(\Phi z) - \|x_\perp\|^2_{P_{\mathrm{full},\epsilon}}$ for every $x \in \mathbb{R}^N$, with $h(\Phi z) = E_{\mathrm{safe}} - z^\top P z$ and $P := \Phi^\top P_{\mathrm{full},\epsilon}\Phi$ (Lemma S2).

Controller. The teacher $\hat{\mathcal{T}}(z) := \Pi_U(\mathcal{N}(v))$ acts on the port vector $v := g_r^\top P z \in \mathbb{R}^m$; Π_U is the radial projection onto U_{adm} and $\mathcal{N}$ is an origin-preserving GroupSort-2 network of even hidden width H whose first layer mixes pairwise sum and difference channels at an angle $\theta_{\mathrm{GS}} \in (0, \pi/4)$, whose hidden layers are orthogonal (isometric) and preserve these two channels, and whose output layer is $W_L = -L_{\mathcal{T}}(W_{L-1}\cdots W_1)^\top$ (S17)–(S18). Then $\mathcal{N}$ is $L_{\mathcal{T}}$-Lipschitz and $\hat{\mathcal{T}}$ is $L_{\mathcal{T},z}$-Lipschitz in z with $L_{\mathcal{T},z} := L_{\mathcal{T}}\|g_r^\top P\|_2$. The student $\Psi_B(z) := \Pi_U(\mathcal{N}_q(v))$ uses the b_B-bit fixed-point realization $\mathcal{N}_q$. The look-up table Ψ_A quantizes v by the mid-tread quantizer Q_{b_A} with $2^{b_A}+1$ levels per axis over the port box of width $D_v := 2\sup_{\mathcal{C}_{r,0}}\|v\|_\infty$, stores the quantized student's value $\Pi_U(\mathcal{N}_q(v_q))$ at each nonzero grid point v_q and outputs zero on the central bin. With latch delays $0 < \tau_A < \tau_B < T_s$, samples at $t_k = kT_s$, $z_k := z(x(t_k))$ and $v_k := g_r^\top P z_k$, the control is

$$u(t) = \begin{cases} \Psi_B(z_{k-1}), & t \in [t_k, t_k + \tau_A), \\ \Psi_A(Q_{b_A}(v_k)), & t \in [t_k + \tau_A, t_k + \tau_B), \\ \Psi_B(z_k), & t \in [t_k + \tau_B, t_{k+1}), \end{cases} \tag{2}$$

with $z_{-1} := z_0$, so that the register holds $u_{B,-1} := \Psi_B(z_0)$ from standby.

The saturation condition (H-sat) is $L_{\mathcal{T},z}\sqrt{K}D_{\mathrm{ODD}}/2 \le u_{\max}$ on the reduced box $\mathcal{Z}_{\mathrm{ODD}} := [-D_{\mathrm{ODD}}/2, D_{\mathrm{ODD}}/2]^K$; under it the teacher is sector-aligned on $\mathcal{Z}_{\mathrm{ODD}}$, $-z^\top P g_r \hat{\mathcal{T}}(z) \ge \eta_{\mathcal{T}}\|g_r^\top P z\|_2^2$ with $\eta_{\mathcal{T}} = L_{\mathcal{T}}\cos(2\theta_{\mathrm{GS}}) > 0$ (Lemma S3). With $B_{\max}$ address bits, S_{bits} bits of memory and b_{out} bits per output, the table fits the hardware if $m(b_A+1) \le B_{\max}$ and $m(b_A+1) + \lceil \log_2(m\, b_{\mathrm{out}})\rceil \le \log_2 S_{\mathrm{bits}}$. Theorem 1 and (7) apply to any $L_{\mathcal{T},z}$-Lipschitz teacher with certified $\delta_{\mathrm{cert}} > 0$; the GroupSort-2 family parameterised by θ_{GS} is the analytical testbed for which Lemma 1 gives the closed form.

Reduced barrier and shell. All reduced constants are taken over $\mathcal{C}_{r,0} := \{z^\top P z \le E_{\mathrm{safe}}\}$: $L_h := 2\sqrt{\lambda_{\max}(P)E_{\mathrm{safe}}}$ bounds $\|\nabla h(\Phi z)\|_2$, $\overline{g}_h$ and $\overline{d}_h$ bound $\|2z^\top P g_r\|_2$ and $\|2z^\top P d_r\|_2$, and $M_z := \sup\|f_r(z) + g_r u + d_r w\|_2 + L_{\perp,P}\varepsilon_{\mathrm{basis},P}$ bounds $\|\dot{z}\|_2$. With the dip Δh_{dip} of Section 3, the tightened reduced barrier, its target set, the reduced safe set and the shell partition are

$$\begin{aligned} &\tilde{h}_r(z) := h(\Phi z) - \varepsilon^2_{\mathrm{basis},P} - \Delta h_{\mathrm{dip}}, \qquad \tilde{\mathcal{C}}_r := \{\tilde{h}_r \ge 0\}, \qquad \mathcal{C}_r := \{h(\Phi z) \ge \varepsilon^2_{\mathrm{basis},P}\}, \\ &\mathcal{C}_{r,\mathrm{act}} := \{z \in \mathcal{C}_r : \tilde{h}_r(z) \le h_{\mathrm{act}}\}, \qquad \mathcal{B}_{\mathrm{int}} := \mathcal{C}_r \setminus \mathcal{C}_{r,\mathrm{act}}. \end{aligned} \tag{3}$$

The tightening by $\varepsilon^2_{\mathrm{basis},P}$ pays for the tail through Lemma S2, and Δh_{dip} pays for what the barrier may lose between samples. The interior core is the ball $\mathcal{B}_{\mathrm{int}} = \{\|z\|_P < r_{\min}\}$ with $r^2_{\min} := E_{\mathrm{safe}} - \varepsilon^2_{\mathrm{basis},P} - \Delta h_{\mathrm{dip}} - h_{\mathrm{act}}$, and the activation level $h_{\mathrm{act}} > L_h M_z T_s$ keeps a trajectory sampled in $\mathcal{B}_{\mathrm{int}}$ out of $\{\tilde{h}_r < 0\}$ for one period, so the budget is needed only on the shell $\mathcal{C}_{r,\mathrm{act}}$.

3. The budget

Along (1), for almost every $t \in [t_k, t_{k+1})$, $\dot{\tilde{h}}_r = \mathcal{L}_{f_r}\tilde{h}_r + \mathcal{L}_{g_r}\tilde{h}_r\, u + \mathcal{L}_{d_r}\tilde{h}_r\, w + \nabla\tilde{h}_r^\top\Delta_\perp$, with $\mathcal{L}_{f_r}\tilde{h}_r(z) = -2z^\top P f_r(z)$, $\mathcal{L}_{g_r}\tilde{h}_r(z) = -2z^\top P g_r$ and $\mathcal{L}_{d_r}\tilde{h}_r(z) = -2z^\top P d_r$. The certified

authority δ_{cert} of (6) bounds this rate plus $\gamma_0\tilde{h}_r$ from below at the sample z_k, under the teacher's command $\hat{\mathcal{T}}(z_k)$, with no disturbance and no tail. Each term below bounds one gap between that ideal rate and the plant's.

Tail coupling. Theorem 1(a) keeps $\|x_\perp\|_{P_{\text{full},\epsilon}} \le \varepsilon_{\text{basis},P}$, so $|\nabla\tilde{h}_r^\top \Delta_\perp| \le L_h L_{\perp,P}\varepsilon_{\text{basis},P}$. The constant $L_{\perp,P}$ is at least the norm of the advective block $P_0^{-1}\Phi^\top P_{\text{full}} A_{\text{adv}} P_\perp$, so the tail term is positive whenever this block is nonzero; the rest of $L_{\perp,P}$ comes from the Lipschitz remainder of F_{nl}.

Quantization, verification slack and lag. The student deviates from the teacher by at most $\varepsilon_{\text{CROWN}}(b_B)$, the bound of Proposition 1. The table deviates by at most $\varepsilon_A(b_A) := \varepsilon_{\text{CROWN}}(b_B) + L_{\mathcal{T}}\sqrt{m}D_v/2^{b_A+1}$, the first summand covering table entries computed by the quantized network; the bound also holds on the central bin. All command errors are measured against the teacher at the sample, $\hat{\mathcal{T}}(z_k)$. During $[t_k, t_k+\tau_A)$ the latched command was computed at z_{k-1}; since $\|\dot{v}\|_2 \le M_v := \|g_r^\top P\|_2 M_z$, it differs from $\hat{\mathcal{T}}(z_k)$ by at most $\varepsilon_{\text{CROWN}}(b_B) + \Delta u_{\text{lag}}$ with $\Delta u_{\text{lag}} := L_{\mathcal{T}} M_v T_s = L_{\mathcal{T},z} M_z T_s$. The three latch phases of (2) therefore carry errors $\varepsilon_{\text{CROWN}}(b_B) + \Delta u_{\text{lag}}$, $\varepsilon_A(b_A)$ and $\varepsilon_{\text{CROWN}}(b_B)$, and the exact integration of $\tilde{h}_r$ with decay rate $\gamma_0 > 0$, the linear class-$\mathcal{K}$ gain of the barrier condition in (6), over one period weights them by

$$w_1 := \frac{e^{\gamma_0\tau_A}-1}{e^{\gamma_0 T_s}-1}, \qquad w_2 := \frac{e^{\gamma_0\tau_B}-e^{\gamma_0\tau_A}}{e^{\gamma_0 T_s}-1}, \qquad w_3 := 1-w_1-w_2. \tag{4}$$

The codesign window $\lfloor \log_2(\sqrt{m}D_v/(2M_vT_s))\rfloor + 1 \le b_A \le \lfloor B_{\max}/m\rfloor - 1$ (S30) keeps the table's grid error below Δu_{lag} at its lower end and is the table capacity at its upper end. Proposition 1 is an instance of differential verification (Paulsen et al., 2020; Zhang et al., 2023b) with CROWN-type relaxations (Zhang et al., 2018; Xu et al., 2020) and adds a closed-form companion in the bit width, so that b_B enters the certificate.

Proposition 1 (Twin-network CROWN bound; proof in Proposition S1) *Let $\mathcal{N}_q$ have per-tensor quantized weights with $|[W_\ell^q - W_\ell]_{ij}| \le s_\ell 2^{-b_B}$, so $\|W_\ell^q - W_\ell\|_2 \le \Delta_{W,\ell}$, bias errors of at most $\Delta_{b,\ell}$, and rounding of input, accumulators and activations modelled as injections bounded by ρ_ℓ, with $\Delta_{W,0} = \Delta_{b,0} = 0$. (i) On the shared ReLU routing graph of the GroupSort-2 comparators, every comparator difference of $\Delta_{\text{net}}(z) := \mathcal{N}_q(g_r^\top Pz) - \mathcal{N}(g_r^\top Pz)$ admits a sound two-sided affine relaxation, exact where the comparator sign is fixed in both networks over a box. (ii) Backward substitution over any partition of $\mathcal{Z}_{\text{ODD}}$ into boxes yields $\varepsilon_{\text{CROWN}}(b_B)$ with $\sup_{\mathcal{Z}_{\text{ODD}}} \|\Psi_B - \hat{\mathcal{T}}\|_2 \le \sup_{\mathcal{Z}_{\text{ODD}}} \|\Delta_{\text{net}}\|_2 \le \varepsilon_{\text{CROWN}}(b_B)$. (iii) With $Y := \|g_r^\top P\|_2\sqrt{K}D_{\text{ODD}}/2$ and widths H_ℓ, $\sup_{\mathcal{Z}_{\text{ODD}}} \|\Delta_{\text{net}}\|_2 \le E_q(b_B) := \sum_{\ell=0}^{L}(\Delta_{W,\ell}Y + \Delta_{b,\ell} + \sqrt{H_\ell}\rho_\ell)\prod_{k>\ell}\|W_k^q\|_2$, and the computed ratio $C_{\text{relax}} := \varepsilon_{\text{CROWN}}(b_B)/E_q(b_B)$ is at most $\sqrt{m}$ when every comparator is sign-stable on every box. (iv) If $\Delta_{b,\ell}, \rho_\ell \le c\,2^{-b_B}$, then for fixed architecture, teacher, scales and partition, $E_q(b_B)$ and $\varepsilon_{\text{CROWN}}(b_B)$ are $O(2^{-b_B})$.*

Hold. With $s := t - t_k$, the state has moved by at most $M_z s$, so the certified rate expression $\mathcal{L}_{f_r}\tilde{h}_r + \mathcal{L}_{g_r}\tilde{h}_r u + \gamma_0\tilde{h}_r$ at $z(t)$ differs from its value at z_k by at most $L_{\text{hold}}s$ with $L_{\text{hold}} := (L_{\mathcal{L}_{f_r}\tilde{h}_r} + L_{\mathcal{L}_{g_r}\tilde{h}_r}u_{\max} + \gamma_0 L_h)M_z$. Integrated with the exponential weight, the hold costs $\varphi(\gamma_0 T_s)L_{\text{hold}}T_s$, where $\varphi(\xi) := ((\xi-1)e^\xi+1)/(\xi(e^\xi-1)) \in (\frac{1}{2}, 1)$. The pointwise deficit that the budget does not absorb inside a period integrates to at most the dip, which (3) reserves,

$$\Delta h_{\text{dip}} := \overline{g}_h(\varepsilon_A(b_A) - \varepsilon_{\text{CROWN}}(b_B))(\tau_B - \tau_A) + \overline{g}_h\,\Delta u_{\text{lag}}\,\tau_A + \tfrac{1}{2}L_{\text{hold}}T_s^2. \tag{5}$$

Adversary. For any admissible $w(t)$, $|\mathcal{L}_{d_r}\tilde{h}_r\, w| \le \overline{d}_h \varepsilon_{\mathrm{adv}}$. The plant and the reduced dynamics (1) are separately affine in u and w, so Isaacs' condition holds for the reduced barrier rate, and the disturbance that minimises that rate pointwise is $w^*_{\mathrm{adv}} = -\varepsilon_{\mathrm{adv}}\, \mathcal{L}_{d_r}\tilde{h}_r^\top / \|\mathcal{L}_{d_r}\tilde{h}_r\|_2$ ($w^*_{\mathrm{adv}} = 0$ where $\mathcal{L}_{d_r}\tilde{h}_r = 0$); its Filippov solutions are Carathéodory solutions for some admissible $w(t)$ (Remark S4).

Authority. The certified authority on the shell is

$$\begin{aligned}\delta_{\mathrm{cert}} &:= \inf_{z\in\mathcal{C}_{r,\mathrm{act}}} \left[\mathcal{L}_{f_r}\tilde{h}_r(z) + \mathcal{L}_{g_r}\tilde{h}_r(z)\hat{\mathcal{T}}(z) + \gamma_0\tilde{h}_r(z)\right] \\ &\le\ \delta_{\max} := \inf_{z\in\mathcal{C}_{r,\mathrm{act}}} \sup_{u\in U_{\mathrm{adm}}} \left[\mathcal{L}_{f_r}\tilde{h}_r + \mathcal{L}_{g_r}\tilde{h}_r u + \gamma_0\tilde{h}_r\right].\end{aligned} \tag{6}$$

On the port null space $\{v = 0\}$ the control term vanishes, so passive dissipation must carry the authority there; the metric is chosen so that it does. The reduced linear part is $A_{r,0} = (J_0 - R_0)P_0$ with J_0 skew, $R_0 \succeq 0$ and $P_0 \succ 0$; write $\mu_{2,P}(A) := \sup_{e\neq 0} e^\top(PA + A^\top P)e/(2e^\top Pe)$. If $R_0 \succ 0$, then $\mu_{2,P_0}(A_{r,0}) \le -\lambda_{\min}(R_0)\lambda_{\min}(P_0) =: -\mu_R$. In class (H), where J_0 is canonical, $P_0 = \mathrm{blkdiag}(\Omega_r^2, I)$ and $R_0 = \mathrm{blkdiag}(0, \gamma I)$, the reduced metric $P_\epsilon := \left[\begin{smallmatrix}\Omega_r^2 & \epsilon I\\ \epsilon I & I\end{smallmatrix}\right]$ with $\epsilon = \gamma/2$ is positive definite, $Q_\epsilon := -(P_\epsilon A_{r,0} + A_{r,0}^\top P_\epsilon) = \gamma P_\epsilon$ and $\mu_{2,P_\epsilon}(A_{r,0}) = -\gamma/2 =: -\mu_R$ (Proposition S2, which also gives the admissible range of ϵ).

Lemma 1 (Class-(H) authority, $m \le K$; proof in Lemma S5) *Let (H0)–(H3) hold, let the reduced model be in class (H) with $R_0 = \mathrm{blkdiag}(0, \gamma I)$, $P = P_\epsilon$, $\epsilon = \gamma/2$, and let the shell be fixed with $r^2_{\min} > 0$. Split $F_{\mathrm{nl}}(x) = A_{\mathrm{adv}}x + n(x)$ with $A_{\mathrm{adv},r} := P_0^{-1}\Phi^\top P_{\mathrm{full}} A_{\mathrm{adv}}\Phi$, $n_r(z) := P_0^{-1}\Phi^\top P_{\mathrm{full}} n(\Phi z)$ and $\|P^{1/2}n_r(z)\|_2 \le s_{\mathrm{nl}}\|z\|_P$ on $\mathcal{C}_{r,\mathrm{act}}$. With $M := \gamma P - (PA_{\mathrm{adv},r} + A_{\mathrm{adv},r}^\top P) + 2\eta_{\mathcal{T}} P g_r g_r^\top P$ and $c_M := \lambda_{\min}(M, P) - 2s_{\mathrm{nl}}$,*

$$\delta_{\mathrm{cert}} \ge \min\left\{c_M r^2_{\min} + \gamma_0 h_{\mathrm{act}},\ \ c_M(r^2_{\min} + h_{\mathrm{act}} + \Delta h_{\mathrm{dip}}) - \gamma_0 \Delta h_{\mathrm{dip}}\right\},$$

the minimum being the first branch exactly when $c_M \ge \gamma_0$.

Collecting the terms gives the budget, the Certified Control Authority Condition (H6):

$$\begin{aligned}\delta_{\mathrm{total}}(K, b_A, b_B, T_s) := \underbrace{L_h L_{\perp,P}\varepsilon_{\mathrm{basis},P}}_{\text{tail}} + \underbrace{\overline{g}_h w_2(\varepsilon_A(b_A) - \varepsilon_{\mathrm{CROWN}}(b_B))}_{\text{table}} + \underbrace{\overline{g}_h \varepsilon_{\mathrm{CROWN}}(b_B)}_{\text{network}} \\ + \underbrace{w_1 \overline{g}_h\, \Delta u_{\mathrm{lag}}}_{\text{lag}} + \underbrace{\varphi(\gamma_0 T_s) L_{\mathrm{hold}} T_s}_{\text{hold}} + \underbrace{\overline{d}_h \varepsilon_{\mathrm{adv}}}_{\text{adversary}} \ \le\ \delta_{\mathrm{cert}},\end{aligned} \tag{7}$$

which is (S33) with the table and network parts separated.

4. Main result

Theorem 1 (Full-order safety under multi-rate quantized control) *Assume (H0) $E_{\mathrm{safe}} < \lambda_{\min}(P_{\mathrm{full},\epsilon})\rho^2_{\mathrm{ODD}}$ and $E_{\mathrm{safe}} < \frac{1}{4}D^2_{\mathrm{ODD}}\lambda_{\min}(P)$; (H1) Assumption 1, with $L_{F_{\mathrm{nl}}}$, $M_\perp$, $L_{\perp,P}$ taken over $\mathcal{X}_{\mathrm{ODD}}$; (H2) Assumption 2 ((H2a)–(H2c)), and in class (H), $\gamma < 2\omega_{\min}$ and $\epsilon = \gamma/2$; (H3) the teacher construction of Section 2* (S17)–(S18) *with $H/2 \ge m$, $G_{\mathrm{port}} := g_r^\top P^2 g_r \succ 0$, and (H-sat); (H4) the shell condition $\varepsilon^2_{\mathrm{basis},P} + \Delta h_{\mathrm{dip}} + L_h M_z T_s < E_{\mathrm{safe}}$ and*

$h_{\mathrm{act}} \in (L_h M_z T_s,\ E_{\mathrm{safe}} - \varepsilon^2_{\mathrm{basis},P} - \Delta h_{\mathrm{dip}})$ *(equivalently* $r^2_{\min} > 0$*); (H5)* $b_A \geq 1$ *satisfies the table capacity conditions of Section 2 (the lower end of the codesign window of Section 3 is optional); (H6) the Certified Control Authority Condition (7),* $\delta_{\mathrm{total}}(K, b_A, b_B, T_s) \leq \delta_{\mathrm{cert}}$ $(\leq \delta_{\max})$*, with* δ_{cert} *defined by (6); (H7)* $x(0) = \Phi z(0) + x_\perp(0)$ *with* $z(0) \in \tilde{\mathcal{C}}_r$ *and* $\|x_\perp(0)\|_{P_{\mathrm{full},\epsilon}} \leq M_\perp/\lambda_{K+1}$*. Then the Carathéodory solution* $x(t)$ *of the plant under the multi-rate law (2) and any measurable* $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}})$ *exists on* $[0,\infty)$ *and satisfies, for all* $t \geq 0$ *and* $k \in \mathbb{N}_0$*: (a)* $\|x_\perp(t)\|_{P_{\mathrm{full},\epsilon}} \leq \varepsilon_{\mathrm{basis},P}$*; (b)* $\tilde{h}_r(z(t_k)) \geq 0$ *and* $\tilde{h}_r(z(t)) \geq -\Delta h_{\mathrm{dip}}$*, so* $z(t) \in \mathcal{C}_r$*; (c)* $h(x(t)) \geq \tilde{h}_r(z(t)) + \Delta h_{\mathrm{dip}} \geq 0$*, so* $x(t) \in \mathcal{C}$*.*

Proof sketch (full proof in the supplement, proof of Theorem S1). Step 1 (existence of the Carathéodory solution and reduction to the first exit of x from $\mathcal{X}_{\mathrm{ODD}}$ or of z from $\mathcal{C}_{r,0}$) and Step 5 (Lemma S2 and (a) split $h(x)$ into the reduced barrier and the tail, which gives (c)) are standard. Step 2, tail invariance: (H2a) makes the $P_{\mathrm{full},\epsilon}$-norm of the tail obey a scalar inequality with no cross term; (H2c) and $M_\perp$ give $\frac{d}{dt}\frac{1}{2}\|x_\perp\|^2_{P_{\mathrm{full},\epsilon}} \leq -\lambda_{K+1}\|x_\perp\|^2_{P_{\mathrm{full},\epsilon}} + M_\perp\|x_\perp\|_{P_{\mathrm{full},\epsilon}}$, which with (H7) yields (a). Step 3, exact integration over one period: if $z_k \in \mathcal{B}_{\mathrm{int}}$, Lipschitz continuity of $\tilde{h}_r$ and $h_{\mathrm{act}} > L_h M_z T_s$ keep $\tilde{h}_r > 0$ over the period. If $z_k \in \mathcal{C}_{r,\mathrm{act}}$, adding and subtracting (6) at z_k bounds $\dot{\tilde{h}}_r + \gamma_0 \tilde{h}_r$ below by δ_{cert} minus the tail and adversary terms, the shift $L_{\mathrm{hold}} s$ and $\overline{g}_h$ times the command error of the current phase. Integrating against $e^{\gamma_0 s}$ over the three latch phases of (2) gives exactly the weights (4), the factor φ and the six terms of (7), so (H6) yields $\tilde{h}_r(z_{k+1}) \geq 0$. Step 4, the dip: within a period the instantaneous deficit is bounded phase by phase, and its integral is at most Δh_{dip} of (5), which gives (b).

Stochastic extension. Corollary S1 of the supplement adds a globally Lipschitz diffusion to the plant and lets the disturbance law drift at Wasserstein-2 rate r_{W_2} over a horizon T; the tail radius is then replaced by an Ornstein–Uhlenbeck envelope $\varepsilon_{\mathrm{basis},\sigma}$, and $\varepsilon_{\mathrm{adv}}$ by $\varepsilon_{\mathrm{adv}} + r_{W_2} T$ in all constants, including δ_{cert}. The Itô correction carries the budget into a stochastic budget $\delta_{\mathrm{total},\sigma} + \delta_{\mathrm{stoch}}(\alpha) + \delta_{\mathrm{drift}} \leq \delta_{\mathrm{cert}}$, in which δ_{stoch} contains $\gamma_0 h_{\mathrm{act}}$. If it closes under the hypotheses of Theorem 1 and the remaining conditions of that corollary, and $T \leq T_{\mathrm{TTL}}$, the smallest of a drift-reserve bound, a Kushner-type bound (Kushner, 1967) and a tail horizon, then $\mathbb{P}(\inf_{t\in[0,T]} h(x(t)) \geq 0) \geq 1 - \alpha$. The experiments use $r_{W_2} = 0$, and the corollary is feasible in regime R1 and infeasible in R2 (Section 5).

5. Numerical certification

Benchmark. The plant is a class-(H) telegraph reaction–diffusion–advection lattice, $\dot{q} = p$, $\dot{p} = -(\nu L_\Delta + \alpha_0 I)q - \gamma p - c\, W_f D_h W_f\, q - \kappa(q - \sin q) + Bu + B_w w$ on 160 nodes (320 states), with L_Δ the discrete Laplacian, $\nu = 0.05$, $\alpha_0 = 1$, $\gamma = 1$, $\kappa = 0.05$, $m = 2$ actuators and one disturbance channel with $\varepsilon_{\mathrm{adv}} = 0.002$. The skew central difference D_h is filtered by $W_f = \Phi_f\, \mathrm{diag}(1, 1, w, w)\Phi_f^\top$ on the first 4 sine modes; the deployed cutoff-mode weight is $w^\star = 2^{-11/2} \approx 0.0221$, selected by the sweep below. The advection stays in F_{nl}, so A_{full} is the pure damped wave, $Q_{\gamma/2} = \gamma P_{\gamma/2}$ holds to 7.039×10^{-15}, and (H2a) holds by eigenmode invariance. The reduced model keeps $K_m = 2$ modes ($K = 4$), fewer than the advection couples, so the tail term is active. The teacher is constructed, not trained, with $L = 3$ layers, $H = 8$ and $\theta_{\mathrm{GS}} = 0.05$; the student's rounding offsets are learned, which lowers $\varepsilon_{\mathrm{CROWN}}(b_B)$ at $b_B = 12$ from 0.004718 (nearest rounding) to 0.004455.

Table 1: The six terms of the budget (7), the authority of Lemma 1 against its brute-force minimum and the utilisation (supplement, Experiments 1 and 3); the budget closes in both regimes.

	R1 (Péclet 1)		R2 (Péclet 20)	
term	value	share	value	share
tail, $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P}$	1.835×10^{-5}	0.006967	0.003666	0.2704
table (LUT), $\overline{g}_h w_2(\varepsilon_A(b_A) - \varepsilon_{\mathrm{CROWN}}(b_B))$	7.392×10^{-6}	0.002806	7.393×10^{-5}	0.005452
network (CROWN), $\overline{g}_h \varepsilon_{\mathrm{CROWN}}(b_B)$	2.172×10^{-4}	0.08246	0.002133	0.1573
lag, $w_1 \overline{g}_h \Delta u_{\mathrm{lag}}$	2.26×10^{-5}	0.008579	1.859×10^{-4}	0.01371
hold, $\varphi(\gamma_0 T_s) L_{\mathrm{hold}} T_s$	0.001474	0.5596	0.006607	0.4872
adversary, $\overline{d}_h \varepsilon_{\mathrm{adv}}$	8.944×10^{-4}	0.3396	8.944×10^{-4}	0.06596
δ_{total}	0.002634		0.01356	
δ_{cert}, lower bound (Lemma 1)	0.03161		0.01456	
δ_{cert}, brute force	0.03164		0.01468	
utilisation $\delta_{\mathrm{total}}/\delta_{\mathrm{cert}}$	0.0833		0.931	

Regime R2 has $c = 1$ (Péclet 20) and open-loop growth rate 0.2925, and 16 of 16 open-loop seeds leave $\mathcal{C}$; regime R1 has $c = 0.05$ (Péclet 1). Both use $E_{\mathrm{safe}} = 0.05$, $B_{\max} = 24$, $\log_2 S_{\mathrm{bits}} = 26$, $b_{\mathrm{out}} = 16$ and $b_B = 12$; R2 (R1) uses $\gamma_0 = 0.35$ (0.3), $u_{\max} = 3.4$ (0.35), $L_{\mathcal{T}} = 5$ (0.5), $T_s = 3 \times 10^{-4}$ (0.002), $\tau_A = 1.5 \times 10^{-5}$ (1×10^{-4}) and $\tau_B = 3 \times 10^{-5}$ (2×10^{-4}). The table width $b_A = 9$ lies in the codesign window $[8, 11]$, and the certificate refuses unless (H0)–(H7) all hold, with (H5) checked on this window.

Soundness of the parts. Over 100016 samples of $\mathcal{Z}_{\mathrm{ODD}}$ and b_B from 3 to 14, the twin-network bound of Proposition 1 exceeds the largest sampled student–teacher gap by a factor of 1.887 to 2.844 and roughly halves with every added bit. The class-(H) bound of Lemma 1 is within one percent of the brute-force minimum in both regimes (Table 1); in all numbers below, δ_{cert} denotes this certified lower bound. Over 16 seeds per regime under the rate-minimising adversary w^*_{adv}, from boundary starts (tail at its largest admissible value) and interior starts, conclusions (a)–(c) of Theorem 1 hold at every integrator node.

Conservatism by term. At the 102636 integrator nodes of the certified R2 runs whose sample lies in $\mathcal{C}_{r,\mathrm{act}}$, each pointwise term of the rate inequality behind (7) is compared with the bound the proof uses: the adversary term reaches 0.8772 of its bound, quantization and lag 0.3729, tail coupling 0.3161 and the hold term 0.07439. The hold term, the largest share of δ_{total} in R2 (Table 1), is the least realised, through $\|z - z_k\|_2$ (at most 0.3205 (R2) and 0.5853 (R1) of $M_z s$) and the Lipschitz constant of the reduced rate (at most 0.2323 and 0.1794). The realised rate exceeds the proof's lower bound by at least 0.6597 δ_{total} (R2) and 11.5 δ_{total} (R1) at every shell node, which suggests that a trajectory-dependent bound in place of M_z and these constants could lengthen the admissible period (medians: Experiment 4b of the supplement).

Ablations. Each ablation alters one component at the deployed $w^\star$, and the full certificate refuses each (Figure 1(a)). Ablation (a) is a projection-based reduced-order certificate without a tail term, at $K_m = 1$: the naive certificate accepts, and 4 of 16 seeds leave the safe set. Ablation (b) is a 2-bit network certified as if it were its teacher: the naive certificate without the quantization term accepts, and 9 of 16 seeds leave the safe set. Ablation (c) holds a single command per period; at the

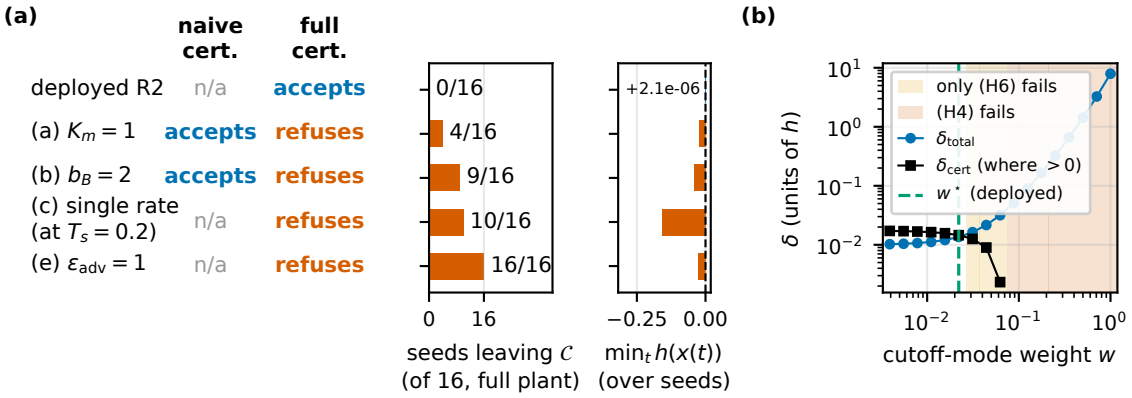


Figure 1: Ablations and the frontier in the certified loop. (a) Naive and full verdicts, the number of the 16 seeds whose full-order trajectory leaves the safe set, and $\min_t h(x(t))$ over the seeds, for the deployed R2 configuration (Experiment 4) and ablations (a), (b), (c) and (e) (Experiment 6, horizon $T = 3$ instead of 5). Row (c) is refused at the deployed period, where none of its seeds exits; its exits are at $T_s = 0.2$, where both loops are uncertified. Where the naive certificate accepts and the full certificate refuses, the plant leaves the safe set. (b) δ_{total} and δ_{cert} (where positive) against the cutoff-mode weight w at Péclet 20, with the bands where only (H6) fails and where (H4) fails; the certificate accepts up to $w^\star$ and refuses above it.

deployed period it has $\delta_{\text{total}} = 0.01702 > \delta_{\text{cert}} = 0.01456$ and is refused, while the multi-rate loop has $\delta_{\text{total}} = 0.01356$, and none of its seeds leaves the safe set there within the horizon. At $T_s = 0.2$, where both loops are uncertified, the single-rate loop exits on 10 of 16 seeds; the multi-rate loop at that period is reported in the supplement. Ablation (e) raises the adversary to $\varepsilon_{\text{adv}} = 1$: all 16 seeds exit, and at the largest certifiable level $\varepsilon^\star_{\text{adv}} = 0.004223$ none does.

Refusals and the frontier. The certificate refuses 6 configurations: a look-up table indexed by z instead of v, whose codesign window is empty, its lower end exceeding its upper end $(9, 5)$; $u_{\max}$ below the (H-sat) bound; unfiltered advection; the plain projector $w = 1$; the longer period $T_s = 5 \times 10^{-4}$, where $\delta_{\text{total}} = 0.01821 > \delta_{\text{cert}} = 0.01455$; and all timing constants scaled by 10. Over 17 values of w from 1 down to 0.003906, with the open loop unsafe at every value, the certificate accepts for $w \le 0.0221$ and refuses above it, through (H6) and then (H4) (Figure 1(b)). For the plain projector, on the scanned grid, the certified advection speeds ($c \le 0.02$, Péclet at most 0.4) and the open-loop-unstable ones ($c \ge 0.5$, Péclet at least 10) are disjoint. The frontier is set by the supremum over the L^2-type domain $\mathcal{X}_{\text{ODD}}$ in $M_\perp$, which charges cutoff-mode advection at full strength. The deadzone result of Proposition S3 holds in R1, where $\mu_R = 0.5$ exceeds the P-norm sector modulus $L_{\text{sec}} = 0.1204$ of $F_{\text{nl},r}$: the mid-tread deadzone silences the table late in the run, while the mid-rise quantizer of ablation (d) chatters at every tested b_A (between 121.6 and 953.1 sign switches per unit time; Experiment 5 of the supplement); in R2, $L_{\text{sec}} = 2.392 > \mu_R$ and it does not apply. Corollary S1 is exercised in R1 at noise level $\sigma^\star = 0.007$: 0 of 2000 paths exit, the Clopper–Pearson upper limit 0.001843 lies below $\alpha = 0.05$, and the horizon bound

$T_{\mathrm{TTL}} = 2022$ exceeds the horizon $T = 2$ by a factor 1011. The corollary is infeasible in R2 because the stochastic budget adds $\gamma_0 h_{\mathrm{act}} = 0.007237$ to δ_{total}, more than the margin 9.994×10^{-4}. ETD2RK step-halving ratios are 4.002 to 4.045. Every constant is computed in float64 from the operators, and the repository's 248 assertions and 65 tests regenerate every number (code: https://github.com/AnuKarthikaS/pgcbf).

6. Limitations and conclusion

The certified plant is the lattice with filtered advection at $w^\star = 0.0221$; unfiltered advection and the plain projector are refused, and certifying full-strength cutoff-mode advection at the unstable Péclet numbers would need a different formulation, for example an H^1-bounded domain, whose effect on a smooth field the filter weight mimics heuristically (Remark S2); the theorem does not have one. The sampling period $T_s = 3 \times 10^{-4}$ is set by the certificate, not by the plant, which is sampled 1.714×10^4 times per slowest plant period in R2; the hold term is 0.4872 of δ_{total}, the price of a continuous-time guarantee with a Lipschitz hold bound (Section 5). On every certified run the minimum of h is attained at the initial instant, and the dip floor $-\Delta h_{\mathrm{dip}}$ is never reached. Theorem 1 assumes exact acquisition of z_k and exact latch instants; rounding of the Path-A input by ε_z adds $L_{\mathcal{T},z}\varepsilon_z$ to $\varepsilon_A(b_A)$ (Remark S3).

The stochastic extension is feasible in R1 only, its horizon bound is 1011 times the tested horizon, and every stochastic run uses $r_{W_2} = 0$. Proposition S3 does not apply in R2 because $L_{\mathrm{sec}} > \mu_R$ there. The evidence is one benchmark class in two regimes. Ablations (a), (b) and (c) are paradigm-level baselines, not reimplementations of published methods; (b) is a 2-bit network and (c) has no naive verdict. Below $w^\star$ the single-rate budget closes (first at $w = 0.01562$), so $w^\star$ is the largest grid value at which every ablation is refused. Under the quadratic barrier, $\theta_{\mathrm{GS}} = 0$ (linear port feedback) maximises $\eta_{\mathcal{T}}$, so on this benchmark the neural teacher is not needed for authority; it is the object whose quantization and verification the budget must charge. A task on which a nonlinear neural policy beats linear port feedback on δ_{cert} is future work. The margin in R2 is $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}} = 9.994 \times 10^{-4}$.

Each of the six gaps between the verified model and the running plant costs one term of (7), and Theorem 1 turns a closed budget into full-order safety for all time; on the lattice the budget closes at utilisation 0.931 and names the hold term as the one to sharpen. The budget is the certificate, and its refusals are results.

Generative AI Use

During the preparation of this manuscript, Anthropic's Claude Opus 5 and GitHub Copilot were used strictly for prose copy-editing, boilerplate code and unit-test generation, and BibTeX formatting. All mathematical formulations, theoretical derivations, proofs, core implementations, and scientific claims are solely the work of the author, who takes full responsibility for the content.

References

Aaron D. Ames, Xiangru Xu, Jessy W. Grizzle, and Paulo Tabuada. Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control*, 62(8): 3861–3876, 2017. doi: 10.1109/TAC.2016.2638961. URL https://doi.org/10.1109/TAC.2016.2638961.

Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *Proceedings of the 18th European Control Conference (ECC)*, pages 3420–3431, 2019. doi: 10.23919/ECC.2019.8796030. URL https://doi.org/10.23919/ECC.2019.8796030.

Cem Anil, James Lucas, and Roger Grosse. Sorting out Lipschitz function approximation. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 291–301, 2019. URL https://proceedings.mlr.press/v97/anil19a.html.

Joseph Breeden, Kunal Garg, and Dimitra Panagou. Control barrier functions in sampled-data systems. *IEEE Control Systems Letters*, 6:367–372, 2022. doi: 10.1109/LCSYS.2021.3076127. URL https://doi.org/10.1109/LCSYS.2021.3076127.

Federico Califano. Passivity-preserving safety-critical control using control barrier functions. *IEEE Control Systems Letters*, 7:1742–1747, 2023. doi: 10.1109/LCSYS.2023.3280082. URL https://doi.org/10.1109/LCSYS.2023.3280082.

Saifon Chaturantabut, Christopher Beattie, and Serkan Gugercin. Structure-preserving model reduction for nonlinear port-Hamiltonian systems. *SIAM Journal on Scientific Computing*, 38 (5):B837–B865, 2016. doi: 10.1137/15M1055085. URL https://doi.org/10.1137/15M1055085.

Andrew Clark. Control barrier functions for stochastic systems. *Automatica*, 130:109688, 2021. doi: 10.1016/j.automatica.2021.109688. URL https://doi.org/10.1016/j.automatica.2021.109688.

Max H. Cohen, Tamas G. Molnar, and Aaron D. Ames. Safety-critical control for autonomous systems: Control barrier functions via reduced-order models. *Annual Reviews in Control*, 57: 100947, 2024. doi: 10.1016/j.arcontrol.2024.100947. URL https://doi.org/10.1016/j.arcontrol.2024.100947.

Max H. Cohen, Noel Csomay-Shanklin, William D. Compton, Tamas G. Molnar, and Aaron D. Ames. Safety-critical controller synthesis with reduced-order models. In *Proceedings of the 2025 American Control Conference (ACC)*, pages 5216–5221, 2025. doi: 10.23919/ACC63710.2025.11108063. URL https://doi.org/10.23919/ACC63710.2025.11108063.

Ryan K. Cosner, Preston Culbertson, and Aaron D. Ames. Bounding stochastic safety: Leveraging Freedman's inequality with discrete-time control barrier functions. *IEEE Control Systems Letters*, 8:1937–1942, 2024. doi: 10.1109/LCSYS.2024.3409105. URL https://doi.org/10.1109/LCSYS.2024.3409105.

Alexander Davydov, Saber Jafarpour, and Francesco Bullo. Non-Euclidean contraction theory for robust nonlinear stability. *IEEE Transactions on Automatic Control*, 67(12):6667–6681, 2022. doi: 10.1109/TAC.2022.3183966. URL https://doi.org/10.1109/TAC.2022.3183966.

Charles Dawson, Sicun Gao, and Chuchu Fan. Safe control with learned certificates: A survey of neural Lyapunov, barrier, and contraction methods for robotics and control. *IEEE Transactions*

on Robotics, 39(3):1749–1767, 2023. doi: 10.1109/TRO.2022.3232542. URL https://doi.org/10.1109/TRO.2022.3232542.

David F. Delchamps. Stabilizing a linear system with quantized state feedback. *IEEE Transactions on Automatic Control*, 35(8):916–924, 1990. doi: 10.1109/9.58500. URL https://doi.org/10.1109/9.58500.

Nicola Elia and Sanjoy K. Mitter. Stabilization of linear systems with limited information. *IEEE Transactions on Automatic Control*, 46(9):1384–1400, 2001. doi: 10.1109/9.948466. URL https://doi.org/10.1109/9.948466.

Mahyar Fazlyab, Alexander Robey, Hamed Hassani, Manfred Morari, and George Pappas. Efficient and accurate estimation of Lipschitz constants for deep neural networks. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/95e1533eb1b20a97777749fb94fdb944-Abstract.html.

Minyue Fu and Lihua Xie. The sector bound approach to quantized feedback control. *IEEE Transactions on Automatic Control*, 50(11):1698–1711, 2005. doi: 10.1109/TAC.2005.858689. URL https://doi.org/10.1109/TAC.2005.858689.

Miroslav Grmela and Hans Christian Öttinger. Dynamics and thermodynamics of complex fluids. I. Development of a general formalism. *Physical Review E*, 56(6):6620–6632, 1997. doi: 10.1103/PhysRevE.56.6620. URL https://doi.org/10.1103/PhysRevE.56.6620.

Serkan Gugercin, Rostyslav V. Polyuga, Christopher Beattie, and Arjan van der Schaft. Structure-preserving tangential interpolation for model reduction of port-Hamiltonian systems. *Automatica*, 48(9):1963–1974, 2012. doi: 10.1016/j.automatica.2012.05.052. URL https://doi.org/10.1016/j.automatica.2012.05.052.

Pushpak Jagtap, Sadegh Soudjani, and Majid Zamani. Formal synthesis of stochastic systems via control barrier certificates. *IEEE Transactions on Automatic Control*, 66(7):3097–3110, 2021. doi: 10.1109/TAC.2020.3013916. URL https://doi.org/10.1109/TAC.2020.3013916.

Mrdjan Jankovic. Robust control barrier functions for constrained stabilization of nonlinear systems. *Automatica*, 96:359–367, 2018. doi: 10.1016/j.automatica.2018.07.004. URL https://doi.org/10.1016/j.automatica.2018.07.004.

Shishir Kolathaya and Aaron D. Ames. Input-to-state safety with control barrier functions. *IEEE Control Systems Letters*, 3(1):108–113, 2019. doi: 10.1109/LCSYS.2018.2853698. URL https://doi.org/10.1109/LCSYS.2018.2853698.

Harold J. Kushner. *Stochastic Stability and Control*, volume 33 of *Mathematics in Science and Engineering*. Academic Press, New York, 1967. doi: 10.1016/S0076-5392(08)X6039-4. URL https://doi.org/10.1016/S0076-5392(08)X6039-4.

Tamas G. Molnar, Ryan K. Cosner, Andrew W. Singletary, Wyatt Ubellacker, and Aaron D. Ames. Model-free safety-critical control for robotic systems. *IEEE Robotics and Automation Letters*, 7

(2):944–951, 2022. doi: 10.1109/LRA.2021.3135569. URL https://doi.org/10.1109/LRA.2021.3135569.

Luyao Niu, Bhaskar Ramasubramanian, Andrew Clark, and Radha Poovendran. Sampling and quantization-aware control barrier functions for safety-critical control of cyber-physical systems. In *Proceedings of the 63rd IEEE Conference on Decision and Control (CDC)*, pages 1637–1644, 2024. doi: 10.1109/CDC56724.2024.10886708. URL https://doi.org/10.1109/CDC56724.2024.10886708.

Patricia Pauli, Anne Koch, Julian Berberich, Paul Kohler, and Frank Allgöwer. Training robust neural networks using Lipschitz bounds. *IEEE Control Systems Letters*, 6:121–126, 2022. doi: 10.1109/LCSYS.2021.3050444. URL https://doi.org/10.1109/LCSYS.2021.3050444.

Brandon Paulsen, Jingbo Wang, and Chao Wang. ReluDiff: Differential verification of deep neural networks. In *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering (ICSE)*, pages 714–726, 2020. doi: 10.1145/3377811.3380337. URL https://doi.org/10.1145/3377811.3380337.

Stephen Prajna, Ali Jadbabaie, and George J. Pappas. A framework for worst-case and stochastic safety verification using barrier certificates. *IEEE Transactions on Automatic Control*, 52(8):1415–1428, 2007. doi: 10.1109/TAC.2007.902736. URL https://doi.org/10.1109/TAC.2007.902736.

Cesar Santoyo, Maxence Dutreix, and Samuel Coogan. A barrier function approach to finite-time stochastic system verification and control. *Automatica*, 125:109439, 2021. doi: 10.1016/j.automatica.2020.109439. URL https://doi.org/10.1016/j.automatica.2020.109439.

Andrew Singletary, Yuxiao Chen, and Aaron D. Ames. Control barrier functions for sampled-data systems with input delays. In *Proceedings of the 59th IEEE Conference on Decision and Control (CDC)*, pages 804–809, 2020. doi: 10.1109/CDC42340.2020.9304281. URL https://doi.org/10.1109/CDC42340.2020.9304281.

Andrew Singletary, Shishir Kolathaya, and Aaron D. Ames. Safety-critical kinematic control of robotic systems. *IEEE Control Systems Letters*, 6:139–144, 2022. doi: 10.1109/LCSYS.2021.3050609. URL https://doi.org/10.1109/LCSYS.2021.3050609.

Oswin So, Andrew Clark, and Chuchu Fan. Almost-sure safety guarantees of stochastic zero-control barrier functions do not hold. arXiv preprint arXiv:2312.02430, 2023. URL https://arxiv.org/abs/2312.02430.

Gustaf Söderlind. The logarithmic norm. History and modern theory. *BIT Numerical Mathematics*, 46(3):631–652, 2006. doi: 10.1007/s10543-006-0069-9. URL https://doi.org/10.1007/s10543-006-0069-9.

Xiao Tan, Ersin Daş, Aaron D. Ames, and Joel W. Burdick. Zero-order control barrier functions for sampled-data systems with state and input dependent safety constraints. In *Proceedings of*

the 2025 American Control Conference (ACC), pages 283–290, 2025. doi: 10.23919/ACC63710.2025.11107720. URL https://doi.org/10.23919/ACC63710.2025.11107720.

Andrew J. Taylor, Victor D. Dorobantu, Ryan K. Cosner, Yisong Yue, and Aaron D. Ames. Safety of sampled-data systems with control barrier functions via approximate discrete time models. In *Proceedings of the 61st IEEE Conference on Decision and Control (CDC)*, pages 7127–7134, 2022. doi: 10.1109/CDC51059.2022.9993226. URL https://doi.org/10.1109/CDC51059.2022.9993226.

Jesus-Pablo Toledo-Zucco, Denis Matignon, Charles Poussot-Vassal, and Yann Le Gorrec. Structure-preserving discretization and model order reduction of boundary-controlled 1D port-Hamiltonian systems. *Systems & Control Letters*, 194:105947, 2024. doi: 10.1016/j.sysconle.2024.105947. URL https://doi.org/10.1016/j.sysconle.2024.105947.

Arjan van der Schaft and Dimitri Jeltsema. Port-Hamiltonian systems theory: An introductory overview. *Foundations and Trends in Systems and Control*, 1(2-3):173–378, 2014. doi: 10.1561/2600000002. URL https://doi.org/10.1561/2600000002.

Shiqi Wang, Huan Zhang, Kaidi Xu, Xue Lin, Suman Jana, Cho-Jui Hsieh, and J. Zico Kolter. Beta-CROWN: Efficient bound propagation with per-neuron split constraints for neural network robustness verification. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, pages 29909–29921, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/fac7fead96dafceaf80c1daffeae82a4-Abstract.html.

Kaidi Xu, Zhouxing Shi, Huan Zhang, Yihan Wang, Kai-Wei Chang, Minlie Huang, Bhavya Kailkhura, Xue Lin, and Cho-Jui Hsieh. Automatic perturbation analysis for scalable certified robustness and beyond. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, pages 1129–1141, 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/0cbc5671ae26f67871cb914d81ef8fc1-Abstract.html.

Lujie Yang, Hongkai Dai, Zhouxing Shi, Cho-Jui Hsieh, Russ Tedrake, and Huan Zhang. Lyapunov-stable neural control for state and output feedback: A novel formulation. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 56033–56046, 2024. URL https://proceedings.mlr.press/v235/yang24f.html.

Hongchao Zhang, Junlin Wu, Yevgeniy Vorobeychik, and Andrew Clark. Exact verification of ReLU neural control barrier functions. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, pages 5685–5705, 2023a. doi: 10.52202/075280-0248. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/120ed726cf129dbeb8375b6f8a0686f8-Abstract-Conference.html.

Huan Zhang, Tsui-Wei Weng, Pin-Yu Chen, Cho-Jui Hsieh, and Luca Daniel. Efficient neural network robustness certification with general activation functions. In *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018. URL https://proceedings.neurips.cc/paper/2018/hash/d04863f100d59b3eb688a11f95b0ae60-Abstract.html.

Yedi Zhang, Fu Song, and Jun Sun. QEBVerif: Quantization error bound verification of neural networks. In *Computer Aided Verification (CAV 2023)*, volume 13965 of *Lecture Notes in Computer Science*, pages 413–437. Springer, 2023b. doi: 10.1007/978-3-031-37703-7_20. URL https://doi.org/10.1007/978-3-031-37703-7_20.

Proceedings of Machine Learning Research vol vvv:1–48, 2027

Supplementary Material for "One Budget for Every Gap: Certifying Reduced-Order, Quantized, Sampled-Data Neural Barrier Controllers on the Full Plant"

Anu Karthika IAMANUKARTHIKA@GMAIL.COM
Independent Researcher, Chennai, India

Abstract

This supplement provides the full mathematical formulation, proofs, numerical validation, and reproducibility record for the main paper. Section S1 states the full-order plant model, the Petrov–Galerkin reduction, the Lyapunov metric, the barrier, and the multi-rate controller schedule. Section S2 defines the spectral tail contraction constants and the GroupSort-2 teacher and quantized student architecture. Section S3 states Proposition S1, Proposition S2, Lemmas S4 and S5, Proposition S3, Theorem S1, Remarks S2–S4, and Corollary S1. Section S4 contains the proofs of all propositions, lemmas, Theorem S1, and Corollary S1. Section S5 reports the numerical validation across Experiments 0–9 with all tables and figures, Section S6 documents the repository layout and execution pipeline, and Section S7 maps each item of the main paper to its counterpart and page in this supplement.

S1. Notation and Standing Assumptions

Consider a nonlinear control-affine dynamical system evolving on a full-order state space $\mathcal{X} \subseteq \mathbb{R}^N$ ($N \gg 1$), governed by the semi-discretized differential equation

$$dx(t) = \Big(F(x(t)) + G\,u(t) + D\,w(t)\Big)dt + \Sigma(x(t))\,dW_t, \tag{S1}$$

where $x(t) \in \mathcal{X}_{\mathrm{ODD}} \subset \mathbb{R}^N$ denotes the state within a compact Operational Design Domain (ODD) containing the origin in its interior, $u(t) \in U_{\mathrm{adm}} := \{u \in \mathbb{R}^m : \|u\|_2 \le u_{\max}\}$ is the norm-bounded control input, $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}}) := \{w \in \mathbb{R}^{n_w} : \|w\|_2 \le \varepsilon_{\mathrm{adv}}\}$ is a bounded, Lebesgue-measurable exogenous or adversarial disturbance, $G \in \mathbb{R}^{N\times m}$ and $D \in \mathbb{R}^{N\times n_w}$ are constant input matrices, W_t is a standard n_W-dimensional Wiener process, and $\Sigma(x) \in \mathbb{R}^{N\times n_W}$ is the diffusion coefficient ($\Sigma \equiv 0$ in the deterministic setting of Theorem S1). All variables are nondimensionalized (positions, momenta, and field amplitudes scaled by fixed reference values), so that Euclidean norms of vectors mixing generalized coordinates and momenta are meaningful and every constant below is a pure number. Let $\rho_{\mathrm{ODD}} := \sup\{\rho > 0 : \{x : \|x\|_2 < \rho\} \subseteq \mathcal{X}_{\mathrm{ODD}}\} > 0$ denote the radius of the largest origin-centred Euclidean ball contained in $\mathcal{X}_{\mathrm{ODD}}$. In the deterministic case, solutions of (S1) are understood in the Carathéodory sense (absolutely continuous, satisfying the integral equation for the given measurable w and the piecewise-constant u of (S9)). The main paper calls δ_{total} the budget and δ_{cert} the certified authority; δ_{cert} is a certified lower bound on the brute-force authority, and in all numbers δ_{cert} denotes this lower bound.

Assumption S1 (Dissipative Linear Port-Hamiltonian Part with Lipschitz Residual) *The full-order drift decomposes into a linear structure operator and a nonlinear residual, $F(x) = (J_{\mathrm{full}} - R_{\mathrm{full}})P_{\mathrm{full}}x + F_{\mathrm{nl}}(x)$ with $F_{\mathrm{nl}}(0) = 0$, where $J_{\mathrm{full}} = -J_{\mathrm{full}}^T \in \mathbb{R}^{N\times N}$ is skew-symmetric, $R_{\mathrm{full}} = R_{\mathrm{full}}^T \succeq 0$ is symmetric positive semi-definite, $P_{\mathrm{full}} = P_{\mathrm{full}}^T \succ 0$ defines the physical energy norm $\|x\|_{P_{\mathrm{full}}} := \sqrt{x^T P_{\mathrm{full}} x}$, F_{nl} is locally Lipschitz on $\mathbb{R}^N$, and $\|F_{\mathrm{nl}}(x_1) - F_{\mathrm{nl}}(x_2)\|_2 \le L_{F_{\mathrm{nl}}}\|x_1 - x_2\|_2$ for all $x_1, x_2 \in \mathcal{X}_{\mathrm{ODD}}$. We write $A_{\mathrm{full}} := (J_{\mathrm{full}} - R_{\mathrm{full}})P_{\mathrm{full}}$.*

© 2027 A. Karthika.

The decomposition carries no structural content beyond linearity of the port-Hamiltonian part; all nonlinear effects are lumped into F_{nl} and enter only through $L_{F_{\mathrm{nl}}}$ and the suprema defined below.

Reduced coordinates (Petrov–Galerkin). Let $\Phi \in \mathbb{R}^{N\times K}$ ($K \ll N$) have full column rank, $\mathcal{V}_r := \mathrm{Im}(\Phi)$, and define

$$\begin{aligned} P_0 &:= \Phi^T P_{\mathrm{full}} \Phi \succ 0, & z(x) &:= P_0^{-1}\Phi^T P_{\mathrm{full}}\, x \in \mathbb{R}^K, \\ P_\perp &:= I_N - \Phi P_0^{-1}\Phi^T P_{\mathrm{full}}, & x_\perp &:= P_\perp x. \end{aligned} \tag{S2}$$

Then $P_\perp^2 = P_\perp$, $x = \Phi z(x) + x_\perp$ exactly, and $\Phi^T P_{\mathrm{full}} x_\perp = \Phi^T P_{\mathrm{full}} x - P_0 z(x) = 0$ for every $x \in \mathbb{R}^N$; $\mathcal{V}_\perp := \mathrm{Im}(P_\perp)$. If Φ is an orthonormal eigenbasis with $P_{\mathrm{full}}\Phi = \Phi P_0$ (mass-normalized modal coordinates), then $P_0^{-1}\Phi^T P_{\mathrm{full}} = \Phi^T$, so (S2) reduces to $z = \Phi^T x$ and $P_\perp = I_N - \Phi\Phi^T$. For second-order Hamiltonian systems ($x = [q^T, p^T]^T \in \mathbb{R}^{2N_m}$), $\Phi = \mathrm{blkdiag}(\Phi_q, \Phi_q)$ with $\Phi_q^T\Phi_q = I_{K_m}$ and $K = 2K_m$ even; $p_r := \Phi_q^T p$ denotes the reduced momenta.

Lyapunov metric and barrier. We consider two classes. *Class (D), strictly dissipative:* $R_{\mathrm{full}} \succ 0$ and $P_{\mathrm{full},\epsilon} := P_{\mathrm{full}}$. *Class (H), underdamped second-order Hamiltonian:* $J_{\mathrm{full}} = \left[\begin{smallmatrix} 0 & I_{Nm} \\ -I_{Nm} & 0 \end{smallmatrix}\right]$, $P_{\mathrm{full}} = \mathrm{blkdiag}(\Omega_{\mathrm{full}}^2, I_{N_m})$, where $\Omega_{\mathrm{full}}^2 = (\Omega_{\mathrm{full}}^2)^T \succ 0$ is the mass-normalized stiffness matrix with eigenvalues $\omega_1^2 \le \cdots \le \omega_{N_m}^2$ and $\omega_{\min} := \omega_1$ the smallest *full-order* natural frequency, $R_{\mathrm{full}} = \mathrm{blkdiag}(0, R_p)$ with momentum damping $R_p = R_p^T \succeq 0$, and a damping level $\gamma > 0$ is fixed with

$$0 < \gamma < 2\omega_{\min}, \qquad \epsilon := \gamma/2, \qquad P_{\mathrm{full},\epsilon} := \begin{bmatrix} \Omega_{\mathrm{full}}^2 & \epsilon I_{N_m} \\ \epsilon I_{N_m} & I_{N_m} \end{bmatrix} \succ 0 \tag{S3}$$

(positive definite because its Schur complement $\Omega_{\mathrm{full}}^2 - \epsilon^2 I \succeq (\omega_{\min}^2 - \epsilon^2) I \succ 0$). The same ϵ is used for the tail and for the reduced metric (there is no separate $\epsilon_\perp$). $\|x\|_{P_{\mathrm{full},\epsilon}}^2 = x^T P_{\mathrm{full},\epsilon} x$ is a Lyapunov metric, not a physical energy. The barrier is defined on all of $\mathbb{R}^N$ as

$$h(x) := E_{\mathrm{safe}} - x^T P_{\mathrm{full},\epsilon}\, x, \qquad \mathcal{C} := \{x \in \mathcal{X}_{\mathrm{ODD}} : h(x) \ge 0\}, \tag{S4}$$

with a threshold $E_{\mathrm{safe}} > 0$ satisfying the ODD containment hypothesis

$$\text{(H0)} \qquad E_{\mathrm{safe}} < \lambda_{\min}(P_{\mathrm{full},\epsilon})\, \rho_{\mathrm{ODD}}^2. \tag{S5}$$

Lemma S1 (ODD containment) *Under (H0),* $\mathcal{C} = \{x \in \mathbb{R}^N : x^T P_{\mathrm{full},\epsilon} x \le E_{\mathrm{safe}}\}$, *and* $\mathcal{C} \subset \mathrm{int}(\mathcal{X}_{\mathrm{ODD}})$.

Proof If $x^T P_{\mathrm{full},\epsilon} x \le E_{\mathrm{safe}}$, then $\lambda_{\min}(P_{\mathrm{full},\epsilon})\|x\|_2^2 \le x^T P_{\mathrm{full},\epsilon} x \le E_{\mathrm{safe}} < \lambda_{\min}(P_{\mathrm{full},\epsilon})\rho_{\mathrm{ODD}}^2$ (Rayleigh quotient), so $\|x\|_2 < \rho_{\mathrm{ODD}}$. The open ball $\{\|x\|_2 < \rho_{\mathrm{ODD}}\}$ is open and contained in $\mathcal{X}_{\mathrm{ODD}}$, hence contained in $\mathrm{int}(\mathcal{X}_{\mathrm{ODD}})$. Thus the ellipsoid lies in $\mathrm{int}(\mathcal{X}_{\mathrm{ODD}})$ and coincides with $\mathcal{C}$. ■

Safety and Multi-Rate Timing Specification. The digital controller operates at sampling period $T_s > 0$ with sample instants $t_k = kT_s$ ($k \in \mathbb{N}_0$) and sampled reduced states $z_k := z(x(t_k))$. The per-axis width $D_{\mathrm{ODD}} > 0$ is a given design constant and $\mathcal{Z}_{\mathrm{ODD}} := [-D_{\mathrm{ODD}}/2, D_{\mathrm{ODD}}/2]^K$. The controller uses a dual-rate execution schedule built from a b_B-bit fixed-point network map $\Psi_B : \mathbb{R}^K \to U_{\mathrm{adm}}$ whose construction and error certificate are given in Section S2 and Proposition S1:

1. **Fast Feedforward LUT Path (Ψ_A):** At each sample instant t_k, parallelized FPGA adder-tree projection acquires $z_k \in \mathcal{Z}_{\mathrm{ODD}}$ and, by the same adder tree, the port vector

$$v_k := g_r^T P\, z_k \in \mathbb{R}^m, \tag{S6}$$

with the reduced input matrix g_r of (S10) and the reduced metric P of (S13) (the Teacher and the Student of Section S2.2 depend on z only through v), within an ultra-short projection sub-interval $\tau_{\mathrm{proj}} < \tau_A$, leaving $\tau_A - \tau_{\mathrm{proj}}$ for quantization and table look-up. Path A is indexed by v_k. Let $D_v := 2\sup_{z\in\mathcal{C}_{r,0}} \|g_r^T P z\|_\infty$ be the per-axis port width over the untightened reduced safe set $\mathcal{C}_{r,0} = \{z^T P z \le E_{\mathrm{safe}}\}$ of (S27); in closed form,

$$D_v = 2\sqrt{E_{\mathrm{safe}}}\, \max_{1\le j\le m} \sqrt{(g_r^T P g_r)_{jj}}, \tag{S7}$$

because $|e_j^T g_r^T P z| = |(P^{1/2} g_r e_j)^T (P^{1/2} z)| \le \|P^{1/2} g_r e_j\|_2\, \|z\|_P$ (Cauchy–Schwarz in the P-inner product, with equality at $z \propto g_r e_j$) and $\|P^{1/2} g_r e_j\|_2^2 = (g_r^T P g_r)_{jj}$. Hence $v_k \in \mathcal{V}_{\mathrm{port}} := [-D_v/2, D_v/2]^m$ whenever $z_k \in \mathcal{C}_{r,0}$; only $z_k \in \mathcal{C}_{r,0}$ is required for the LUT domain, which Step 0 of Section S4.4 ensures on $[0, t^*)$. The quantizer Q_{b_A} is the mid-tread lattice with $2^{b_A}+1$ levels $\{j\Delta_{q,A} : j = -2^{b_A-1}, \dots, 2^{b_A-1}\}$ per port axis, step $\Delta_{q,A} := D_v/2^{b_A}$ ($b_A \ge 1$), and half-open bins $[(j-\frac{1}{2})\Delta_{q,A}, (j+\frac{1}{2})\Delta_{q,A})$ (the outermost bins closed at $\pm D_v/2$; indices saturate at the outermost level for $\|v\|_\infty > D_v/2$, a case that does not occur under Theorem S1), so that $\|Q_{b_A}(v) - v\|_\infty \le \Delta_{q,A}/2$ for every $v \in \mathcal{V}_{\mathrm{port}}$ and the zero bin is the open deadzone $\mathcal{B}_0 := \{v \in \mathbb{R}^m : \|v\|_\infty < \Delta_{q,A}/2\}$. The table stores the quantized student's value $\Psi_A(v_q) := \Pi_U(\mathcal{N}_q(v_q))$ for lattice points $v_q \ne 0$ (evaluated offline, matching `controller.py psi_A, lut_space="port"`; by Proposition S1, its deviation from $\hat{\mathcal{T}}_v(v_q)$ of (S18) is at most $\varepsilon_{\mathrm{CROWN}}(b_B)$, reserved in (S29)) and $\Psi_A(0) := 0$ (deadzone; only Path A is dead-zoned), and latches $u_{A,k} = \Psi_A(Q_{b_A}(v_k))$ at latency $\tau_A < T_s$. With $B_{\max}$ address bits, on-chip memory S_{bits}, and b_{out} bits per output component, the address-width and storage conditions are

$$m(b_A+1) \le B_{\max}, \qquad m(b_A+1) + \lceil \log_2(m\, b_{\mathrm{out}}) \rceil \le \log_2 S_{\mathrm{bits}}, \tag{S8}$$

since $\lceil \log_2(2^{b_A}+1) \rceil = b_A + 1$ address bits are needed per port axis and the table holds $2^{m(b_A+1)}$ words of $m\, b_{\mathrm{out}}$ bits. Any decode latency of the table (e.g., multi-cycle access at large depth) is part of τ_A.

2. **Fine Quantized Neural Path (Ψ_B):** A b_B-bit ($b_B > b_A$) quantized neural network executes across the multi-layer matrix-vector pipeline and commits a refined command $u_{B,k} = \Psi_B(z_k)$ at latency $\tau_B \in (\tau_A, T_s)$. The piecewise-constant control applied to the physical plant over $t \in [t_k, t_{k+1})$ is

$$u(t) = \begin{cases} u_{B,k-1}, & t \in [t_k,\ t_k + \tau_A), \\ u_{A,k}, & t \in [t_k + \tau_A,\ t_k + \tau_B), \\ u_{B,k}, & t \in [t_k + \tau_B,\ t_k + T_s), \end{cases} \tag{S9}$$

where, prior to closed-loop release at $t_0 = 0$, the plant is held in standby at rest ($x(t) \equiv x(0)$ for $t \in [-T_s, 0]$) and the actuator register is loaded causally with $u_{B,-1} := \Psi_B(z(x(-T_s))) = \Psi_B(z_0)$.

S2. Reduction, Quantized Synthesis and the Budget Terms

S2.1. Petrov–Galerkin Projection and Spectral Tail Contraction

Applying $z(\cdot)$ of (S2) to (S1) (with $\Sigma \equiv 0$) and using $x = \Phi z + x_\perp$ gives the exact reduced dynamics

$$\dot{z}(t) = \underbrace{\big[A_{r,0} z + F_{\mathrm{nl},r}(z)\big]}_{=:f_r(z)} + \underbrace{P_0^{-1}\Phi^T P_{\mathrm{full}} G}_{=:g_r} u(t) + \underbrace{P_0^{-1}\Phi^T P_{\mathrm{full}} D}_{=:d_r} w(t) + \Delta_\perp(z, x_\perp), \tag{S10}$$

where $A_{r,0} := (J_0 - R_0)P_0$ is the open-loop reduced operator with $J_0 := P_0^{-1}\Phi^T P_{\mathrm{full}} J_{\mathrm{full}} P_{\mathrm{full}}\Phi P_0^{-1} = -J_0^T$ and $R_0 := P_0^{-1}\Phi^T P_{\mathrm{full}} R_{\mathrm{full}} P_{\mathrm{full}}\Phi P_0^{-1} = R_0^T \succeq 0$ (so that $P_0^{-1}\Phi^T P_{\mathrm{full}} A_{\mathrm{full}}\Phi = (J_0 - R_0)P_0$ identically), $F_{\mathrm{nl},r}(z) := P_0^{-1}\Phi^T P_{\mathrm{full}} F_{\mathrm{nl}}(\Phi z)$, and

$$\Delta_\perp(z, x_\perp) := A_{r\perp} x_\perp + P_0^{-1}\Phi^T P_{\mathrm{full}}\big[F_{\mathrm{nl}}(\Phi z + x_\perp) - F_{\mathrm{nl}}(\Phi z)\big], \qquad A_{r\perp} := P_0^{-1}\Phi^T P_{\mathrm{full}} A_{\mathrm{full}} P_\perp.$$

In the eigenbasis case these reduce to $J_0 = \Phi^T J_{\mathrm{full}}\Phi$, $R_0 = \Phi^T R_{\mathrm{full}}\Phi$, $g_r = \Phi^T G$, $d_r = \Phi^T D$; in class (H) with orthonormal Φ_q, J_0 is canonical and $P_0 = \mathrm{blkdiag}(\Phi_q^T \Omega_{\mathrm{full}}^2 \Phi_q, I_{K_m})$. The cross-coupling vanishes, $A_{r\perp} = 0$, if and only if $\mathcal{V}_r$ is invariant under the P_{full}-adjoint $P_{\mathrm{full}}^{-1} A_{\mathrm{full}}^T P_{\mathrm{full}}$ (because $\ker P_\perp^T = P_{\mathrm{full}}\mathcal{V}_r$ and $A_{r\perp} = 0 \iff P_\perp^T A_{\mathrm{full}}^T P_{\mathrm{full}}\Phi = 0$); when $P_{\mathrm{full}}\mathcal{V}_r = \mathcal{V}_r$ this is $A_{\mathrm{full}}^T \mathcal{V}_r \subseteq \mathcal{V}_r$. Invariance of $\mathcal{V}_r$ under A_{full} itself is neither necessary nor sufficient. $A_{r\perp} \neq 0$ captures non-commuting advective or elastic cross-coupling.

Assumption S2 (Exponentially Dissipative Complement in the Unified Metric) *With $P_{\mathrm{full},\epsilon}$ as in (S4)–(S3):*

(H2a) Tail $P_{\mathrm{full},\epsilon}$-orthogonality: $\Phi^T P_{\mathrm{full},\epsilon} P_\perp = 0$. *(In class (D) this holds automatically by (S2). In class (H) with $\Phi = \mathrm{blkdiag}(\Phi_q, \Phi_q)$, $\Phi_q^T\Phi_q = I$, it holds if and only if $\mathrm{Im}(\Phi_q)$ is Ω_{full}^2-invariant, e.g. for any set of mass-normalized eigenmodes; see Lemma S7 in Section S4.2.)*

(H2b) Tail damping (class (H)): *the momentum-damping block restricted to the tail, $R_{p,\perp} := (I - \Phi_q\Phi_q^T)R_p(I - \Phi_q\Phi_q^T)$, satisfies $v^T R_{p,\perp} v > \epsilon\|v\|_2^2$ for all nonzero $v \in \mathrm{Im}(I - \Phi_q\Phi_q^T)$. (This is necessary for (S11); general SPD R_p violating it yields an indefinite tail dissipation form.)*

(H2c) Spectral gap: *there is $\lambda_{K+1} > 0$ with*

$$-x_\perp^T\Big(P_{\mathrm{full},\epsilon} A_{\mathrm{full}} + A_{\mathrm{full}}^T P_{\mathrm{full},\epsilon}\Big)x_\perp \geq 2\,\lambda_{K+1}\|x_\perp\|_{P_{\mathrm{full},\epsilon}}^2 \quad \forall x_\perp \in \mathcal{V}_\perp, \tag{S11}$$

with uniformly bounded tail excitation over the ODD

$$M_\perp := \sup_{(x,u,w)\in\mathcal{X}_{\mathrm{ODD}}\times U_{\mathrm{adm}}\times\mathcal{B}(\varepsilon_{\mathrm{adv}})} \big\|P_\perp\big[A_{\mathrm{full}}\Phi z(x) + F_{\mathrm{nl}}(x) + Gu + Dw\big]\big\|_{P_{\mathrm{full},\epsilon}} < \infty,\ z(x) \text{ in (S2)}. \tag{S12}$$

$M_\perp$ is finite because $\mathcal{X}_{\mathrm{ODD}}$ is compact and the bracket is continuous. Only upper bounds are used below, so $M_\perp$ may be replaced by any larger number; we take $M_\perp > 0$ without loss of generality (if the supremum vanishes, any positive value for which the hypotheses of Theorem S1 hold may be used).

Remark S1 (Tail rates for the two benchmark classes) (i) *Uniformly damped lattice* (class (H), $R_p = \gamma I$, $\mathrm{Im}\Phi_q$ spanned by eigenmodes): A_{full} and $P_{\mathrm{full},\epsilon}$ decouple into 2×2 modal blocks, and the largest admissible rate of mode i is the pencil rate $r(\epsilon, \omega_i) = \frac{\gamma}{2} - \frac{\omega_i |\gamma - 2\epsilon|}{2\sqrt{\omega_i^2 - \epsilon^2}} \le \frac{\gamma}{2}$ (Lemma S6); hence $\lambda_{K+1} := \min_{i > K_m} r(\epsilon, \omega_i)$, which equals $\gamma/2$ for every tail mode at $\epsilon = \gamma/2$ (and (H2b) holds since $\gamma > \gamma/2$). The tail rate does *not* grow with K; truncation enters only through $M_\perp$ and $L_{\perp,P}$. (ii) *Reaction–diffusion–advection* $\partial_t x = \nu \partial_{\xi\xi} x - c_{\mathrm{adv}} \partial_\xi x - \alpha_0 x +$ (nonlinear) on $(0, \ell)$ with Dirichlet conditions (class (D), $P_{\mathrm{full}} = I$ in L^2, advection skew): with Φ the first K sine modes, $\lambda_{K+1} = \alpha_0 + \nu\big((K+1)\pi/\ell\big)^2$ for the continuum and $\lambda_{K+1} = \alpha_0 + \nu \frac{4}{h^2} \sin^2 \frac{(K+1)\pi h}{2\ell}$ for the central finite-difference semi-discretization with mesh $h = \ell/(N+1)$ (exact discrete sine eigenvalues; the FD value is smaller). Both require $\lambda_{K+1} > 0$, which may fail for $\alpha_0 < 0$; α_0 is retained. See Remark S2.

Lemma S2 (Quadratic splitting) *Under (H2a), for every $x \in \mathbb{R}^N$ with $z = z(x)$, $x_\perp = P_\perp x$: $\Phi^T P_{\mathrm{full},\epsilon} x_\perp = 0$, $\|x\|^2_{P_{\mathrm{full},\epsilon}} = z^T P z + \|x_\perp\|^2_{P_{\mathrm{full},\epsilon}}$, and*

$$h(\Phi z + x_\perp) = h(\Phi z) - \|x_\perp\|^2_{P_{\mathrm{full},\epsilon}}, \qquad h(\Phi z) = E_{\mathrm{safe}} - z^T P z, \qquad P := \Phi^T P_{\mathrm{full},\epsilon} \Phi. \tag{S13}$$

Proof $\Phi^T P_{\mathrm{full},\epsilon} x_\perp = \Phi^T P_{\mathrm{full},\epsilon} P_\perp x = 0$ by (H2a). Expanding,

$$\begin{aligned}(\Phi z + x_\perp)^T P_{\mathrm{full},\epsilon} (\Phi z + x_\perp) &= z^T \Phi^T P_{\mathrm{full},\epsilon} \Phi z + 2 z^T (\Phi^T P_{\mathrm{full},\epsilon} x_\perp) + x_\perp^T P_{\mathrm{full},\epsilon} x_\perp \\ &= z^T P z + \|x_\perp\|^2_{P_{\mathrm{full},\epsilon}};\end{aligned}$$

subtract from E_{safe}. ∎

The reduced metric is $P = P_0$ in class (D) and $P = P_\epsilon := \begin{bmatrix} \Omega_r^2 & \epsilon I \\ \epsilon I & I \end{bmatrix}$ in class (H) with eigenmode Φ_q ($\Omega_r = \mathrm{diag}(\omega_1, \ldots, \omega_{K_m})$). Under (H0) and Lemma S2, $x \in \mathcal{C}$ implies $\Phi z(x) \in \mathcal{C}$ (since $\|\Phi z\|^2_{P_{\mathrm{full},\epsilon}} \le \|x\|^2_{P_{\mathrm{full},\epsilon}}$). We define the $P_{\mathrm{full},\epsilon}$-norm tail coupling constant $L_{\perp,P}$ as the smallest constant such that

$$\big\| P_0^{-1} \Phi^T P_{\mathrm{full}} \big[A_{\mathrm{full}} x_\perp + F_{\mathrm{nl}}(\Phi z + x_\perp) - F_{\mathrm{nl}}(\Phi z) \big] \big\|_2 = \|\Delta_\perp(z, x_\perp)\|_2 \le L_{\perp,P} \, \|x_\perp\|_{P_{\mathrm{full},\epsilon}} \tag{S14}$$

for all $x = \Phi z + x_\perp \in \mathcal{X}_{\mathrm{ODD}}$ with $\Phi z \in \mathcal{X}_{\mathrm{ODD}}$. It is finite: $L_{\perp,P} \le \|A_{r\perp} P_{\mathrm{full},\epsilon}^{-1/2}\|_2 + \|P_0^{-1} \Phi^T P_{\mathrm{full}}\|_2 \, L_{F_{\mathrm{nl}}} \, \lambda_{\min}(P_{\mathrm{full},\epsilon})^{-1/2}$ (using $A_{r\perp} x_\perp = P_0^{-1} \Phi^T P_{\mathrm{full}} A_{\mathrm{full}} x_\perp$ because $P_\perp x_\perp = x_\perp$, and Assumption S1); the defined supremum (S14), not this upper bound, is what enters the certificate. Finally, the $P_{\mathrm{full},\epsilon}$-norm tail radius is

$$\varepsilon_{\mathrm{basis},P} := M_\perp / \lambda_{K+1}. \tag{S15}$$

S2.2. Isometric GroupSort-2 Teacher and Quantized Student

The continuous network $\mathcal{N} : \mathbb{R}^m \to \mathbb{R}^m$ acts on the port output $v := g_r^T P z \in \mathbb{R}^m$. It is an origin-preserving ($b_\ell = 0$) L-layer network ($L \ge 2$) of hidden width H (even, $H/2 \ge m$) with pairwise GroupSort-2 (MaxMin) activations implemented via the exact ReLU routing identity

$$\sigma_{\mathrm{GS2}}(y_{2i-1}, y_{2i}) = \begin{bmatrix} \max(y_{2i-1}, y_{2i}) \\ \min(y_{2i-1}, y_{2i}) \end{bmatrix} = \begin{bmatrix} y_{2i} + \mathrm{ReLU}(y_{2i-1} - y_{2i}) \\ y_{2i-1} - \mathrm{ReLU}(y_{2i-1} - y_{2i}) \end{bmatrix}. \tag{S16}$$

σ_{GS2} is norm-preserving ($\|\sigma_{\mathrm{GS2}}(y)\|_2 = \|y\|_2$) and non-expansive ($\|\sigma_{\mathrm{GS2}}(y) - \sigma_{\mathrm{GS2}}(\tilde{y})\|_2 \le \|y - \tilde{y}\|_2$). Let $U_{\mathrm{sum}} := \frac{1}{\sqrt{2}} \sum_{i=1}^{H/2} (e_{2i-1} + e_{2i}) e_i^T$ and $U_{\mathrm{diff}} := \frac{1}{\sqrt{2}} \sum_{i=1}^{H/2} (e_{2i-1} - e_{2i}) e_i^T \in \mathbb{R}^{H \times H/2}$, whose ranges are exactly $\mathrm{span}\{e_{2i-1} + e_{2i}\}$ and $\mathrm{span}\{e_{2i-1} - e_{2i}\}$, aligned with the GS2 pairing ($U_{\mathrm{sum}}^T U_{\mathrm{sum}} = U_{\mathrm{diff}}^T U_{\mathrm{diff}} = I_{H/2}$, $U_{\mathrm{sum}}^T U_{\mathrm{diff}} = 0$, $[U_{\mathrm{sum}}\ U_{\mathrm{diff}}] \in O(H)$). Fix a mixing angle $\theta_{\mathrm{GS}} \in (0, \pi/4)$, $Q_{\mathrm{mix}} \in O(H/2)$, a semi-orthogonal input matrix $W_{\mathrm{in}} \in \mathbb{R}^{(H/2) \times m}$ ($W_{\mathrm{in}}^T W_{\mathrm{in}} = I_m$), and set

$$
\begin{aligned}
&W_1 = \cos(\theta_{\mathrm{GS}}) U_{\mathrm{sum}} W_{\mathrm{in}} + \sin(\theta_{\mathrm{GS}}) U_{\mathrm{diff}} Q_{\mathrm{mix}} W_{\mathrm{in}} \in \mathbb{R}^{H \times m}, \qquad W_1^T W_1 = I_m, \\
&W_\ell = U_{\mathrm{sum}} Q_{\mathrm{sum},\ell} U_{\mathrm{sum}}^T + U_{\mathrm{diff}} Q_{\mathrm{diff},\ell} U_{\mathrm{diff}}^T \in O(H)\ (2 \le \ell \le L-1,\ Q_{\mathrm{sum},\ell}, Q_{\mathrm{diff},\ell} \in O(H/2)), \\
&W_L = -L_{\mathcal{T}} (W_{L-1} \cdots W_1)^T \in \mathbb{R}^{m \times H}, \qquad \mathcal{N}(v) := W_L\, \sigma_{\mathrm{GS2}}(W_{L-1} \cdots \sigma_{\mathrm{GS2}}(W_1 v)).
\end{aligned}
\tag{S17}
$$

Hence $\|W_\ell\|_2 \le 1$ for $\ell < L$, $\|W_L\|_2 = L_{\mathcal{T}}$, $\mathcal{N}(0) = 0$, and $\mathcal{N}$ is $L_{\mathcal{T}}$-Lipschitz in ℓ_2. The Teacher and the quantized Student are

$$
\hat{\mathcal{T}}(z) := \Pi_U\big(\mathcal{N}(g_r^T P z)\big), \qquad \Psi_B(z) := \Pi_U\big(\mathcal{N}_q(g_r^T P z)\big), \qquad \Pi_U(a) := a \min\big(1, u_{\max}/\|a\|_2\big),
\tag{S18}
$$

where $\mathcal{N}_q$ is the b_B-bit fixed-point realization of $\mathcal{N}$ specified in Proposition S1 and Π_U is the (radial = metric) projection onto U_{adm}, which satisfies $\|\Pi_U a - \Pi_U b\|_2 \le \|a - b\|_2$. Consequently $\hat{\mathcal{T}}(0) = 0$, $\hat{\mathcal{T}}$ and Ψ_B take values in U_{adm}, every Lipschitz constant and every bound on $\|\Psi_B - \hat{\mathcal{T}}\|_2$ obtained for the unprojected maps carries over, and $\hat{\mathcal{T}}$ is $L_{\mathcal{T},z}$-Lipschitz in z with

$$
L_{\mathcal{T},z} := L_{\mathcal{T}}\, \|g_r^T P\|_2.
\tag{S19}
$$

We also write $\hat{\mathcal{T}}_v(v) := \Pi_U(\mathcal{N}(v))$ for the Teacher as a function of the port vector, so that $\hat{\mathcal{T}}(z) = \hat{\mathcal{T}}_v(g_r^T P z)$, $\hat{\mathcal{T}}_v(0) = 0$ and $\hat{\mathcal{T}}_v$ is $L_{\mathcal{T}}$-Lipschitz on $\mathbb{R}^m$ (non-expansiveness of Π_U); Path A of Section S1 tabulates $\hat{\mathcal{T}}_v$ on the port lattice. We assume the port Gram matrix $G_{\mathrm{port}} := g_r^T P^2 g_r \succ 0$ (full column rank of g_r), so the port map $z \mapsto g_r^T P z$ is onto $\mathbb{R}^m$ and its null space has dimension exactly $K - m$, and the saturation hypothesis

$$
\text{(H-sat)} \qquad L_{\mathcal{T},z} \sqrt{K}\, D_{\mathrm{ODD}}/2 \le u_{\max},
\tag{S20}
$$

under which $\|\mathcal{N}(g_r^T P z)\|_2 \le L_{\mathcal{T},z} \|z\|_2 \le L_{\mathcal{T},z} \sqrt{K} D_{\mathrm{ODD}}/2 \le u_{\max}$ on $\mathcal{Z}_{\mathrm{ODD}}$, so Π_U acts as the identity on the image $\mathcal{N}(g_r^T P\, \mathcal{Z}_{\mathrm{ODD}})$. (Without (H-sat), Lemma S3 still holds with $\eta_{\mathcal{T}}$ replaced by $\eta_{\mathcal{T}} \min(1, u_{\max}/(L_{\mathcal{T}} \|v\|_2))$.)

Lemma S3 (Sector alignment; proof in Section S4.3) *For the construction* (S17)–(S18) *under (H-sat), every $z \in \mathcal{Z}_{\mathrm{ODD}}$ satisfies $-z^T P g_r \hat{\mathcal{T}}(z) \ge \eta_{\mathcal{T}} \|g_r^T P z\|_2^2$ with $\eta_{\mathcal{T}} = L_{\mathcal{T}} \cos(2\theta_{\mathrm{GS}}) > 0$; in particular $z^T P g_r \hat{\mathcal{T}}(z) \le 0$.*

While the linear sum channel U_{sum} enforces this baseline P-dissipative floor, the rectified difference channel U_{diff} splits the port space by first-layer comparator signs into at most $2^{H/2}$ polyhedral cones (all linear regions of $\mathcal{N}$ are cones because $\mathcal{N}$ is positively homogeneous), providing piecewise-linear performance shaping and actuator allocation beyond a static linear gain $u = -k_{\mathrm{lin}}\, g_r^T P z$ (which avoids saturation on $\mathcal{Z}_{\mathrm{ODD}}$ only if $k_{\mathrm{lin}} \|g_r^T P\|_2 \sqrt{K} D_{\mathrm{ODD}}/2 \le u_{\max}$). The control term vanishes on the $(K-m)$-dimensional subspace $\{g_r^T P z = 0\}$, which meets every level set of $h(\Phi z)$; there safety must be carried by passive dissipation (condition (S38) below).

S3. Main Theoretical Guarantees

S3.1. Coupled Twin-Network Verification and Structure-Preserving Contraction

Proposition S1 (Coupled Twin-Network CROWN Verification Bound) *Let $\mathcal{N}$ be the network (S17) with layer widths $H_0 := m$, $H_\ell := H$ ($1 \le \ell \le L-1$), $H_L := m$, weights satisfying $\|W_\ell\|_2 \le 1$ ($\ell < L$), $\|W_L\|_2 = L_{\mathcal{T}}$, and zero biases. Let $\mathcal{N}_q$ be its fixed-point realization with weights W_ℓ^q obtained by per-tensor quantization with scale $s_\ell > 0$ such that $|[W_\ell^q - W_\ell]_{ij}| \le s_\ell 2^{-b_B}$, hence $\|W_\ell^q - W_\ell\|_2 \le \Delta_{W,\ell} \le \sqrt{H_\ell H_{\ell-1}}\, s_\ell\, 2^{-b_B}$, biases b_ℓ^q with $\|b_\ell^q\|_2 \le \Delta_{b,\ell}$, and rounding of the input, accumulators and activations modelled as injections $e_0 \in [-\rho_0, \rho_0]^{H_0}$ (input $v + e_0$) and $e_\ell \in [-\rho_\ell, \rho_\ell]^{H_\ell}$ added to each quantized pre-activation, $y_\ell^q = W_\ell^q \sigma_{\mathrm{GS2}}(y_{\ell-1}^q) + b_\ell^q + e_\ell$ (with $\sigma_{\mathrm{GS2}}(y_0^q) := v + e_0$). Write y_ℓ, y_ℓ^q for the pre-activations as functions of z (through $v = g_r^T P z$) and of the injections. Over any sub-hyperrectangle $\mathcal{Z}_s = [\underline{z}_s, \overline{z}_s] \subset \mathcal{Z}_{\mathrm{ODD}}$, let $d_{\ell,i}(z) := y_{\ell,2i-1}(z) - y_{\ell,2i}(z) \in [\underline{d}_{\ell,i}, \overline{d}_{\ell,i}]$ and $d_{\ell,i}^q(z) := y_{\ell,2i-1}^q(z) - y_{\ell,2i}^q(z) \in [\underline{d}_{\ell,i}^q, \overline{d}_{\ell,i}^q]$ denote the layer-ℓ comparators on the shared ReLU routing graph (S16), with perturbation $\delta d_{\ell,i}(z) := d_{\ell,i}^q(z) - d_{\ell,i}(z) \in [\underline{\delta}_{\ell,i}, \overline{\delta}_{\ell,i}]$, all intervals being produced by the same backward pass over $\mathcal{Z}_s$ and the injection boxes. Let $\mathcal{I}_\ell^+(\mathcal{Z}_s) := \{i : \underline{d}_{\ell,i} \ge 0,\ \underline{d}_{\ell,i}^q \ge 0\}$, $\mathcal{I}_\ell^-(\mathcal{Z}_s) := \{i : \overline{d}_{\ell,i} \le 0,\ \overline{d}_{\ell,i}^q \le 0\}$, and $\mathcal{I}_\ell^*(\mathcal{Z}_s) := \{1, \dots, H/2\} \setminus (\mathcal{I}_\ell^+ \cup \mathcal{I}_\ell^-)$ (which contains every comparator whose sorting order can flip under quantization). Then:*

(i) In the coupled difference graph $\Delta_{\mathrm{net}}(z) := \mathcal{N}_q(g_r^T P z) - \mathcal{N}(g_r^T P z)$, every comparator difference $\nu_{\ell,i}(z) := \mathrm{ReLU}(d_{\ell,i}^q(z)) - \mathrm{ReLU}(d_{\ell,i}(z))$ admits the sound two-sided affine relaxation in $\delta d_{\ell,i}(z)$

$$\alpha_{\ell,i,s}^L\, \delta d_{\ell,i}(z) + \beta_{\ell,i,s}^L \le \nu_{\ell,i}(z) \le \alpha_{\ell,i,s}^U\, \delta d_{\ell,i}(z) + \beta_{\ell,i,s}^U, \tag{S21}$$

with non-negative slopes $\alpha_{\ell,i,s}^L, \alpha_{\ell,i,s}^U \in [0,1]$, exact (zero-slack) coefficients $\alpha^L = \alpha^U = 1$, $\beta^L = \beta^U = 0$ on $\mathcal{I}_\ell^+(\mathcal{Z}_s)$ and $\alpha^L = \alpha^U = 0$, $\beta^L = \beta^U = 0$ on $\mathcal{I}_\ell^-(\mathcal{Z}_s)$, and, for $i \in \mathcal{I}_\ell^*(\mathcal{Z}_s)$, the coefficients (all depending on s through $[\underline{\delta}_{\ell,i}, \overline{\delta}_{\ell,i}]$)

$$(\alpha_{\ell,i,s}^L, \beta_{\ell,i,s}^L, \alpha_{\ell,i,s}^U, \beta_{\ell,i,s}^U) = \begin{cases} (0,\ 0,\ 1,\ 0), & \text{if } \underline{\delta}_{\ell,i} \ge 0, \\ (1,\ 0,\ 0,\ 0), & \text{if } \overline{\delta}_{\ell,i} \le 0, \\ \frac{1}{\overline{\delta}_{\ell,i} - \underline{\delta}_{\ell,i}} \big(-\underline{\delta}_{\ell,i},\ \underline{\delta}_{\ell,i}\, \overline{\delta}_{\ell,i},\ \overline{\delta}_{\ell,i},\ -\underline{\delta}_{\ell,i}\, \overline{\delta}_{\ell,i} \big), & \text{if } \underline{\delta}_{\ell,i} < 0 < \overline{\delta}_{\ell,i}. \end{cases} \tag{S22}$$

The pointwise envelope $\min(0, \delta d) \le \nu \le \max(0, \delta d)$ underlying (S22) is attained (upper side at $d_{\ell,i} = 0$, lower side at $d_{\ell,i} = -\delta d_{\ell,i}$); the secant chord is exact at the endpoints of $[\underline{\delta}_{\ell,i}, \overline{\delta}_{\ell,i}]$.

(ii) Propagating (S21) by backward linear substitution over $\mathcal{Z}_s$ (for the upper bound of each output row, choosing (α^U, β^U) for comparator i where its accumulated coefficient $c_{\ell,i} := [\Lambda W_{\ell+1}^q]_{\cdot,2i-1} - [\Lambda W_{\ell+1}^q]_{\cdot,2i}$ is non-negative and (α^L, β^L) otherwise, and the reverse for the lower bound; Λ is the backward coefficient matrix accumulated from the output) yields component-wise bounds $[\Delta_{\mathrm{net}}(z)]_j \in [\underline{D}_{s,j}, \overline{D}_{s,j}]$, $j = 1, \dots, m$, and for any fixed partition

$\mathcal{Z}_{\mathrm{ODD}} = \bigcup_{s=1}^{M_{\mathrm{sub}}} \mathcal{Z}_s$ *of the* whole *domain into* M_{sub} *boxes,*

$$\sup_{z\in\mathcal{Z}_{\mathrm{ODD}}} \|\Psi_B(z) - \hat{\mathcal{T}}(z)\|_2 \le \sup_{z\in\mathcal{Z}_{\mathrm{ODD}}} \|\Delta_{\mathrm{net}}(z)\|_2 \le \max_{1\le s\le M_{\mathrm{sub}}} \left(\sum_{j=1}^{m} \max\big(|\underline{D}_{s,j}|^2,\, |\overline{D}_{s,j}|^2\big)\right)^{1/2} =: \varepsilon_{\mathrm{CROWN}}(b_B). \tag{S23}$$

(iii) (Explicit constant.) With $Y := \|g_r^T P\|_2 \sqrt{K} D_{\mathrm{ODD}}/2$ *(so* $\|v\|_2 \le Y$ *on* $\mathcal{Z}_{\mathrm{ODD}}$*) and the convention* $\Delta_{W,0} = \Delta_{b,0} = 0$*,*

$$\sup_{z\in\mathcal{Z}_{\mathrm{ODD}}} \|\Delta_{\mathrm{net}}(z)\|_2 \le E_q(b_B) := \sum_{\ell=0}^{L} \Big(\Delta_{W,\ell}\, Y + \Delta_{b,\ell} + \sqrt{H_\ell}\,\rho_\ell\Big) \prod_{k>\ell} \|W_k^q\|_2, \qquad \varepsilon_{\mathrm{CROWN}}(b_B) \le C_{\mathrm{relax}}\, E_q(b_B), \tag{S24}$$

where $\prod_{k>\ell} \|W_k^q\|_2 \le (L_{\mathcal{T}} + \Delta_{W,L}) \prod_{k<L}(1+\Delta_{W,k})$ *for* $\ell < L$ *(the empty product for* $\ell = L$ *is* 1*), and* $C_{\mathrm{relax}} := \varepsilon_{\mathrm{CROWN}}(b_B)/E_q(b_B) \in (0,\infty)$ *is the (computable) relaxation factor of the architecture and partition;* $C_{\mathrm{relax}} \le \sqrt{m}$ *whenever every comparator lies in* $\mathcal{I}_\ell^+ \cup \mathcal{I}_\ell^-$ *on every box. Since both* $\varepsilon_{\mathrm{CROWN}}(b_B)$ *and* $E_q(b_B)$ *are sound,* $\min(\varepsilon_{\mathrm{CROWN}}(b_B), E_q(b_B))$ *may be used in place of* $\varepsilon_{\mathrm{CROWN}}(b_B)$ *throughout.*

(iv) (Rate.) If $\Delta_{b,\ell}, \rho_\ell \le c\, 2^{-b_B}$ *for a constant* c*, then for fixed architecture, teacher weights, scales* s_ℓ *and partition,* $E_q(b_B) = O(2^{-b_B})$ *and* $\varepsilon_{\mathrm{CROWN}}(b_B) = O(2^{-b_B})$*.*

Proposition S2 (Structure-Preserving Contraction via P-Weighted Matrix Measure) *Consider the linearized reduced error dynamics* $\dot{e} = A_{r,0} e$ *with* $A_{r,0} = (J_0 - R_0)P_0$*, where* $J_0 = -J_0^T$*,* $R_0 = R_0^T \succeq 0$*, and* $P_0 = P_0^T \succ 0$*.*

1. ***Strictly Dissipative Case (*** $R_0 \succ 0$***):*** *In the energy metric* $\|e\|_{P_0} := \sqrt{e^T P_0 e}$*, the conservative skew-symmetric term vanishes identically (*$(P_0 e)^T J_0 (P_0 e) = 0$ *for all* $e \in \mathbb{R}^K$*), and the* P_0*-weighted Dahlquist matrix measure satisfies*

$$\mu_{2,P_0}(A_{r,0}) := \sup_{e\neq 0} \frac{e^T (P_0 A_{r,0} + A_{r,0}^T P_0) e}{2\, e^T P_0 e} \le -\lambda_{\min}(R_0)\, \lambda_{\min}(P_0) =: -\mu_R < 0. \tag{S25}$$

2. ***Underdamped Hamiltonian Case (*** $R_0 = \mathrm{blkdiag}(0, \gamma I_{K_m})$***,*** $\gamma > 0$***):*** *For canonical second-order systems with* $J_0 = \left[\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right]$ *and* $P_0 = \mathrm{blkdiag}(\Omega_r^2, I_{K_m})$ *(*$\Omega_r = \mathrm{diag}(\omega_1, \ldots, \omega_{K_m}) \succ 0$*), the cross-weighted symmetric metric* $P_\epsilon := \left[\begin{smallmatrix} \Omega_r^2 & \epsilon I \\ \epsilon I & I \end{smallmatrix}\right]$ *satisfies* $P_\epsilon \succ 0$ *and* $Q_\epsilon := -(P_\epsilon A_{r,0} + A_{r,0}^T P_\epsilon) \succ 0$ *for every* $\epsilon \in \big(0, \frac{4\gamma\omega_{\min}^2}{\gamma^2 + 4\omega_{\min}^2}\big)$*. The choice* $\epsilon = \gamma/2$ *lies in this interval if and only if* $\gamma < 2\omega_{\min}$*; for it,* $Q_{\gamma/2} = \gamma P_{\gamma/2}$ *and*

$$\mu_{2,P_{\gamma/2}}(A_{r,0}) = -\gamma/2 =: -\mu_R, \qquad \|e(t)\|_{P_{\gamma/2}} = e^{-\gamma t/2} \|e(0)\|_{P_{\gamma/2}}. \tag{S26}$$

Moreover $(1 - \epsilon/\omega_{\min}) P_0 \preceq P_\epsilon \preceq (1 + \epsilon/\omega_{\min}) P_0$*.*

Consequently, in both cases, an ellipsoidal error tube $\mathcal{E}_P(r) = \{e \in \mathbb{R}^K : \|e\|_P \le r\}$ *contracts strictly at rate* $e^{-\mu_R \Delta t} < 1$*, whereas axis-aligned interval (*ℓ_∞*) propagation incurs artificial conservative wrapping growth* $\mu_\infty(A_{r,0}) = \max_i \big((A_{r,0})_{ii} + \sum_{j\neq i} |(A_{r,0})_{ij}|\big) > 0$ *whenever off-diagonal skew-symmetric couplings dominate diagonal damping.*

S3.2. Sampled-Data Constants and the Tightened Reduced Barrier

All reduced constants are evaluated on the *untightened* reduced safe set

$$\mathcal{C}_{r,0} := \{z \in \mathcal{Z}_{\mathrm{ODD}} : h(\Phi z) \geq 0\} = \{z \in \mathbb{R}^K : z^T P z \leq E_{\mathrm{safe}}\} \subset \mathrm{int}(\mathcal{Z}_{\mathrm{ODD}}), \tag{S27}$$

where the equality and the inclusion follow from the containment condition $E_{\mathrm{safe}} < \frac{1}{4} D_{\mathrm{ODD}}^2 \lambda_{\min}(P)$, assumed together with (H0): $z^T P z \leq E_{\mathrm{safe}}$ gives $\|z\|_\infty \leq \|z\|_2 \leq (E_{\mathrm{safe}}/\lambda_{\min}(P))^{1/2} < D_{\mathrm{ODD}}/2$. Since $\nabla_z h(\Phi z) = \Phi^T \nabla_x h(\Phi z) = -2Pz$ does not depend on any constant level shift, we define on $\mathcal{C}_{r,0}$ the Lie derivatives $\mathcal{L}_{f_r}\tilde{h}_r(z) := -2z^T P f_r(z)$, $\mathcal{L}_{g_r}\tilde{h}_r(z) := -2z^T P g_r$, $\mathcal{L}_{d_r}\tilde{h}_r(z) := -2z^T P d_r$ (valid for any $\tilde{h}_r$ of the form $h(\Phi z) - \text{const}$), and the constants

$$\begin{aligned} L_h &:= \sup_{z \in \mathcal{C}_{r,0}} \|2Pz\|_2 = 2\sqrt{\lambda_{\max}(P) E_{\mathrm{safe}}},\ \bar{g}_h := \sup_{z \in \mathcal{C}_{r,0}} \|\mathcal{L}_{g_r}\tilde{h}_r(z)\|_2,\ \bar{d}_h := \sup_{z \in \mathcal{C}_{r,0}} \|\mathcal{L}_{d_r}\tilde{h}_r(z)\|_2, \\ M_z &:= \sup_{(z,u,w) \in \mathcal{C}_{r,0} \times U_{\mathrm{adm}} \times \mathcal{B}(\varepsilon_{\mathrm{adv}})} \|f_r(z) + g_r u + d_r w\|_2 + L_{\perp,P}\, \varepsilon_{\mathrm{basis},P}, \end{aligned} \tag{S28}$$

and $L_{\mathcal{L}_{f_r}\tilde{h}_r}$, $L_{\mathcal{L}_{g_r}\tilde{h}_r}$ the Lipschitz constants of $\mathcal{L}_{f_r}\tilde{h}_r$ and $\mathcal{L}_{g_r}\tilde{h}_r$ on $\mathcal{C}_{r,0}$ (the latter equals $2\|g_r^T P\|_2$). Not one of these depends on Δh_{dip}, h_{act} or the shell, and $\varepsilon_{\mathrm{basis},P}$ depends only on $M_\perp$ and λ_{K+1}, so the definitions are acyclic. The $P_{\mathrm{full},\epsilon}$-norm form of the tail tolerance removes $\lambda_{\min}(P_{\mathrm{full},\epsilon})$ entirely from the certificate.

Path-A error and co-design window. By Proposition S1, the Path-B QNN satisfies $\|\Psi_B(z) - \hat{\mathcal{T}}(z)\|_2 \leq \varepsilon_{\mathrm{CROWN}}(b_B)$ on $\mathcal{Z}_{\mathrm{ODD}}$. Path A is indexed by the port vector $v = g_r^T P z$ of (S6); since $\hat{\mathcal{T}}(z) = \hat{\mathcal{T}}_v(v)$ with $\hat{\mathcal{T}}_v$ $L_{\mathcal{T}}$-Lipschitz in v, and $\|Q_{b_A}(v) - v\|_2 \leq \sqrt{m}\,\|Q_{b_A}(v) - v\|_\infty \leq \sqrt{m}\,\Delta_{q,A}/2 = \sqrt{m} D_v / 2^{b_A+1}$ for $z \in \mathcal{C}_{r,0}$, the Path-A LUT $\Psi_A(Q_{b_A}(v))$ satisfies, for $Q_{b_A}(v) \neq 0$,

$$\begin{aligned} \|\Psi_A(Q_{b_A}(v)) - \hat{\mathcal{T}}(z)\|_2 &\leq \|\Psi_A(Q_{b_A}(v)) - \hat{\mathcal{T}}_v(Q_{b_A}(v))\|_2 + \|\hat{\mathcal{T}}_v(Q_{b_A}(v)) - \hat{\mathcal{T}}_v(v)\|_2 \\ &\leq \varepsilon_{\mathrm{CROWN}}(b_B) + L_{\mathcal{T}} \frac{\sqrt{m}\, D_v}{2^{b_A+1}} =: \varepsilon_A(b_A), \end{aligned} \tag{S29}$$

where the first term is at most $\varepsilon_{\mathrm{CROWN}}(b_B)$ because $\Psi_A(v_q) = \Pi_U(\mathcal{N}_q(v_q))$ stores the quantized student (Proposition S1), while for $Q_{b_A}(v) = 0$ (deadzone, $\Psi_A = 0$, $\|v\|_\infty < \Delta_{q,A}/2$) $\|0 - \hat{\mathcal{T}}(z)\|_2 = \|\hat{\mathcal{T}}_v(0) - \hat{\mathcal{T}}_v(v)\|_2 \leq L_{\mathcal{T}}\|v\|_2 \leq L_{\mathcal{T}}\sqrt{m} D_v / 2^{b_A+1} \leq \varepsilon_A(b_A)$. Let $M_v := \|g_r^T P\|_2\, M_z$ (so that $\|\dot{v}\|_2 \leq M_v$ along the reduced dynamics on $\mathcal{C}_{r,0}$) and define $\Delta u_{\mathrm{lag}} := L_{\mathcal{T}} M_v T_s$; this constant is invariant under the port-space substitution, $L_{\mathcal{T}} M_v T_s = L_{\mathcal{T}} \|g_r^T P\|_2 M_z T_s = L_{\mathcal{T},z} M_z T_s$ by (S19), so every downstream quantity built from Δu_{lag} (Δh_{dip}, δ_{total}, $\bar{\varepsilon}_q$, r_{ss}) is unchanged in value. All command errors on $[t_k, t_{k+1})$ are measured against $\hat{\mathcal{T}}(z_k)$ at the current sample z_k: on $[t_k, t_k + \tau_A)$, with $u_{B,-1} := \Psi_B(z_0)$, $\|u_{B,-1} - \hat{\mathcal{T}}(z_0)\|_2 \leq \varepsilon_{\mathrm{CROWN}}(b_B)$ at $k = 0$ and $\|u_{B,k-1} - \hat{\mathcal{T}}(z_k)\|_2 \leq \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}$ for $k \geq 1$. The co-design window is

$$\left\lfloor \log_2 \left(\frac{\sqrt{m}\, D_v}{2 M_v T_s} \right) \right\rfloor + 1 \;\leq\; b_A \;\leq\; \left\lfloor \frac{B_{\max}}{m} \right\rfloor - 1, \tag{S30}$$

together with the storage condition in (S8). The upper bound is the address-width condition of (S8). The lower bound is a design choice that is not used in the proof of Theorem S1 (which holds for every $b_A \geq 1$); it ensures $\varepsilon_{\mathrm{CROWN}}(b_B) < \varepsilon_A(b_A) < \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}$: indeed $2^{b_A} \geq 2^{\lfloor \log_2 y \rfloor + 1} > y$ with $y = \sqrt{m} D_v / (2 M_v T_s)$ gives $L_{\mathcal{T}}\sqrt{m} D_v / 2^{b_A+1} < L_{\mathcal{T}} M_v T_s = \Delta u_{\mathrm{lag}}$, and $L_{\mathcal{T}}\sqrt{m} D_v > 0$ ($D_v > 0$ because g_r has full column rank) gives the left inequality.

Exponential weights, hold factor, dip buffer and margin. For sampling period T_s, delays $0 < \tau_A < \tau_B < T_s$ and a linear barrier decay rate $\gamma_0 > 0$ (the rate used in the certificate (S37) below), define the normalized exponential sub-interval weights

$$w_1(\gamma_0) := \frac{e^{\gamma_0\tau_A} - 1}{e^{\gamma_0 T_s} - 1}, \quad w_2(\gamma_0) := \frac{e^{\gamma_0\tau_B} - e^{\gamma_0\tau_A}}{e^{\gamma_0 T_s} - 1}, \quad w_3(\gamma_0) := \frac{e^{\gamma_0 T_s} - e^{\gamma_0\tau_B}}{e^{\gamma_0 T_s} - 1}, \tag{S31}$$

the dimensionless hold-integral factor $\varphi(\xi) := \frac{(\xi-1)e^{\xi}+1}{\xi(e^{\xi}-1)} \in (\frac{1}{2}, 1)$,[1] the hold-rate constant $L_{\text{hold}} := (L_{\mathcal{L}_{f_r}\tilde{h}_r} + L_{\mathcal{L}_{g_r}\tilde{h}_r} u_{\max} + \gamma_0 L_h) M_z$, the inter-sample continuous-time dip buffer

$$\Delta h_{\text{dip}} := \overline{g}_h \big(\varepsilon_A(b_A) - \varepsilon_{\text{CROWN}}(b_B)\big)(\tau_B - \tau_A) + \overline{g}_h \, \Delta u_{\text{lag}} \, \tau_A + \frac{1}{2} L_{\text{hold}} T_s^2, \tag{S32}$$

and the multi-rate sampled-data budget

$$\begin{aligned}\delta_{\text{total}}(K, b_A, b_B, T_s) := \underbrace{L_h L_{\perp,P} \, \varepsilon_{\text{basis},P}}_{\text{ROM Tail Coupling}} + \underbrace{\overline{g}_h \Big[w_2(\gamma_0) \, \varepsilon_A(b_A) + \big(1 - w_2(\gamma_0)\big) \varepsilon_{\text{CROWN}}(b_B) \Big]}_{\text{Exponentially Weighted Multi-Rate Quantization Error}} \\ + \underbrace{w_1(\gamma_0) \, \overline{g}_h \, \Delta u_{\text{lag}}}_{\text{Pre-Arrival Lag Rate}} + \underbrace{\overline{d}_h \, \varepsilon_{\text{adv}}}_{\text{Adversary}} + \underbrace{\varphi(\gamma_0 T_s) \, L_{\text{hold}} \, T_s}_{\text{Sampled-Data Hold Rate}} .\end{aligned} \tag{S33}$$

Tightened barrier, shell partition and certificate. With the decay rate γ_0 of (S31), the level-set-tightened reduced barrier, the tightened target set, and the un-dipped reduced safe set are

$$\begin{aligned}\tilde{h}_r(z) &:= h(\Phi z) - \varepsilon_{\text{basis},P}^2 - \Delta h_{\text{dip}}, \\ \tilde{\mathcal{C}}_r := \{z \in \mathcal{Z}_{\text{ODD}} : \tilde{h}_r(z) \geq 0\}, &\qquad \mathcal{C}_r := \{z \in \mathcal{Z}_{\text{ODD}} : h(\Phi z) \geq \varepsilon_{\text{basis},P}^2\},\end{aligned} \tag{S34}$$

so $\tilde{\mathcal{C}}_r \subseteq \mathcal{C}_r \subseteq \mathcal{C}_{r,0}$, and $\mathcal{C}_r \subset \text{int}(\mathcal{C}_{r,0}) = \{z^T P z < E_{\text{safe}}\}$ because $\varepsilon_{\text{basis},P} > 0$. The tail-tolerance hypothesis is

$$\varepsilon_{\text{basis},P}^2 + \Delta h_{\text{dip}} + L_h M_z T_s < E_{\text{safe}} = \sup_{z \in \mathcal{Z}_{\text{ODD}}} h(\Phi z), \tag{S35}$$

which is exactly the condition for the activation-threshold interval $h_{\text{act}} \in (L_h M_z T_s, \; E_{\text{safe}} - \varepsilon_{\text{basis},P}^2 - \Delta h_{\text{dip}})$ to be non-empty. For such h_{act}, partition $\mathcal{C}_r$ into the active boundary shell and the passive interior core

$$\mathcal{C}_{r,\text{act}} := \{z \in \mathcal{C}_r : \tilde{h}_r(z) \leq h_{\text{act}}\}, \qquad \mathcal{B}_{\text{int}} := \{z \in \mathcal{C}_r : \tilde{h}_r(z) > h_{\text{act}}\} = \{z : \|z\|_P < r_{\min}\}, \tag{S36}$$

where $r_{\min}^2 := E_{\text{safe}} - \varepsilon_{\text{basis},P}^2 - \Delta h_{\text{dip}} - h_{\text{act}} > 0$, so that $\|z\|_P \geq r_{\min}$ on $\mathcal{C}_{r,\text{act}}$. Because $h_{\text{act}} > L_h M_z T_s$, no trajectory segment that starts at a sample point $z_k \in \mathcal{B}_{\text{int}}$ can reach $\{\tilde{h}_r < 0\}$ within one period (Step 2 of Section S4.4). The Teacher is certified over a partition of $\mathcal{Z}_{\text{ODD}}$ (Proposition S1(ii)), and its certified CBF margin is *defined* as

$$\delta_{\text{cert}} := \inf_{z \in \mathcal{C}_{r,\text{act}}} \Big[\mathcal{L}_{f_r} \tilde{h}_r(z) + \mathcal{L}_{g_r} \tilde{h}_r(z) \hat{\mathcal{T}}(z) + \gamma_0 \, \tilde{h}_r(z) \Big]. \tag{S37}$$

1. For implementation near $\xi \to 0$ use the series $\varphi(\xi) = \frac{1}{2} + \frac{\xi}{12} - \frac{\xi^3}{720} + O(\xi^5)$ to avoid cancellation in floating point.

Because $\hat{\mathcal{T}}(z) \in U_{\mathrm{adm}}$, automatically $\delta_{\mathrm{cert}} \le \delta_{\max} := \inf_{z\in\mathcal{C}_{r,\mathrm{act}}} \sup_{u\in U_{\mathrm{adm}}} [\mathcal{L}_{f_r}\tilde{h}_r(z) + \mathcal{L}_{g_r}\tilde{h}_r(z)u + \gamma_0\tilde{h}_r(z)]$. On the port null space the control term vanishes ($\mathcal{L}_{g_r}\tilde{h}_r(z) = -2(g_r^T P z)^T = 0$), so $\delta_{\mathrm{cert}} \ge \delta$ entails the necessary null-space condition

$$z^T Q z - 2z^T P F_{\mathrm{nl},r}(z) + \gamma_0\tilde{h}_r(z) \ge \delta \quad \forall z \in \mathcal{C}_{r,\mathrm{act}} \text{ with } g_r^T P z = 0, \quad Q := -(PA_{r,0} + A_{r,0}^T P), \tag{S38}$$

i.e., passive dissipation ($z^T Q z = \gamma\|z\|_P^2$ in class (H) with $R_0 = \mathrm{blkdiag}(0, \gamma I_{K_m})$ at $\epsilon = \gamma/2$, by Proposition S2) must carry the margin there; with $P = P_0$ in class (H), $z^T Q_0 z = 2\gamma\|p_r\|_2^2$ vanishes on $\{p_r = 0\}$, which is why P_0 is generally not certifiable under momentum actuation and P_ϵ is used.

Lemma S4 (Closed-form certificate bound for full-rank modal actuation) *Let (H0), (H-sat), $P = P_0 = I_K$, $\Phi^T\Phi = I_K$, $P_{\mathrm{full}}\Phi = \Phi$, $g_r = I_K$ ($m = K$) hold, and $c := 2(\lambda_{\min}(R_0) - L_{F_{\mathrm{nl}}} + \eta_{\mathcal{T}})$. Then*

$$\delta_{\mathrm{cert}} \ge \min\Big\{ c\, r_{\min}^2 + \gamma_0 h_{\mathrm{act}},\ \ c\big(r_{\min}^2 + h_{\mathrm{act}} + \Delta h_{\mathrm{dip}}\big) - \gamma_0 \Delta h_{\mathrm{dip}} \Big\}, \tag{S39}$$

and the minimum equals the first branch if and only if $c \ge \gamma_0$.

Proof For $z \in \mathcal{C}_{r,\mathrm{act}}$, $\mathcal{L}_{f_r}\tilde{h}_r(z) = -2z^T(J_0 - R_0)z - 2z^T F_{\mathrm{nl},r}(z) = 2z^T R_0 z - 2z^T\Phi^T F_{\mathrm{nl}}(\Phi z)$ (skew term vanishes). Since $0, \Phi z \in \mathcal{C} \subset \mathcal{X}_{\mathrm{ODD}}$ (Lemmas S2 and S1) and $F_{\mathrm{nl}}(0) = 0$, Cauchy–Schwarz and Assumption S1 give $|2z^T\Phi^T F_{\mathrm{nl}}(\Phi z)| \le 2\|z\|_2 L_{F_{\mathrm{nl}}}\|\Phi z\|_2 = 2L_{F_{\mathrm{nl}}}\|z\|_2^2$; Rayleigh gives $2z^T R_0 z \ge 2\lambda_{\min}(R_0)\|z\|_2^2$; Lemma S3 with $v = z$ gives $-2z^T\hat{\mathcal{T}}(z) \ge 2\eta_{\mathcal{T}}\|z\|_2^2$. Hence the bracket in (S37) is at least $c\rho + \gamma_0\tilde{h}_r(z)$ with $\rho := \|z\|_2^2$. On $\mathcal{C}_{r,\mathrm{act}}$, $\tilde{h}_r(z) = r_{\min}^2 + h_{\mathrm{act}} - \rho$ and $\rho \in [r_{\min}^2, r_{\min}^2 + h_{\mathrm{act}} + \Delta h_{\mathrm{dip}}]$, so the lower bound $(c - \gamma_0)\rho + \gamma_0(r_{\min}^2 + h_{\mathrm{act}})$ is affine in ρ and its infimum is attained at an endpoint: $\rho = r_{\min}^2$ gives the first branch, $\rho = r_{\min}^2 + h_{\mathrm{act}} + \Delta h_{\mathrm{dip}}$ the second. Their difference (first minus second) is $-(c - \gamma_0)(h_{\mathrm{act}} + \Delta h_{\mathrm{dip}})$, which is ≤ 0 iff $c \ge \gamma_0$. ■

Lemma S5 (Class-(H) certified margin, $m \le K$; proof in Section S4.3) *Let (H0)–(H3) hold, let the ROM be in class (H) with $R_0 = \mathrm{blkdiag}(0, \gamma I_{K_m})$, $P = P_\epsilon$ and $\epsilon = \gamma/2$, so that $Q := -(PA_{r,0} + A_{r,0}^T P) = \gamma P$ (Proposition S2), and let the shell $\mathcal{C}_{r,\mathrm{act}}$ of (S36) be fixed ($r_{\min}^2 > 0$). Split the residual of Assumption S1 on $\mathcal{X}_{\mathrm{ODD}}$ into its linear and nonlinear parts, $F_{\mathrm{nl}}(x) = A_{\mathrm{adv}}x + n(x)$, so that $F_{\mathrm{nl},r}(z) = A_{\mathrm{adv},r}z + n_r(z)$ with $A_{\mathrm{adv},r} := P_0^{-1}\Phi^T P_{\mathrm{full}} A_{\mathrm{adv}}\Phi \in \mathbb{R}^{K\times K}$ the linear (advective) part and $n_r(z) := P_0^{-1}\Phi^T P_{\mathrm{full}}\, n(\Phi z)$ the remainder, and assume $\|P^{1/2} n_r(z)\|_2 \le s_{\mathrm{nl}}\|z\|_P$ for all $z \in \mathcal{C}_{r,\mathrm{act}}$. Define also the advective cross-coupling block $A_{\mathrm{adv},r,\perp} := P_0^{-1}\Phi^T P_{\mathrm{full}} A_{\mathrm{adv}} P_\perp$ (the advective part of $\Delta_\perp$ in (S10); the open-loop operator $A_{r,0}$ and the identity $Q = \gamma P$ are unaffected, since the advection stays inside F_{nl}). Define*

$$M := \gamma P - \big(PA_{\mathrm{adv},r} + A_{\mathrm{adv},r}^T P\big) + 2\eta_{\mathcal{T}}\, P g_r g_r^T P, \qquad c_M := \lambda_{\min}(M, P) - 2s_{\mathrm{nl}}, \tag{S40}$$

where $\lambda_{\min}(M, P)$ is the smallest generalized eigenvalue of the pencil (M, P) (equivalently, $\lambda_{\min}(P^{-1/2}MP^{-1/2})$). Then for every $z \in \mathcal{C}_{r,\mathrm{act}}$,

$$\mathcal{L}_{f_r}\tilde{h}_r(z) + \mathcal{L}_{g_r}\tilde{h}_r(z)\hat{\mathcal{T}}(z) + \gamma_0\tilde{h}_r(z) \ \ge\ c_M\|z\|_P^2 + \gamma_0\tilde{h}_r(z), \tag{S41}$$

and consequently

$$\delta_{\text{cert}} \geq \min\Big\{c_M\, r_{\min}^2 + \gamma_0 h_{\text{act}},\;\; c_M\big(r_{\min}^2 + h_{\text{act}} + \Delta h_{\text{dip}}\big) - \gamma_0 \Delta h_{\text{dip}}\Big\}, \tag{S42}$$

the minimum being the first branch if and only if $c_M \geq \gamma_0$. The lemma does not require $c_M > 0$; $c_M > 0$ makes the first branch positive (the second branch is positive iff $c_M(r_{\min}^2 + h_{\text{act}} + \Delta h_{\text{dip}}) > \gamma_0 \Delta h_{\text{dip}}$). When $P = I_K$, $g_r = I_K$ ($m = K$) and $A_{\text{adv},r} = 0$, with the dissipation identity $z^T Q z = \gamma \|z\|_P^2$ replaced by the Rayleigh bound $z^T Q z = 2 z^T R_0 z \geq 2\lambda_{\min}(R_0)\|z\|_2^2$ of Lemma S4, the same argument gives $s_{\text{nl}} = L_{F_{\text{nl}}}$ and $c_M = 2\lambda_{\min}(R_0) + 2\eta_{\mathcal{T}} - 2L_{F_{\text{nl}}} = c$, i.e. (S42) reduces to (S39).

If A_{adv} is the linearization of F_{nl} at the origin ($\|n(x)\|_2 = o(\|x\|_2)$) and $A_{r\perp} = 0$ (eigenmode invariance, cf. Lemma S7), then evaluating (S14) at $z = 0$, $x_\perp \to 0$ shows that $L_{\perp,P} \geq \sup_{0 \neq x_\perp \in \mathcal{V}_\perp} \|A_{\text{adv},r,\perp} x_\perp\|_2 / \|x_\perp\|_{P_{\text{full},\epsilon}}$, the $P_{\text{full},\epsilon}$-to-ℓ_2 operator norm of the advective block; hence the ROM tail-coupling term $L_h L_{\perp,P} \varepsilon_{\text{basis},P}$ of (S33) is strictly positive whenever $A_{\text{adv},r,\perp} \neq 0$, even though $A_{r\perp}$ (built from A_{full} alone) vanishes by eigenmode invariance.

S3.3. Deadzone Practical Stability and Full-State Sampled-Data CBF Invariance

Let $\bar{\varepsilon}_q := \max\big(\varepsilon_A(b_A),\, \varepsilon_{\text{CROWN}}(b_B) + \Delta u_{\text{lag}}\big) + \Delta u_{\text{lag}}$, and, when F_{nl} is C^1 on a neighbourhood of $\Phi\mathcal{C}_{r,0}$, let $L_{\text{sec}} := \sup_{z \in \mathcal{C}_{r,0}} \|P^{1/2} \nabla F_{\text{nl},r}(z) P^{-1/2}\|_2$ be the P-norm sector modulus.

Proposition S3 (Deadzone-Sector Limit-Cycle Elimination) *Let $P \in \{P_0, P_\epsilon\}$ and $\mu_R > 0$ be as in Proposition S2 (so $\mu_{2,P}(A_{r,0}) \leq -\mu_R$), let $\hat{\mathcal{T}}$ be the Teacher (S18) under (H-sat) (so $\hat{\mathcal{T}}(0) = 0$ and $z^T P g_r \hat{\mathcal{T}}(z) \leq 0$ on $\mathcal{Z}_{\text{ODD}}$ by Lemma S3), let F_{nl} be C^1 near $\Phi\mathcal{C}_{r,0}$, and assume (i) $\mu_R > L_{\text{sec}}$; (ii) $\|u(t) - \hat{\mathcal{T}}(z(t))\|_2 \leq \bar{\varepsilon}_q$ for all t in an interval I on which a solution of the reduced closed loop with $w \equiv 0$, $x_\perp \equiv 0$ satisfies $z(t) \in \mathcal{C}_{r,0}$. Then, with*

$$\mathcal{B}_{\text{ss}}(r_{\text{ss}}) := \left\{ z \in \mathbb{R}^K : \|z\|_P \leq r_{\text{ss}} := \frac{\sqrt{\lambda_{\max}(P)}\, \|g_r\|_2\, \bar{\varepsilon}_q}{\mu_R - L_{\text{sec}}} \right\}, \tag{S43}$$

(a) $D^+\|z(t)\|_P \leq -(\mu_R - L_{\text{sec}})\|z(t)\|_P + \sqrt{\lambda_{\max}(P)}\|g_r\|_2 \bar{\varepsilon}_q$ for all $t \in I$ (upper right Dini derivative; at $z = 0$ as well); (b) $\|z(t)\|_P \leq \max(\|z(t_0)\|_P, r_{\text{ss}})$ for $t \geq t_0$ in I, so $\mathcal{B}_{\text{ss}}(r_{\text{ss}}) \cap \mathcal{C}_{r,0}$ is forward invariant, and $\|z(t)\|_P > r_{\text{ss}}$ forces strict decrease; (c) no periodic orbit (in particular no quantization-induced limit cycle) contained in $\mathcal{C}_{r,0}$ meets $\mathcal{C}_{r,0} \setminus \mathcal{B}_{\text{ss}}(r_{\text{ss}})$. Under the hypotheses of Theorem S1, (ii) holds along the closed loop, including when the sampled port vector $v_k = g_r^T P z_k$ lies in the Path-A deadzone $\mathcal{B}_0 = \{v : \|v\|_\infty < \Delta_{q,A}/2\}$: there $\Psi_A = 0$ and $\|0 - \hat{\mathcal{T}}(z_k)\|_2 = \|\hat{\mathcal{T}}_v(0) - \hat{\mathcal{T}}_v(v_k)\|_2 \leq L_{\mathcal{T}} \sqrt{m}\, \Delta_{q,A}/2 \leq \varepsilon_A(b_A) \leq \bar{\varepsilon}_q$ by (S29), so $\bar{\varepsilon}_q$ keeps its form with the port constants D_v, M_v, $L_{\mathcal{T}}$ entering through $\varepsilon_A(b_A)$ and Δu_{lag} (Section S4.3). If moreover

$$\textit{(H-dz)} \qquad r_{\text{ss}} < r_{\min}, \tag{S44}$$

then $\mathcal{B}_{\text{ss}}(r_{\text{ss}}) \subset \mathcal{B}_{\text{int}}$.

Theorem S1 (Full-State Continuous-Time Safety Under Multi-Rate Quantized Control) *Assume, with every constant evaluated as defined above,*

(H0) ODD containment (S5) and $E_{\text{safe}} < \frac{1}{4} D_{\text{ODD}}^2 \lambda_{\min}(P)$*;*

(H1) Assumption S1, with $L_{F_{\text{nl}}}$*,* $M_\perp$*,* $L_{\perp,P}$ *taken over* $\mathcal{X}_{\text{ODD}}$*;*

(H2) Assumption S2 ((H2a)–(H2c)); in class (H), $\gamma < 2\omega_{\min}$ *and* $\epsilon = \gamma/2$*;*

(H3) the Teacher construction (S17)–(S18) with $H/2 \geq m$*,* $G_{\text{port}} = g_r^T P^2 g_r \succ 0$*, and (H-sat) (S20);*

(H4) the tail tolerance (S35) and $h_{\text{act}} \in (L_h M_z T_s,\ E_{\text{safe}} - \varepsilon_{\text{basis},P}^2 - \Delta h_{\text{dip}})$ *(equivalently* $r_{\min}^2 > 0$*);*

(H5) $b_A \geq 1$ *satisfies the port-space LUT capacity condition (S8) (the lower bound of (S30) is optional);*

(H6) the Certified Control Authority Condition

$$\delta_{\text{total}}(K, b_A, b_B, T_s) \leq \delta_{\text{cert}} \ \big(\leq \delta_{\max} \big), \tag{S45}$$

with δ_{cert} *defined by (S37);*

(H7) initial conditions $x(0) = \Phi z(0) + x_\perp(0)$ *with* $z(0) \in \tilde{\mathcal{C}}_r$ *and* $\|x_\perp(0)\|_{P_{\text{full},\epsilon}} \leq M_\perp/\lambda_{K+1}$*.*

Then the Carathéodory solution $x(t)$ *of (S1) (*$\Sigma \equiv 0$*) under the multi-rate control law (S9) and any measurable* $w(t) \in \mathcal{B}(\varepsilon_{\text{adv}})$ *exists on* $[0, \infty)$ *and satisfies, for all* $t \geq 0$ *and* $k \in \mathbb{N}_0$*: (a)* $\|x_\perp(t)\|_{P_{\text{full},\epsilon}} \leq \varepsilon_{\text{basis},P}$*; (b)* $\tilde{h}_r(z(t_k)) \geq 0$ *and* $\tilde{h}_r(z(t)) \geq -\Delta h_{\text{dip}}$*, i.e.* $z(t) \in \mathcal{C}_r$*; (c)* $h(x(t)) \geq \tilde{h}_r(z(t)) + \Delta h_{\text{dip}} \geq 0$*, i.e.* $x(t) \in \mathcal{C}$*.*

Remark S2 (Spectral cutoff for RDA systems) This remark is not part of Theorem S1. For the RDA class of Remark S1(ii) let $\mathrm{Pe} := c_{\text{adv}}\ell/\nu$, a tail tolerance $\bar{\varepsilon} > 0$, and a coupling budget $\bar{\delta}_\perp > 0$. Define $K_{\min}(\mathrm{Pe})$ as the smallest K with both $\varepsilon_{\text{basis},P}(K) \leq \bar{\varepsilon}$ and $L_{\perp,P}(K)\,\varepsilon_{\text{basis},P}(K) \leq \bar{\delta}_\perp$. With the continuum rate, $\varepsilon_{\text{basis},P}(K) = M_\perp(K)/(\alpha_0 + \nu(K+1)^2\pi^2/\ell^2)$, so the tail-tolerance component is the exact inversion $\alpha_0 + \nu\pi^2(K+1)^2/\ell^2 \geq M_\perp(K)/\bar{\varepsilon}$. *$H_0^1$-bounded sets* ($M_\perp$ independent of K): $(K+1) \geq \frac{\ell}{\pi}\sqrt{(M_\perp/\bar{\varepsilon} - \alpha_0)_+/\nu}$, which is $\frac{\ell}{\pi}\sqrt{M_\perp/(\nu\bar{\varepsilon})}$ for $\alpha_0 = 0$ and scales as $O(\sqrt{\mathrm{Pe}})$ when $M_\perp \propto c_{\text{adv}}$. *$L^2$-bounded sets* ($M_\perp = c_0 c_{\text{adv}}(K+1)$): the inequality is quadratic in $K+1$ with no square-root law; for $\alpha_0 = 0$ it reads $K+1 \geq c_0 \mathrm{Pe}\,\ell/(\pi^2\bar{\varepsilon})$ ($= c_0\mathrm{Pe}/(\pi^2\bar{\varepsilon})$ for unit length), and for $\alpha_0 \neq 0$, $K+1$ must exceed the larger root $\frac{\ell^2}{2\pi^2\nu}\big[\frac{c_0 c_{\text{adv}}}{\bar{\varepsilon}} + \big((\frac{c_0 c_{\text{adv}}}{\bar{\varepsilon}})^2 - \frac{4\alpha_0\nu\pi^2}{\ell^2}\big)_+^{1/2}\big]$. *Coupling component:* integrating by parts against the sine modes, $\|\Phi^T\partial_\xi x_\perp\|_2 \leq \frac{K\pi}{\ell}\|x_\perp\|_2$, so $L_{\perp,P}(K) \leq c_{\text{adv}}K\pi/\ell + L_{F_{\text{nl}}}$; hence on H_0^1-bounded sets $L_{\perp,P}\varepsilon_{\text{basis},P} \lesssim c_{\text{adv}}M_\perp\ell/(\pi\nu K) \propto \mathrm{Pe}/K$, requiring $K = O(\mathrm{Pe})$, while on L^2-bounded sets the product saturates at a K-independent level $\propto c_{\text{adv}}\mathrm{Pe}$, so the coupling budget is a condition on $(\mathrm{Pe}, \bar{\delta}_\perp)$ that increasing K cannot repair. The overall cutoff scaling is therefore $O(\mathrm{Pe})$ (when achievable), and $O(\sqrt{\mathrm{Pe}})$ describes only the tail-tolerance component. Since advection is skew in L^2 but not in H_0^1, and $x^T P_{\text{full}} J_{\text{full}} P_{\text{full}} x \equiv 0$, the advective cross-coupling cancels exactly in the full-energy derivative; a full-state barrier could exploit this (future work).

Remark S3 (LUT indexing variants and measurement rounding) (i) *State-space LUT* (special case $m = K$, $g_r^T P$ invertible): $z \mapsto v = g_r^T P z$ is then a bijection of $\mathbb{R}^K$, and Path A may

index the table by $Q_{b_A}(z)$ directly, with K, D_{ODD}, M_z and $L_{\mathcal{T},z}$ in place of m, D_v, M_v and $L_{\mathcal{T}}$ in (S8), (S29) and (S30) (the deadzone becomes $\{z : \|z\|_\infty < D_{\mathrm{ODD}}/2^{b_A+1}\}$, on which $\|\hat{\mathcal{T}}(0) - \hat{\mathcal{T}}(z)\|_2 \le L_{\mathcal{T},z}\|z\|_2 \le L_{\mathcal{T},z}\sqrt{K}D_{\mathrm{ODD}}/2^{b_A+1}$); the window then ensures $L_{\mathcal{T},z}\sqrt{K}D_{\mathrm{ODD}}/2^{b_A+1} < L_{\mathcal{T},z}M_zT_s = \Delta u_{\mathrm{lag}}$. (ii) *Hashed boundary-shell indexing:* for a fixed b_A (after Δh_{dip} and $\mathcal{C}_{r,\mathrm{act}}$ are fixed, so no definitional cycle arises), let N_{shell} be the number of lattice cells meeting $\mathcal{C}_{r,\mathrm{act}}$; store only these under a collision-free (perfect) hash built offline, with $\lceil \log_2 N_{\mathrm{shell}} \rceil \le B_{\max}$ address bits plus key-check bits per entry so that misses occur only for queries outside the shell. On a miss (in particular for $z_k \in \mathcal{B}_{\mathrm{int}}$) Path A does not fire and the register keeps $u_{B,k-1}$ on $[t_k + \tau_A, t_k + \tau_B)$, with error $\le \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}$; this is harmless because the certificate is not invoked when $z_k \in \mathcal{B}_{\mathrm{int}}$ (Step 2, Section S4.4), and $\bar{\varepsilon}_q$ already covers it. (iii) *Measurement rounding:* Theorem S1 assumes exact acquisition of z_k; input rounding for Path B is modelled by e_0 in Proposition S1. If the Path-A port vector is computed from a rounded state $\hat{z}_k = z_k + e_z$ with $\|e_z\|_2 \le \varepsilon_z$, so that $\hat{v}_k = v_k + g_r^T P e_z$, replace $\varepsilon_A(b_A)$ by $\varepsilon_A(b_A) + \varepsilon_{\mathrm{meas}}$, $\varepsilon_{\mathrm{meas}} := L_{\mathcal{T}}\|g_r^T P\|_2\varepsilon_z = L_{\mathcal{T},z}\varepsilon_z$.

Remark S4 (Relation to zero-sum differential safety games) Because (S1) and (S10) are separately affine in $u \in U_{\mathrm{adm}}$ and $w \in \mathcal{B}(\varepsilon_{\mathrm{adv}})$, the continuous-time Hamilton–Jacobi–Isaacs (HJI) safety Hamiltonian satisfies Isaacs' minimax condition identically ($\sup_{u\in U_{\mathrm{adm}}} \inf_{w\in\mathcal{B}(\varepsilon_{\mathrm{adv}})} \dot{\tilde{h}}_r = \inf_{w\in\mathcal{B}(\varepsilon_{\mathrm{adv}})} \sup_{u\in U_{\mathrm{adm}}} \dot{\tilde{h}}_r$). Consequently, Theorem S1 certifies winning safety invariance for an asymmetric two-player zero-sum differential game in which a continuous-time, full-information adversary applies the pointwise worst-case disturbance $w^*_{\mathrm{adv}}(x(t)) = -\varepsilon_{\mathrm{adv}}\,\mathcal{L}_{d_r}\tilde{h}_r(z(t))^T/\|\mathcal{L}_{d_r}\tilde{h}_r(z(t))\|_2$ when $\mathcal{L}_{d_r}\tilde{h}_r(z(t)) \neq 0$ (and $w^*_{\mathrm{adv}}(x(t)) = 0$ otherwise, plus worst-case tail excitation within $M_\perp$) inside the inter-sample window $[t_k, t_{k+1})$ against the quantized, multi-rate sampled-data defender (S9). The feedback w^*_{adv} is discontinuous, so closed-loop solutions are understood in the Filippov sense; since the Filippov regularization takes convex combinations of values in the convex ball $\mathcal{B}(\varepsilon_{\mathrm{adv}})$, every Filippov solution is a Carathéodory solution for some measurable $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}})$, which is all the proof uses. Under the distributional drift of Corollary S1 the ball is $\mathcal{B}(\varepsilon_{\mathrm{adv}}(T))$.

Corollary S1 (Stochastic Itô Sampled-Data Safety and Wasserstein-2 Horizon) *Consider the stochastic system (S1) over a finite horizon $T = N_T T_s > 0$ with Σ globally Lipschitz (so strong solutions exist and are unique under the adapted control (S9)), projected diffusion $\Sigma_r(x) := P_0^{-1}\Phi^T P_{\mathrm{full}}\Sigma(x)$ and tail diffusion $\Sigma_\perp(x) := P_\perp\Sigma(x)$, bounded over the safe set by $\overline{\sigma}_r := \sup_{x\in\mathcal{C}}\|\Sigma_r(x)\|_2 < \infty$ and by the trace-class (Hilbert–Schmidt in the $P_{\mathrm{full},\epsilon}$-metric) bound $\overline{\sigma}_\perp^2 := \sup_{x\in\mathcal{C}}\|P_{\mathrm{full},\epsilon}^{1/2}P_\perp\Sigma(x)\|_F^2 < \infty$.*[2] *The exogenous disturbance decomposes as $w(t) = w_{\mathrm{adv}}(t) + \int_\Xi \psi(\xi)\,d\mu_t(\xi)$, where w_{adv} is progressively measurable with $w_{\mathrm{adv}}(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}})$, (Ξ, d_Ξ) is a Polish space, $\psi : \Xi \to \mathbb{R}^{n_w}$ is 1-Lipschitz from d_Ξ to ℓ_2 with $\int_\Xi \psi\,d\mu_0 = 0$, and the laws μ_t have finite second moments and drift at Wasserstein-2 rate $W_2(\mu_t, \mu_0) \le r_{W_2}t$. Set $\varepsilon_{\mathrm{adv}}(T) := \varepsilon_{\mathrm{adv}} + r_{W_2}T$ and the drift-enlarged constants*

$$M_\perp(T) := M_\perp + \|P_{\mathrm{full},\epsilon}^{1/2}P_\perp D\|_2\, r_{W_2}T, \qquad \varepsilon_{\mathrm{basis},P}(T) := M_\perp(T)/\lambda_{K+1}, \qquad \eta := \frac{\lambda_{K+1}}{2\overline{\sigma}_\perp^2},$$

2. The bound is valid but N-dependent: for spatially white noise $\|P_{\mathrm{full},\epsilon}^{1/2}P_\perp\Sigma\|_F = \sigma\sqrt{N-K}$ when $P_{\mathrm{full},\epsilon} = I$; mode-weighted functionals give $\sum_k \sigma_k^2/(2\lambda_k)$ instead. Space–time white noise in one dimension violates the hypothesis in the continuum limit.

the continuous-time $(1-\alpha/4)$ *Ornstein–Uhlenbeck tail envelope (in the* $P_{\mathrm{full},\epsilon}$*-norm)*

$$\varepsilon_{\mathrm{basis},\sigma}(K,\alpha,T) := \varepsilon_{\mathrm{basis},P}(T) + \Delta_\sigma,$$
$$\big(1+\tfrac{1}{2}\lambda_{K+1}T\big)e^{-\eta\Delta_\sigma^2} = \tfrac{\alpha}{4}, \;\; \textit{i.e.} \;\; \Delta_\sigma = \overline{\sigma}_\perp\sqrt{\frac{2\ln\big(4(1+\frac{1}{2}\lambda_{K+1}T)/\alpha\big)}{\lambda_{K+1}}},$$

and, globally replacing $\varepsilon_{\mathrm{basis},P}$ *by* $\varepsilon_{\mathrm{basis},\sigma}$ *and* $\varepsilon_{\mathrm{adv}}$ *by* $\varepsilon_{\mathrm{adv}}(T)$ *in* M_z *(so* $M_z(T) \le M_z + \|d_r\|_2 r_{W_2}T + L_{\perp,P}(\varepsilon_{\mathrm{basis},\sigma} - \varepsilon_{\mathrm{basis},P})$*),* L_{hold}*,* Δu_{lag}*, the window* (S30)*,* (S32)*,* (S34)*,* (S36)*,* $r_{\min}$ *and* (S37)*, but keeping* $\overline{d}_h\varepsilon_{\mathrm{adv}}$ *in the adversary term, denote the resulting margin* (S33) *by* $\delta_{\mathrm{total},\sigma}$ *(it reduces to* δ_{total} *when* $\overline{\sigma}_\perp = 0$*,* $r_{W_2} = 0$*). Let* $T_\perp := \sup\{T' > 0 : \varepsilon_{\mathrm{basis},\sigma}(T')^2 + \Delta h_{\mathrm{dip}}(T') + L_h M_z(T')T_s < E_{\mathrm{safe}}\}$*. Suppose* h *is as in* (S4) *(so* $h \in C^\infty$*),* $\mathrm{tr}\big(\Sigma_r(x)^T\nabla_z^2\tilde{h}_r(z)\Sigma_r(x)\big) \ge -2\,\delta_{\mathrm{It\hat{o}}}$ *and* $\sigma_h := \sup_{x\in\mathcal{C}}\|\nabla_z\tilde{h}_r(z)^T\Sigma_r(x)\|_2 < \infty$*,* $\overline{h}_r := h_{\mathrm{act}}$ *with* $h_{\mathrm{act}} > L_h(M_z(T)T_s + r_{\mathrm{BM}}(\alpha))$*,* $\|x_\perp(0)\|_{P_{\mathrm{full},\epsilon}} \le M_\perp(T)/\lambda_{K+1}$*, and* $z(0) \in \mathrm{int}(\tilde{\mathcal{C}}_r)$ *(non-empty) is deterministic. Define the inter-sample Brownian excursion radius* $r_{\mathrm{BM}}(\alpha) := \overline{\sigma}_r\sqrt{2KT_s\ln(8KN_T/\alpha)}$*, the pointwise multi-rate excess* $\Delta\delta_{\mathrm{pw}} := \overline{g}_h\max(\varepsilon_A(b_A), \varepsilon_{\mathrm{CROWN}}(b_B)+\Delta u_{\mathrm{lag}}) - \overline{g}_h[w_2\varepsilon_A(b_A)+(1-w_2)\varepsilon_{\mathrm{CROWN}}(b_B)+w_1\Delta u_{\mathrm{lag}}] + (1-\varphi(\gamma_0 T_s))L_{\mathrm{hold}}T_s + (L_{\mathcal{L}_{f_r}\tilde{h}_r} + L_{\mathcal{L}_{g_r}\tilde{h}_r}u_{\max} + \gamma_0 L_h + \overline{g}_h L_{\mathcal{T},z})r_{\mathrm{BM}}(\alpha)$*, the drift process* $\mu_B(t) := \nabla\tilde{h}_r(z_t)^T[f_r(z_t) + g_r u(t) + d_r w(t) + \Delta_\perp(z_t, x_{\perp,t})]$*, the reduced activation level* $h_{\mathrm{act}}^- := h_{\mathrm{act}} - L_h(M_z(T)T_s + r_{\mathrm{BM}}(\alpha)) > 0$*, the reference level* $h_0 \in (0, h_{\mathrm{act}}]$*, the Kushner exponent* $\theta_{\mathrm{K}} := \frac{\ln(2/\alpha)}{h_0}$*, the worst-case negative interior drift*

$$\underline{\mu}_{\mathrm{int}}^- := \sup\Big\{\big(-\nabla\tilde{h}_r(z)^T[f_r(z) + g_r u + d_r w + \Delta_\perp(z, x_\perp)]\big)_+ : z \in \mathcal{C}_{r,0},\ \tilde{h}_r(z) > h_{\mathrm{act}}^-,\ u \in U_{\mathrm{adm}},\ \|w\|_2 \le \varepsilon_{\mathrm{adv}}(T),\ x_\perp \in \mathcal{V}_\perp,\ \|x_\perp\|_{P_{\mathrm{full},\epsilon}} \le \varepsilon_{\mathrm{basis},\sigma},\ \Phi z + x_\perp \in \mathcal{C}\Big\},$$

the passive interior generator bound $c_{\mathrm{int}} := \theta_{\mathrm{K}}\big(\underline{\mu}_{\mathrm{int}}^- + \delta_{\mathrm{It\hat{o}}} + \frac{1}{2}\theta_{\mathrm{K}}\sigma_h^2\big)e^{-\theta_{\mathrm{K}}(h_{\mathrm{act}}^- + \Delta h_{\mathrm{dip}})}$*, and*

$$\delta_{\mathrm{stoch}}(\alpha) := \Delta\delta_{\mathrm{pw}} + \gamma_0\overline{h}_r + \delta_{\mathrm{It\hat{o}}} + \frac{\ln(2/\alpha)}{2\,h_0}\,\sigma_h^2. \tag{S46}$$

If, in addition to the hypotheses of Theorem S1 with the above replacements, $\delta_{\mathrm{total},\sigma} + \delta_{\mathrm{stoch}}(\alpha) + \delta_{\mathrm{drift}} \le \delta_{\mathrm{cert}}$ *with drift reserve* $\delta_{\mathrm{drift}} > 0$*, the initial level* $\beta_0 := \tilde{h}_r(z(0)) + \Delta h_{\mathrm{dip}}$ *satisfies* $\beta_0 > h_0$*, and*

$$T \le T_{\mathrm{TTL}} := \min\Big\{\frac{\delta_{\mathrm{drift}}}{\overline{d}_h\, r_{W_2}},\ \frac{\alpha\big(1-e^{-\theta_{\mathrm{K}}(\beta_0-h_0)}\big)}{2c_{\mathrm{int}}},\ T_\perp\Big\} \tag{S47}$$

(terms with zero denominator read $+\infty$*; the second bound is equivalent to* $\beta_0 \ge h_0 - \theta_{\mathrm{K}}^{-1}\ln(1-2c_{\mathrm{int}}T/\alpha)$ *with a positive logarithm argument), then the full-order trajectory satisfies* $\mathbb{P}\big(\inf_{t\in[0,T]} h(x(t)) \ge 0\big) \ge 1-\alpha$*.*

S4. Proofs

S4.1. Proof of Proposition S1 (Coupled Twin-Network CROWN)

Let $y_\ell(z)$ and $y_\ell^q(z)$ denote the pre-activation vectors at layer ℓ of $\mathcal{N}$ and $\mathcal{N}_q$ evaluated at the port input $v = g_r^T P z$, with post-activations $s_\ell := \sigma_{\mathrm{GS2}}(y_\ell)$, $s_\ell^q := \sigma_{\mathrm{GS2}}(y_\ell^q)$ ($s_0 := v$, $s_0^q := v + e_0$) and

difference $\delta y_\ell(z) := y^q_\ell(z) - y_\ell(z)$. Write $\Delta W_\ell := W^q_\ell - W_\ell$ and $\Delta b_\ell := b^q_\ell$ (the teacher biases vanish). At $\ell = 1$, $\delta y_1(z) = \Delta W_1 v + W^q_1 e_0 + \Delta b_1 + e_1$ is an exact affine function of (z, e_0, e_1), because $v = g_r^T P z$ is linear in z. By the shared ReLU routing identity (S16), the post-activation difference for each pair $(2i-1, 2i)$ is

$$\sigma_{\mathrm{GS2}}(y^q_{\ell,2i-1}, y^q_{\ell,2i}) - \sigma_{\mathrm{GS2}}(y_{\ell,2i-1}, y_{\ell,2i}) = \begin{bmatrix} \delta y_{\ell,2i}(z) + \nu_{\ell,i}(z) \\ \delta y_{\ell,2i-1}(z) - \nu_{\ell,i}(z) \end{bmatrix}, \tag{S48}$$

where $\nu_{\ell,i}(z) := \mathrm{ReLU}(d_{\ell,i}(z)+\delta d_{\ell,i}(z)) - \mathrm{ReLU}(d_{\ell,i}(z))$ and $\delta d_{\ell,i}(z) = \delta y_{\ell,2i-1}(z) - \delta y_{\ell,2i}(z) \in [\underline{\delta}_{\ell,i}, \overline{\delta}_{\ell,i}]$. On any sub-hyperrectangle $\mathcal{Z}_s \subset \mathcal{Z}_{\mathrm{ODD}}$:

1. For $i \in \mathcal{I}^+_\ell(\mathcal{Z}_s)$ ($\underline{d}_{\ell,i} \geq 0$ and $\underline{d}^q_{\ell,i} \geq 0$), $\nu_{\ell,i}(z) = 1 \cdot \delta d_{\ell,i}(z) + 0$ identically on $\mathcal{Z}_s$ ($\alpha^L_{\ell,i} = \alpha^U_{\ell,i} = 1$, $\beta^L_{\ell,i} = \beta^U_{\ell,i} = 0$).

2. For $i \in \mathcal{I}^-_\ell(\mathcal{Z}_s)$ ($\overline{d}_{\ell,i} \leq 0$ and $\overline{d}^q_{\ell,i} \leq 0$), $\nu_{\ell,i}(z) = 0 \cdot \delta d_{\ell,i}(z) + 0$ identically on $\mathcal{Z}_s$ ($\alpha^L_{\ell,i} = \alpha^U_{\ell,i} = 0$, $\beta^L_{\ell,i} = \beta^U_{\ell,i} = 0$).

3. For $i \in \mathcal{I}^*_\ell(\mathcal{Z}_s)$ (sign-indeterminate, including any comparator where quantization flips the sorting order between $d_{\ell,i}$ and $d^q_{\ell,i}$), monotonicity and 1-Lipschitz continuity of ReLU imply the universal pointwise envelope

$$\min(0, \delta d_{\ell,i}) = -\mathrm{ReLU}(-\delta d_{\ell,i}) \leq \nu_{\ell,i}(z) \leq \mathrm{ReLU}(\delta d_{\ell,i}) = \max(0, \delta d_{\ell,i}) \quad \forall d_{\ell,i} \in \mathbb{R},$$

   whose upper envelope is attained at $d_{\ell,i}(z) = 0$ (where $\nu_{\ell,i} = \mathrm{ReLU}(\delta d_{\ell,i})$) and whose lower envelope is attained at $d_{\ell,i}(z) = -\delta d_{\ell,i}$, so both envelopes are attained. If $\underline{\delta}_{\ell,i} \geq 0$, then $\min(0, \delta d_{\ell,i}) = 0$ and $\max(0, \delta d_{\ell,i}) = \delta d_{\ell,i}$ are already exact linear functions; similarly, if $\overline{\delta}_{\ell,i} \leq 0$, then $\min(0, \delta d_{\ell,i}) = \delta d_{\ell,i}$ and $\max(0, \delta d_{\ell,i}) = 0$ are exact linear functions. When $\underline{\delta}_{\ell,i} < 0 < \overline{\delta}_{\ell,i}$, using $\max(0, \delta d_{\ell,i}) = \mathrm{ReLU}(\delta d_{\ell,i})$ and $\min(0, \delta d_{\ell,i}) = \delta d_{\ell,i} - \mathrm{ReLU}(\delta d_{\ell,i})$ and bounding $\mathrm{ReLU}(\delta d_{\ell,i})$ from above by its secant chord $\alpha^U_{\ell,i}\delta d_{\ell,i} + \beta^U_{\ell,i}$ (exact at the endpoints $\underline{\delta}_{\ell,i}, \overline{\delta}_{\ell,i}$, and an upper bound in between by convexity of ReLU) and from below by 0 yields both the affine upper bound $\alpha^U_{\ell,i}\delta d_{\ell,i} + \beta^U_{\ell,i}$ and the affine lower bound $\delta d_{\ell,i} - (\alpha^U_{\ell,i}\delta d_{\ell,i} + \beta^U_{\ell,i}) = (1 - \alpha^U_{\ell,i})\delta d_{\ell,i} - \beta^U_{\ell,i} = \alpha^L_{\ell,i}\delta d_{\ell,i} + \beta^L_{\ell,i}$, establishing (S21)–(S22).

Layer propagation and part (ii). At layer $\ell+1$, decompose $\delta y_{\ell+1}(z) = \Delta W_{\ell+1}\sigma_{\mathrm{GS2}}(y_\ell(z)) + \Delta b_{\ell+1} + e_{\ell+1} + W^q_{\ell+1}[\sigma_{\mathrm{GS2}}(y^q_\ell(z)) - \sigma_{\mathrm{GS2}}(y_\ell(z))]$. The first term is the single teacher network multiplied by the fixed matrix $\Delta W_{\ell+1}$; it is relaxed by single-network CROWN secant bounds on $[\underline{d}_{\ell,i}, \overline{d}_{\ell,i}]$ for teacher comparators that are not sign-stable on $\mathcal{Z}_s$, and is exactly affine otherwise. The terms $\Delta b_{\ell+1}$ and $e_{\ell+1}$ are constants and box-bounded injections, respectively. In the last term, substitute (S48): by (S48), the coefficient of $\nu_{\ell,i}$ in any row of $\Lambda W^q_{\ell+1}[\sigma_{\mathrm{GS2}}(y^q_\ell) - \sigma_{\mathrm{GS2}}(y_\ell)]$, where Λ is the coefficient matrix accumulated by the backward pass from the output, is $c_{\ell,i} := [\Lambda W^q_{\ell+1}]_{\cdot,2i-1} - [\Lambda W^q_{\ell+1}]_{\cdot,2i}$. For the upper bound of a row, replace $\nu_{\ell,i}$ by $\alpha^U\delta d_{\ell,i} + \beta^U$ where $c_{\ell,i} \geq 0$ and by $\alpha^L\delta d_{\ell,i} + \beta^L$ where $c_{\ell,i} < 0$; for the lower bound reverse the choice. By (S21) each replacement is sound (multiplying an inequality by a non-negative, resp. negative, coefficient preserves, resp. reverses, it), and since $\delta d_{\ell,i}$ is itself a linear combination of δy_ℓ, the substitution yields affine upper and lower bounds in $(z, e_0, \ldots, e_L)$ that are recursively expressed through layer ℓ. Note that $\alpha^L_{\ell,i}, \alpha^U_{\ell,i} \in [0, 1]$ and $|\beta^{U/L}_{\ell,i}| = \frac{-\underline{\delta}_{\ell,i}\overline{\delta}_{\ell,i}}{\overline{\delta}_{\ell,i} - \underline{\delta}_{\ell,i}} \leq \min(|\underline{\delta}_{\ell,i}|, \overline{\delta}_{\ell,i}) \leq \overline{\delta}_{\ell,i} - \underline{\delta}_{\ell,i}$. Iterating to $\ell = 1$, where everything is

affine, and concretizing the resulting affine forms over $\mathcal{Z}_s$ and the injection boxes (exact maximization of an affine function over a product of boxes) gives $[\Delta_{\mathrm{net}}(z)]_j \in [\underline{D}_{s,j}, \overline{D}_{s,j}]$ for all $z \in \mathcal{Z}_s$ and all admissible injections. Hence $\|\Delta_{\mathrm{net}}(z)\|_2^2 = \sum_j [\Delta_{\mathrm{net}}(z)]_j^2 \le \sum_j \max(|\underline{D}_{s,j}|^2, |\overline{D}_{s,j}|^2)$ on $\mathcal{Z}_s$; since the boxes cover $\mathcal{Z}_{\mathrm{ODD}}$, taking the maximum over s bounds the supremum over $\mathcal{Z}_{\mathrm{ODD}}$. Finally $\|\Psi_B(z) - \hat{\mathcal{T}}(z)\|_2 = \|\Pi_U(\mathcal{N}_q(v)) - \Pi_U(\mathcal{N}(v))\|_2 \le \|\Delta_{\mathrm{net}}(z)\|_2$ by non-expansiveness of the metric projection Π_U onto the closed convex set U_{adm}. This proves (S23).

Part (iii). Let $\delta_\ell := \sup \|\delta y_\ell\|_2$ over $z \in \mathcal{Z}_{\mathrm{ODD}}$ and admissible injections, and $\delta_0 := \sup \|s_0^q - s_0\|_2 = \sup \|e_0\|_2 \le \sqrt{H_0}\rho_0$. For $1 \le \ell \le L$, $\delta y_\ell = \Delta W_\ell s_{\ell-1} + W_\ell^q(s_{\ell-1}^q - s_{\ell-1}) + \Delta b_\ell + e_\ell$. Since σ_{GS2} is norm-preserving and $\|W_k\|_2 \le 1$ for $k < L$, $\|s_{\ell-1}\|_2 \le \|v\|_2 \le \|g_r^T P\|_2 \|z\|_2 \le Y$ for $\ell - 1 \le L - 1$ (using $\|z\|_2 \le \sqrt{K} D_{\mathrm{ODD}}/2$ on $\mathcal{Z}_{\mathrm{ODD}}$); since σ_{GS2} is non-expansive, $\|s_{\ell-1}^q - s_{\ell-1}\|_2 \le \|\delta y_{\ell-1}\|_2$ for $\ell \ge 2$. The triangle inequality and $\|e_\ell\|_2 \le \sqrt{H_\ell}\rho_\ell$ (box $\subset$ ball) give $\delta_\ell \le \Delta_{W,\ell} Y + \Delta_{b,\ell} + \sqrt{H_\ell}\rho_\ell + \|W_\ell^q\|_2 \, \delta_{\ell-1}$. Unrolling this linear recursion from δ_0 to δ_L and noting $\Delta_{\mathrm{net}} = \delta y_L$ (no output activation) yields $\sup \|\Delta_{\mathrm{net}}\|_2 \le E_q(b_B)$ as in (S24); $\|W_k^q\|_2 \le \|W_k\|_2 + \Delta_{W,k}$ gives the product bound. The entrywise bound $|[\Delta W_\ell]_{ij}| \le s_\ell 2^{-b_B}$ gives $\|\Delta W_\ell\|_2 \le \|\Delta W_\ell\|_F \le \sqrt{H_\ell H_{\ell-1}} s_\ell 2^{-b_B}$. The relation $\varepsilon_{\mathrm{CROWN}}(b_B) \le C_{\mathrm{relax}} E_q(b_B)$ holds by definition of C_{relax} (set $C_{\mathrm{relax}} := 1$ if $E_q = 0$, in which case all perturbations vanish and $\varepsilon_{\mathrm{CROWN}}(b_B) = 0$); its content is the bound in the exact case: if every comparator of both networks is sign-stable on $\mathcal{Z}_s$, all relaxations are exact, Δ_{net} is exactly affine on $\mathcal{Z}_s \times$ (injection boxes), and the concretization returns its exact component ranges, so $\max(|\underline{D}_{s,j}|, |\overline{D}_{s,j}|) = \sup |[\Delta_{\mathrm{net}}]_j| \le \sup \|\Delta_{\mathrm{net}}\|_2 \le E_q(b_B)$, whence $\varepsilon_{\mathrm{CROWN}}(b_B) \le \sqrt{m}\, E_q(b_B)$. In general $C_{\mathrm{relax}} = 1$ is false: for $m = 2$, a 45° rotation applied to an input box $[-\rho, \rho]^2$ has exact ℓ_2 range $\sqrt{2}\rho$, while the component-wise ℓ_2 assembly returns 2ρ.

Part (iv). Let $\kappa := 2^{-b_B} \le 1$. All perturbation sources are $O(\kappa)$: $\|\Delta W_\ell\|_2 \le \sqrt{H_\ell H_{\ell-1}} s_\ell \kappa$, $\Delta_{b,\ell}, \rho_\ell \le c\kappa$. For E_q the claim is immediate from (S24) since the products are bounded by $(L_{\mathcal{T}} + \sqrt{m H} s_L) \prod_{k<L}(1 + H s_k)$ uniformly in $\kappa \le 1$. For $\varepsilon_{\mathrm{CROWN}}(b_B)$, argue by induction on the layer index that every interval half-width and every endpoint produced by the twin backward pass is at most $C_\ell \kappa$ with C_ℓ independent of κ and of the partition: the backward coefficient matrices are products of W_k^q (entries bounded uniformly in $\kappa \le 1$) and diagonal slope matrices with entries in $[0, 1]$, hence bounded by a κ-independent constant; the quantities they multiply are $\Delta W_k s_{k-1}$ ($\le \|\Delta W_k\|_2 Y$), Δb_k, e_k, and offsets β with $|\beta_{k,i}| \le \overline{\delta}_{k,i} - \underline{\delta}_{k,i} \le 2 C_k \kappa$ by the induction hypothesis (the map $(\underline{\delta}, \overline{\delta}) \mapsto (\alpha, \beta)$ is positively homogeneous of degree 0 in α and degree 1 in β). Summing finitely many such terms gives $C_{\ell+1}\kappa$, and the output assembly is $\le \sqrt{m} C_L \kappa$. ∎

S4.2. Proof of Proposition S2 (P-Weighted Matrix Measure Contraction)

Part 1 ($R_0 \succ 0$)**:** Let $V(e) = \frac{1}{2} e^T P_0 e$ and define $\varpi := P_0^{1/2} e$, so $V(e) = \frac{1}{2}\|\varpi\|_2^2$ and $P_0 e = P_0^{1/2}\varpi$. Differentiating along $\dot{e} = (J_0 - R_0) P_0 e$ gives

$$\dot{V}(e) = (P_0 e)^T J_0 (P_0 e) - (P_0 e)^T R_0 (P_0 e).$$

Since $J_0^T = -J_0$, the scalar $(P_0 e)^T J_0 (P_0 e)$ equals its own negative, hence $(P_0 e)^T J_0 (P_0 e) = 0$ identically for all $e \in \mathbb{R}^K$. For the dissipative term,

$$\begin{aligned}(P_0 e)^T R_0 (P_0 e) &\ge \lambda_{\min}(R_0)\, \|P_0 e\|_2^2 = \lambda_{\min}(R_0)\, \varpi^T P_0 \varpi \\ &\ge \lambda_{\min}(R_0)\, \lambda_{\min}(P_0)\, \|\varpi\|_2^2 = 2\, \lambda_{\min}(R_0)\, \lambda_{\min}(P_0)\, V(e),\end{aligned}$$

proving (S25).

Part 2 ($R_0 = \mathrm{blkdiag}(0, \gamma I_{K_m})$): Here $A_{r,0} = (J_0 - R_0)P_0 = \begin{bmatrix} 0 & I \\ -\Omega_r^2 & -\gamma I \end{bmatrix}$. For $P_\epsilon = \begin{bmatrix} \Omega_r^2 & \epsilon I \\ \epsilon I & I \end{bmatrix}$, the Schur complement $\Omega_r^2 - \epsilon^2 I \succ 0$ holds for all $\epsilon \in (0, \omega_{\min})$. Direct block multiplication of $Q_\epsilon := -(P_\epsilon A_{r,0} + A_{r,0}^T P_\epsilon)$ cancels the off-diagonal $\pm\Omega_r^2$ blocks and yields

$$Q_\epsilon = \begin{bmatrix} 2\epsilon\Omega_r^2 & \epsilon\gamma I \\ \epsilon\gamma I & 2(\gamma - \epsilon)I \end{bmatrix},$$

which decouples mode-by-mode into 2×2 symmetric blocks $Q_{\epsilon,i} = \begin{bmatrix} 2\epsilon\omega_i^2 & \epsilon\gamma \\ \epsilon\gamma & 2(\gamma-\epsilon) \end{bmatrix}$ for $i = 1, \ldots, K_m$. By Sylvester's criterion, $Q_{\epsilon,i} \succ 0$ if and only if $2\epsilon\omega_i^2 > 0$ and

$$\det(Q_{\epsilon,i}) = \epsilon\Big(4\gamma\omega_i^2 - \epsilon\big(\gamma^2 + 4\omega_i^2\big)\Big) > 0 \iff 0 < \epsilon < \frac{4\gamma\omega_i^2}{\gamma^2 + 4\omega_i^2}.$$

Because $f(\omega^2) = \frac{4\gamma\omega^2}{\gamma^2+4\omega^2}$ is strictly increasing in $\omega^2 > 0$, the intersection over $i = 1, \ldots, K_m$ is governed by $\omega_{\min} = \min_i \omega_i$ and satisfies $\frac{4\gamma\omega_{\min}^2}{\gamma^2+4\omega_{\min}^2} \le \frac{4\gamma\omega_{\min}^2}{4\gamma\omega_{\min}} = \omega_{\min}$, so both $P_\epsilon \succ 0$ and $Q_\epsilon \succ 0$ hold for all $\epsilon \in \big(0, \frac{4\gamma\omega_{\min}^2}{\gamma^2+4\omega_{\min}^2}\big)$.

The choice $\epsilon = \gamma/2$. $\frac{\gamma}{2} < \frac{4\gamma\omega_{\min}^2}{\gamma^2+4\omega_{\min}^2} \iff \gamma^2 + 4\omega_{\min}^2 < 8\omega_{\min}^2 \iff \gamma < 2\omega_{\min}$ (divide by $\gamma/2 > 0$). At $\epsilon = \gamma/2$, $Q_{\gamma/2} = \begin{bmatrix} \gamma\Omega_r^2 & (\gamma^2/2)I \\ (\gamma^2/2)I & \gamma I \end{bmatrix} = \gamma P_{\gamma/2}$ (Pencil-Rate identity). Hence for every $e \neq 0$, $e^T(P_{\gamma/2}A_{r,0} + A_{r,0}^T P_{\gamma/2})e/(2e^T P_{\gamma/2} e) = -\gamma/2$, so the supremum in the definition of $\mu_{2,P_{\gamma/2}}$ equals $-\gamma/2$, and $\frac{d}{dt}\|e\|_{P_{\gamma/2}}^2 = -e^T Q_{\gamma/2} e = -\gamma\|e\|_{P_{\gamma/2}}^2$ integrates to (S26). Since P_ϵ and Q_ϵ do not depend on any reduced quantity other than Ω_r, the same identity holds on the full-order momentum-damped lattice with Ω_{full} in place of Ω_r.

Metric sandwich. For each mode, $2|q_i p_i| \le \omega_i q_i^2 + p_i^2/\omega_i$ (AM–GM), so $|2\epsilon q_i p_i| \le \frac{\epsilon}{\omega_i}(\omega_i^2 q_i^2 + p_i^2) \le \frac{\epsilon}{\omega_{\min}}(\omega_i^2 q_i^2 + p_i^2)$. Summing over i, $e^T P_\epsilon e = e^T P_0 e + 2\epsilon q^T p \in [(1 - \epsilon/\omega_{\min}), (1 + \epsilon/\omega_{\min})]\, e^T P_0 e$. ■

Lemma S6 (Pencil rate) *For one mode with $\omega > 0$, $\gamma > 0$, $\epsilon \in (0, \omega)$, $A_\omega = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -\gamma \end{bmatrix}$, $P_{\epsilon,\omega} = \begin{bmatrix} \omega^2 & \epsilon \\ \epsilon & 1 \end{bmatrix}$ and $Q = -(P_{\epsilon,\omega}A_\omega + A_\omega^T P_{\epsilon,\omega})$, the largest λ with $Q \succeq 2\lambda P_{\epsilon,\omega}$ is $r(\epsilon, \omega) = \frac{\gamma}{2} - \frac{\omega|\gamma - 2\epsilon|}{2\sqrt{\omega^2 - \epsilon^2}} \le \frac{\gamma}{2}$, with equality iff $\epsilon = \gamma/2$.*

Proof With $t := \gamma/2 - \lambda$ and $a := \gamma/2 - \epsilon$, $Q - 2\lambda P_{\epsilon,\omega} = \begin{bmatrix} 2\omega^2(t-a) & 2\epsilon t \\ 2\epsilon t & 2(t+a) \end{bmatrix}$ (entry by entry: $2\epsilon\omega^2 - 2\lambda\omega^2$, $\epsilon\gamma - 2\lambda\epsilon$, $2(\gamma - \epsilon) - 2\lambda$), with determinant $4[\omega^2(t^2 - a^2) - \epsilon^2 t^2] = 4[(\omega^2 - \epsilon^2)t^2 - \omega^2 a^2]$. A symmetric 2×2 matrix is PSD iff its diagonal entries and determinant are non-negative, i.e. iff $t \ge |a|$ and $t \ge \omega|a|/\sqrt{\omega^2 - \epsilon^2}$; the second implies the first since $\omega/\sqrt{\omega^2 - \epsilon^2} \ge 1$. Hence $\lambda \le \gamma/2 - \omega|a|/\sqrt{\omega^2 - \epsilon^2} = r(\epsilon, \omega)$. ■

Lemma S7 (Characterization of (H2a) in class (H)) *Let $\Phi = \mathrm{blkdiag}(\Phi_q, \Phi_q)$, $\Phi_q^T\Phi_q = I$, $S := \Phi_q^T \Omega_{\text{full}}^2 \Phi_q$. Then $P_\perp = \mathrm{blkdiag}(I - \Phi_q S^{-1}\Phi_q^T\Omega_{\text{full}}^2,\ I - \Phi_q\Phi_q^T)$, and $\Phi^T P_{\text{full},\epsilon} P_\perp$ has zero diagonal blocks, zero $(1,2)$ block, and $(2,1)$ block $\epsilon(\Phi_q^T - S^{-1}\Phi_q^T\Omega_{\text{full}}^2)$. Hence (H2a) holds iff $\Omega_{\text{full}}^2\Phi_q = \Phi_q S$, i.e. iff $\mathrm{Im}(\Phi_q)$ is Ω_{full}^2-invariant; in that case $P_{\text{full}}\Phi = \Phi P_0$, $z = \Phi^T x$, and $P = P_\epsilon$ with $\Omega_r^2 = S$.*

Proof $P_0 = \mathrm{blkdiag}(S, I)$ and $P_0^{-1}\Phi^T P_{\mathrm{full}} = \mathrm{blkdiag}(S^{-1}\Phi_q^T\Omega_{\mathrm{full}}^2, \Phi_q^T)$, which gives $P_\perp$. Multiplying $\Phi^T P_{\mathrm{full},\epsilon} = \begin{bmatrix} \Phi_q^T\Omega_{\mathrm{full}}^2 & \epsilon\Phi_q^T \\ \epsilon\Phi_q^T & \Phi_q^T \end{bmatrix}$ by $P_\perp$: the $(1,1)$ block is $\Phi_q^T\Omega_{\mathrm{full}}^2 - SS^{-1}\Phi_q^T\Omega_{\mathrm{full}}^2 = 0$, the $(1,2)$ and $(2,2)$ blocks are multiples of $\Phi_q^T(I - \Phi_q\Phi_q^T) = 0$, and the $(2,1)$ block is $\epsilon\Phi_q^T(I - \Phi_q S^{-1}\Phi_q^T\Omega_{\mathrm{full}}^2) = \epsilon(\Phi_q^T - S^{-1}\Phi_q^T\Omega_{\mathrm{full}}^2)$. It vanishes iff $\Phi_q^T\Omega_{\mathrm{full}}^2 = S\Phi_q^T$, i.e. (transposing) $\Omega_{\mathrm{full}}^2\Phi_q = \Phi_q S$. Conversely, if $\Omega_{\mathrm{full}}^2\Phi_q = \Phi_q M$ for some M, then $S = \Phi_q^T\Phi_q M = M$. The final statements follow by substitution; diagonalizing S by an orthogonal change of reduced coordinates gives Ω_r diagonal. ∎

Necessity of (H2b). In class (H), $A_{\mathrm{full}} = \begin{bmatrix} 0 & I \\ -\Omega_{\mathrm{full}}^2 & -R_p \end{bmatrix}$ and block multiplication gives $-(P_{\mathrm{full},\epsilon}A_{\mathrm{full}} + A_{\mathrm{full}}^T P_{\mathrm{full},\epsilon}) = \begin{bmatrix} 2\epsilon\Omega_{\mathrm{full}}^2 & \epsilon R_p \\ \epsilon R_p & 2(R_p - \epsilon I) \end{bmatrix}$. Evaluating (S11) at $x_\perp = (0, p_\perp)$ with $p_\perp \in \mathrm{Im}(I - \Phi_q\Phi_q^T)$ (an element of $\mathcal{V}_\perp$ under (H2a)) gives $2p_\perp^T(R_p - \epsilon I)p_\perp \geq 2\lambda_{K+1}\|p_\perp\|_2^2 > 0$, which is (H2b).

S4.3. Proofs of Lemma S3, Proposition S3 and Lemma S5 (Sector Routing, Deadzone Practical Stability, Class-(H) Margin)

Sector alignment (Lemma S3). To verify the constructive sector bound for the L-layer origin-preserving ($b_\ell = 0$) Teacher acting on the port vector $v := g_r^T P z \in \mathbb{R}^m$ (which equals $z \in \mathbb{R}^K$ whenever $m = K$ and $g_r^T P = I_K$), let $W_1 = \cos(\theta_{\mathrm{GS}})U_{\mathrm{sum}}W_{\mathrm{in}} + \sin(\theta_{\mathrm{GS}})U_{\mathrm{diff}}Q_{\mathrm{mix}}W_{\mathrm{in}} \in \mathbb{R}^{H\times m}$ with $W_{\mathrm{in}}^T W_{\mathrm{in}} = I_m$, let intermediate hidden layers $W_\ell = U_{\mathrm{sum}}Q_{\mathrm{sum},\ell}U_{\mathrm{sum}}^T + U_{\mathrm{diff}}Q_{\mathrm{diff},\ell}U_{\mathrm{diff}}^T \in O(H)$ ($2 \leq \ell \leq L-1$, with $Q_{\mathrm{sum},\ell}, Q_{\mathrm{diff},\ell} \in O(H/2)$) be block-diagonal orthogonal, and set $W_L = -L_{\mathcal{T}}(W_{L-1}\cdots W_1)^T \in \mathbb{R}^{m\times H}$. Since $\max(a,b) + \min(a,b) = a + b$ and $\max(a,b) - \min(a,b) = |a - b|$, the orthonormal pairwise sum and difference matrices $U_{\mathrm{sum}}, U_{\mathrm{diff}}$ satisfy $U_{\mathrm{sum}}^T\sigma_{\mathrm{GS2}}(y) = U_{\mathrm{sum}}^T y$ and $\|U_{\mathrm{diff}}^T\sigma_{\mathrm{GS2}}(y)\|_2 = \|U_{\mathrm{diff}}^T y\|_2$ for all $y \in \mathbb{R}^H$. By induction across $\ell = 1, \ldots, L-1$, the final hidden post-activation $s_{L-1}(v) := \sigma_{\mathrm{GS2}}(y_{L-1}(v))$ satisfies $U_{\mathrm{sum}}^T s_{L-1}(v) = \cos(\theta_{\mathrm{GS}})Q_{\mathrm{sum},L-1}\cdots Q_{\mathrm{sum},2}W_{\mathrm{in}}v$ and $\|U_{\mathrm{diff}}^T s_{L-1}(v)\|_2 = \sin(\theta_{\mathrm{GS}})\|v\|_2$. Defining $\tilde{Q}_{\mathrm{diff}} := Q_{\mathrm{diff},L-1}\cdots Q_{\mathrm{diff},2}Q_{\mathrm{mix}} \in O(H/2)$, decomposing $W_{L-1}\cdots W_1 v$ in the orthonormal basis $[U_{\mathrm{sum}}\ U_{\mathrm{diff}}]$, and applying Cauchy–Schwarz in $\mathbb{R}^{H/2}$ yields, under (H-sat) (so that Π_U acts as the identity and $\hat{\mathcal{T}}(z) = \mathcal{N}(v)$),

$$
\begin{aligned}
-v^T\mathcal{N}(v) &= L_{\mathcal{T}}\Big(\cos^2(\theta_{\mathrm{GS}})\|W_{\mathrm{in}}v\|_2^2 + \sin(\theta_{\mathrm{GS}})(\tilde{Q}_{\mathrm{diff}}W_{\mathrm{in}}v)^T U_{\mathrm{diff}}^T s_{L-1}(v)\Big) \\
&\geq L_{\mathcal{T}}\big(\cos^2\theta_{\mathrm{GS}} - \sin^2\theta_{\mathrm{GS}}\big)\|v\|_2^2 = \eta_{\mathcal{T}}\|g_r^T P z\|_2^2 \geq 0,
\end{aligned}
$$

and $-z^T P g_r\hat{\mathcal{T}}(z) = -v^T\mathcal{N}(v)$. Without (H-sat), $\Pi_U(a) = \lambda a$ with $\lambda = \min(1, u_{\max}/\|a\|_2) \in (0,1]$ and $\|\mathcal{N}(v)\|_2 \leq L_{\mathcal{T}}\|v\|_2$, so $-v^T\Pi_U(\mathcal{N}(v)) = -\lambda v^T\mathcal{N}(v) \geq \eta_{\mathcal{T}}\min(1, u_{\max}/(L_{\mathcal{T}}\|v\|_2))\|v\|_2^2$. ∎

Lemma S8 (P-norm sector bound of the reduced residual) *If F_{nl} is C^1 on a neighbourhood of $\Phi\mathcal{C}_{r,0}$, then $\|F_{\mathrm{nl},r}(z)\|_P \leq L_{\mathrm{sec}}\|z\|_P$ for all $z \in \mathcal{C}_{r,0}$.*

Proof $\mathcal{C}_{r,0}$ is a convex ellipsoid containing 0, so the segment $\{sz : s \in [0,1]\}$ lies in $\mathcal{C}_{r,0}$, and $F_{\mathrm{nl},r}(0) = 0$. By the integral mean-value theorem, $F_{\mathrm{nl},r}(z) = \int_0^1 \nabla F_{\mathrm{nl},r}(sz)\, z\, ds$, hence $P^{1/2}F_{\mathrm{nl},r}(z) = \int_0^1 \big(P^{1/2}\nabla F_{\mathrm{nl},r}(sz)P^{-1/2}\big)P^{1/2}z\, ds$ and $\|F_{\mathrm{nl},r}(z)\|_P \leq \sup_s \|P^{1/2}\nabla F_{\mathrm{nl},r}(sz)$

$P^{-1/2}\|_2\,\|z\|_P \le L_{\mathrm{sec}}\|z\|_P$. (Only the sector ratio $\sup_{z\in\mathcal{C}_{r,0}\setminus\{0\}} \|F_{\mathrm{nl},r}(z)\|_P/\|z\|_P \le L_{\mathrm{sec}}$ is used below, and it may replace L_{sec}.) ■

Proof of Proposition S3. With $w \equiv 0$, $x_\perp \equiv 0$, the closed-loop reduced trajectory satisfies $\dot z = A_{r,0}z + F_{\mathrm{nl},r}(z) + g_r\hat{\mathcal{T}}(z) + g_r[u(t) - \hat{\mathcal{T}}(z(t))]$ wherever $\dot z$ exists; since u is piecewise constant and the right-hand side is continuous in z, z is piecewise C^1 and has a right derivative $\dot z^+(t)$ everywhere. Let $\rho(t) := \|z(t)\|_P$. For $z(t) \neq 0$, $\|\cdot\|_P$ is differentiable at $z(t)$ and the upper right Dini derivative satisfies $D^+\rho(t) = z^TP\dot z^+/\|z\|_P$. By the definition of $\mu_{2,P}$, $z^TPA_{r,0}z \le -\mu_R\|z\|_P^2$; by Cauchy–Schwarz in the P-inner product and Lemma S8, $z^TPF_{\mathrm{nl},r}(z) \le \|z\|_P\|F_{\mathrm{nl},r}(z)\|_P \le L_{\mathrm{sec}}\|z\|_P^2$; by Lemma S3, $z^TPg_r\hat{\mathcal{T}}(z) \le 0$; and by Cauchy–Schwarz in the P-metric ($\|P^{1/2}g_ra\|_2 \le \sqrt{\lambda_{\max}(P)}\|g_r\|_2\|a\|_2$) and hypothesis (ii),

$$D^+\|z\|_P \le -\big(\mu_R - L_{\mathrm{sec}}\big)\|z\|_P + \sqrt{\lambda_{\max}(P)}\,\|g_r\|_2\,\bar\varepsilon_q.$$

For $z(t) = 0$: $A_{r,0}0 = 0$, $F_{\mathrm{nl},r}(0) = 0$, $\hat{\mathcal{T}}(0) = 0$, so $\dot z^+(t) = g_ru(t) = g_r[u(t) - \hat{\mathcal{T}}(0)]$ and $D^+\rho(t) = \limsup_{h\downarrow 0}\|z(t+h)\|_P/h = \|\dot z^+(t)\|_P \le \sqrt{\lambda_{\max}(P)}\|g_r\|_2\bar\varepsilon_q$, which is the same bound at $\rho = 0$. This proves (a); no case distinction for the Path-A deadzone $\mathcal{B}_0 = \{v \in \mathbb{R}^m : \|v\|_\infty < \Delta_{q,A}/2\}$ (a set in port space, $v = g_r^TPz$) is needed: if $Q_{b_A}(v_k) = 0$ then $\Psi_A = 0$, and since $\hat{\mathcal{T}}$ depends on z only through v, with $\hat{\mathcal{T}}_v(0) = 0$ and $\hat{\mathcal{T}}_v$ $L_{\mathcal{T}}$-Lipschitz, $\|0 - \hat{\mathcal{T}}(z_k)\|_2 = \|\hat{\mathcal{T}}_v(0) - \hat{\mathcal{T}}_v(v_k)\|_2 \le L_{\mathcal{T}}\|v_k\|_2 \le L_{\mathcal{T}}\sqrt{m}\,\Delta_{q,A}/2 = L_{\mathcal{T}}\sqrt{m}D_v/2^{b_A+1} \le \varepsilon_A(b_A)$, so $\bar\varepsilon_q = \max(\varepsilon_A(b_A), \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}) + \Delta u_{\mathrm{lag}}$ already covers $\Psi_A(0) = 0$ (see (S29)) and keeps its form with the port constants D_v, M_v, $L_{\mathcal{T}}$; r_{ss} in (S43) and (H-dz) are unchanged in form.

(b) Let $c := \max(\rho(t_0), r_{\mathrm{ss}})$ and suppose $\rho(t_1) > c$ for some $t_1 > t_0$ in I. Let $t^\sharp := \sup\{t \in [t_0, t_1] : \rho(t) \le c\}$; by continuity $\rho(t^\sharp) = c$ and $\rho > c \ge r_{\mathrm{ss}}$ on $(t^\sharp, t_1]$, where (a) gives $D^+\rho < 0$. A continuous function with $D^+\rho \le 0$ on an interval is non-increasing there, so $\rho(t_1) \le \rho(t^\sharp) = c$, a contradiction. Strict decrease when $\rho > r_{\mathrm{ss}}$ follows from $D^+\rho < 0$ by the same monotonicity lemma.

(c) Suppose a periodic solution of period $T_{\mathrm{per}} > 0$ lies in $\mathcal{C}_{r,0}$ and $\rho(t_0) > r_{\mathrm{ss}}$ for some t_0. If $\rho > r_{\mathrm{ss}}$ on $[t_0, t_0 + T_{\mathrm{per}}]$, strict decrease gives $\rho(t_0 + T_{\mathrm{per}}) < \rho(t_0)$; otherwise $\rho(t') \le r_{\mathrm{ss}}$ for some $t' \in (t_0, t_0 + T_{\mathrm{per}}]$, and (b) applied from t' gives $\rho(t_0 + T_{\mathrm{per}}) \le r_{\mathrm{ss}} < \rho(t_0)$. Both contradict periodicity.

Validity of (ii) under Theorem S1: by the e_u bounds of Step 2 in Section S4.4 and Lipschitz continuity of $\hat{\mathcal{T}}$,

$$\|u(t) - \hat{\mathcal{T}}(z_k)\|_2 \le \max(\varepsilon_A(b_A), \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}),$$
$$\|\hat{\mathcal{T}}(z_k) - \hat{\mathcal{T}}(z(t))\|_2 \le L_{\mathcal{T},z}\|z(t) - z_k\|_2 \le L_{\mathcal{T},z}M_zT_s = \Delta u_{\mathrm{lag}}$$

(recall $\Delta u_{\mathrm{lag}} = L_{\mathcal{T}}M_vT_s = L_{\mathcal{T},z}M_zT_s$ by (S19)), for $t \in [t_k, t_{k+1})$, so $\|u(t) - \hat{\mathcal{T}}(z(t))\|_2 \le \bar\varepsilon_q$ for all $t \ge 0$.

Finally, under (H-dz), $\|z\|_P \le r_{\mathrm{ss}} < r_{\min}$ gives $\tilde h_r(z) = r_{\min}^2 + h_{\mathrm{act}} - \|z\|_P^2 > h_{\mathrm{act}}$, and $h(\Phi z) > \varepsilon_{\mathrm{basis},P}^2$, so $z \in \mathcal{B}_{\mathrm{int}}$. ■

Proof of Lemma S5. Fix $z \in \mathcal{C}_{r,\mathrm{act}}$. Since $\mathcal{C}_{r,\mathrm{act}} \subset \mathcal{C}_{r,0} \subset \mathcal{Z}_{\mathrm{ODD}}$ (by (S27) under (H0)), Lemma S3 applies at z. By (S10) and the split $F_{\mathrm{nl},r}(z) = A_{\mathrm{adv},r}z + n_r(z)$,

$$\mathcal{L}_{f_r}\tilde h_r(z) = -2z^TP\big(A_{r,0}z + A_{\mathrm{adv},r}z + n_r(z)\big) = z^TQz - z^T\big(PA_{\mathrm{adv},r} + A_{\mathrm{adv},r}^TP\big)z - 2z^TPn_r(z),$$

where $-2z^TPA_{r,0}z = -z^T(PA_{r,0} + A_{r,0}^TP)z = z^TQz$ and likewise $-2z^TPA_{\mathrm{adv},r}z = -z^T(PA_{\mathrm{adv},r} + A_{\mathrm{adv},r}^TP)z$ (a scalar equals its transpose). (i) *Dissipation (Proposition S2):* in class (H) with $R_0 = \mathrm{blkdiag}(0, \gamma I_{K_m})$, $P = P_{\gamma/2}$, the identity $Q_{\gamma/2} = \gamma P_{\gamma/2}$ gives $z^TQz = \gamma\, z^TPz$, so the first two terms equal $z^T[\gamma P - (PA_{\mathrm{adv},r} + A_{\mathrm{adv},r}^TP)]z$. (ii) *Teacher (Lemma S3 with* $v = g_r^TPz$*):* $\mathcal{L}_{g_r}\tilde{h}_r(z)\hat{\mathcal{T}}(z) = -2z^TPg_r\hat{\mathcal{T}}(z) \geq 2\eta_{\mathcal{T}}\|g_r^TPz\|_2^2 = 2\eta_{\mathcal{T}}\, z^TPg_rg_r^TPz$. (iii) *Remainder (Cauchy–Schwarz in the* P*-inner product):* $|2z^TPn_r(z)| = 2|(P^{1/2}z)^T(P^{1/2}n_r(z))| \leq 2\|z\|_P\|P^{1/2}n_r(z)\|_2 \leq 2s_{\mathrm{nl}}\|z\|_P^2$. Summing (i)–(iii),

$$\mathcal{L}_{f_r}\tilde{h}_r(z) + \mathcal{L}_{g_r}\tilde{h}_r(z)\hat{\mathcal{T}}(z) \geq z^TMz - 2s_{\mathrm{nl}}\|z\|_P^2$$

with M as in (S40). (iv) *Rayleigh quotient for the pencil* (M, P)*:* with $y := P^{1/2}z$, $z^TMz = y^T(P^{-1/2}MP^{-1/2})y \geq \lambda_{\min}(P^{-1/2}MP^{-1/2})\|y\|_2^2 = \lambda_{\min}(M,P)\|z\|_P^2$, because $P^{-1/2}MP^{-1/2}$ is symmetric and its eigenvalues are exactly the generalized eigenvalues of the pencil (M, P) ($Mx = \lambda Px \iff (P^{-1/2}MP^{-1/2})(P^{1/2}x) = \lambda(P^{1/2}x)$). Adding $\gamma_0\tilde{h}_r(z)$ to both sides yields (S41). (v) *Affine-in-*ρ *endpoint argument:* on $\mathcal{C}_{r,\mathrm{act}}$, $\tilde{h}_r(z) = E_{\mathrm{safe}} - \varepsilon_{\mathrm{basis},P}^2 - \Delta h_{\mathrm{dip}} - \|z\|_P^2 = r_{\min}^2 + h_{\mathrm{act}} - \rho$ with $\rho := \|z\|_P^2$, and $-\Delta h_{\mathrm{dip}} \leq \tilde{h}_r(z) \leq h_{\mathrm{act}}$ (since $\mathcal{C}_{r,\mathrm{act}} \subseteq \mathcal{C}_r = \{\tilde{h}_r \geq -\Delta h_{\mathrm{dip}}\}$ by (S34)–(S36)), i.e. $\rho \in [r_{\min}^2,\ r_{\min}^2 + h_{\mathrm{act}} + \Delta h_{\mathrm{dip}}]$. The right-hand side of (S41) equals $(c_M - \gamma_0)\rho + \gamma_0(r_{\min}^2 + h_{\mathrm{act}})$, affine in ρ, so its infimum over the interval is attained at an endpoint: $\rho = r_{\min}^2$ gives the first branch of (S42) and $\rho = r_{\min}^2 + h_{\mathrm{act}} + \Delta h_{\mathrm{dip}}$ the second; taking the infimum over $z \in \mathcal{C}_{r,\mathrm{act}}$ in (S37) proves (S42). The difference (first minus second) is $-(c_M - \gamma_0)(h_{\mathrm{act}} + \Delta h_{\mathrm{dip}})$, which is ≤ 0 iff $c_M \geq \gamma_0$. Positivity of c_M is used nowhere in (i)–(v). *Reduction:* only step (i) is specific to class (H); if instead $P = I_K$, $g_r = I_K$ ($m = K$), $A_{\mathrm{adv},r} = 0$ and $z^TQz = 2z^TR_0z \geq 2\lambda_{\min}(R_0)\|z\|_2^2$ is used in (i), then (ii)–(v) go through verbatim with $s_{\mathrm{nl}} = L_{F_{\mathrm{nl}}}$ (as $\|n_r(z)\|_2 = \|\Phi^TF_{\mathrm{nl}}(\Phi z)\|_2 \leq L_{F_{\mathrm{nl}}}\|z\|_2$) and $c_M = 2\lambda_{\min}(R_0) + 2\eta_{\mathcal{T}} - 2L_{F_{\mathrm{nl}}} = c$, which is Lemma S4. ■

S4.4. Proof of Theorem S1 and Corollary S1

Step 0 (Existence and joint first exit). The closed-loop right-hand side $F(x) + Gu(t) + Dw(t)$ is locally Lipschitz in x (Assumption S1), and measurable and locally bounded in t (u is piecewise constant and determined causally by the samples $z(x(t_k))$; w is measurable and bounded). By the Carathéodory existence and uniqueness theorem, applied successively on $[t_k, t_{k+1})$, there is a unique absolutely continuous solution on a maximal interval $[0, t_{\max})$, and $\|x(t)\| \to \infty$ as $t \uparrow t_{\max}$ if $t_{\max} < \infty$. At $t = 0$, (H7) gives $z(0) \in \tilde{\mathcal{C}}_r \subseteq \mathcal{C}_r \subset \mathrm{int}(\mathcal{C}_{r,0})$ and, by Lemma S2, $h(x(0)) = h(\Phi z(0)) - \|x_\perp(0)\|_{P_{\mathrm{full},\epsilon}}^2 \geq \tilde{h}_r(z(0)) + \Delta h_{\mathrm{dip}} \geq 0$, so $x(0) \in \mathcal{C} \subset \mathrm{int}(\mathcal{X}_{\mathrm{ODD}})$ (Lemma S1). Define

$$t^* := \sup\{t \in [0, t_{\max}) : x(s) \in \mathcal{X}_{\mathrm{ODD}} \text{ and } z(s) \in \mathcal{C}_{r,0} \text{ for all } s \in [0,t]\} > 0.$$

On $[0, t^*)$, every supremum defining $M_\perp$, $L_{\perp,P}$ (note $z \in \mathcal{C}_{r,0} \Rightarrow \Phi z \in \mathcal{C} \subset \mathcal{X}_{\mathrm{ODD}}$ by Lemma S1), M_z, L_h, $\overline{g}_h$, $\overline{d}_h$, $L_{\mathcal{L}_{f_r}\tilde{h}_r}$, $L_{\mathcal{L}_{g_r}\tilde{h}_r}$ applies, because $u(t) \in U_{\mathrm{adm}}$ (Ψ_A, Ψ_B take values in U_{adm}) and $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}})$. Steps 1–2 below, carried out on $[0, t^*)$, give $\|x_\perp(t)\|_{P_{\mathrm{full},\epsilon}} \leq \varepsilon_{\mathrm{basis},P}$, $z(t) \in \mathcal{C}_r$ and $x(t) \in \mathcal{C}$ for all $t \in [0, t^*)$. Suppose $t^* < t_{\max}$. By continuity and closedness, $x(t^*) \in \mathcal{C} \subset \mathrm{int}(\mathcal{X}_{\mathrm{ODD}})$ and $z(t^*) \in \mathcal{C}_r \subset \mathrm{int}(\mathcal{C}_{r,0})$, so there is $\delta > 0$ with $x(s) \in \mathcal{X}_{\mathrm{ODD}}$ and $z(s) \in \mathcal{C}_{r,0}$ on $[0, t^* + \delta]$, contradicting the definition of t^*. Hence $t^* = t_{\max}$, $x(t)$ stays in the compact set $\mathcal{C}$ on $[0, t_{\max})$, and therefore $t_{\max} = \infty$.

Step 1 (Uniform Spectral Tail Bound in the Unified $P_{\text{full},\epsilon}$-Norm): Using $P_{\text{full},\epsilon}$-orthogonality ($\Phi^T P_{\text{full},\epsilon} P_\perp = 0 \implies P_\perp^T P_{\text{full},\epsilon} x_\perp = P_{\text{full},\epsilon} x_\perp$, because $P_\perp^T P_{\text{full},\epsilon} x_\perp = P_{\text{full},\epsilon} x_\perp - P_{\text{full}} \Phi P_0^{-1} (\Phi^T P_{\text{full},\epsilon} x_\perp)$ and $\Phi^T P_{\text{full},\epsilon} x_\perp = \Phi^T P_{\text{full},\epsilon} P_\perp x = 0$) to project (S1) (with $\Sigma \equiv 0$ and $A_{\text{full}} = (J_{\text{full}} - R_{\text{full}}) P_{\text{full}}$) onto $\mathcal{V}_\perp = \text{Im}(P_\perp)$, i.e. $\dot{x}_\perp = P_\perp \dot{x} = P_\perp A_{\text{full}} x_\perp + P_\perp [A_{\text{full}} \Phi z + F_{\text{nl}}(x) + Gu + Dw]$, and differentiating the unified tail Lyapunov function $V_\perp(x_\perp) := \frac{1}{2} \|x_\perp(t)\|^2_{P_{\text{full},\epsilon}} = \frac{1}{2} x_\perp^T P_{\text{full},\epsilon} x_\perp$ (absolutely continuous in t) gives, for a.e. t,

$$\frac{d}{dt} V_\perp(x_\perp) = \frac{1}{2} x_\perp^T \Big(P_{\text{full},\epsilon} A_{\text{full}} + A_{\text{full}}^T P_{\text{full},\epsilon} \Big) x_\perp + x_\perp^T P_{\text{full},\epsilon} P_\perp \Big(A_{\text{full}} \Phi z + F_{\text{nl}}(x) + Gu + Dw \Big).$$

The quadratic term is bounded by (S11), i.e. hypothesis (H2c) (in class (D), skew-symmetry $J_{\text{full}}^T = -J_{\text{full}}$ yields $(P_{\text{full}} x_\perp)^T J_{\text{full}} (P_{\text{full}} x_\perp) = 0$ identically on $\mathcal{V}_\perp$ without requiring J_{full} and R_{full} to commute, so (H2c) reduces to a condition on R_{full}; in class (H) the rate is computed by Lemma S6). Applying (S11) and Cauchy–Schwarz in the $P_{\text{full},\epsilon}$-inner product together with (S12) yields

$$\|x_\perp\|_{P_{\text{full},\epsilon}} \frac{d}{dt} \|x_\perp\|_{P_{\text{full},\epsilon}} \le -\lambda_{K+1} \|x_\perp\|^2_{P_{\text{full},\epsilon}} + M_\perp \|x_\perp\|_{P_{\text{full},\epsilon}} \implies$$
$$\|x_\perp(t)\|_{P_{\text{full},\epsilon}} \le \max\left(\|x_\perp(0)\|_{P_{\text{full},\epsilon}}, \frac{M_\perp}{\lambda_{K+1}} \right) \quad \forall t \in [0, t^*).$$

(The implication is the comparison lemma applied to $V_\perp$: with $c := \max(\|x_\perp(0)\|_{P_{\text{full},\epsilon}}, M_\perp/\lambda_{K+1})$, if $\|x_\perp(t_1)\|_{P_{\text{full},\epsilon}} > c$, then on the last interval $(t^\sharp, t_1]$ on which $\|x_\perp\|_{P_{\text{full},\epsilon}} > c$ we have $\dot{V}_\perp \le \|x_\perp\|_{P_{\text{full},\epsilon}} (-\lambda_{K+1} \|x_\perp\|_{P_{\text{full},\epsilon}} + M_\perp) < 0$ a.e., hence $V_\perp(t_1) < V_\perp(t^\sharp) = c^2/2$, a contradiction.) By (H7), $\|x_\perp(t)\|_{P_{\text{full},\epsilon}} \le \varepsilon_{\text{basis},P}$ on $[0, t^*)$.

Step 2 (Exact Exponentially Weighted Integration Over $[t_k, t_k + T_s]$): Let $s := t - t_k \in [0, T_s)$ and suppose by induction that $z_k \in \tilde{\mathcal{C}}_r \subseteq \mathcal{C}_r = \mathcal{C}_{r,\text{act}} \cup \mathcal{B}_{\text{int}}$. Along (S10), $\|\Delta_\perp(z(t_k + s), x_\perp(t_k + s))\|_2 \le L_{\perp,P} \varepsilon_{\text{basis},P}$ by (S14) and Step 1, hence $\|\dot{z}\|_2 \le M_z$ and $\|z(t_k + s) - z_k\|_2 \le M_z s \le M_z T_s$. Since $\mathcal{C}_{r,0}$ is convex and $\|\nabla \tilde{h}_r\|_2 = \|2Pz\|_2 \le L_h$ there, $\tilde{h}_r$ is L_h-Lipschitz on $\mathcal{C}_{r,0}$. If $z_k \in \mathcal{B}_{\text{int}}$ (so $\tilde{h}_r(z_k) > h_{\text{act}} > L_h M_z T_s$), then L_h-Lipschitz continuity of $\tilde{h}_r$ immediately ensures $\tilde{h}_r(z(t_k + s)) \ge \tilde{h}_r(z_k) - L_h M_z s > h_{\text{act}} - L_h M_z T_s > 0$ for all $s \in [0, T_s]$, so $z(t_{k+1}) \in \tilde{\mathcal{C}}_r$ and $\tilde{h}_r(z(t_k + s)) > 0 \ge -\Delta h_{\text{dip}}$ hold automatically. Otherwise, $z_k \in \mathcal{C}_{r,\text{act}} \cap \tilde{\mathcal{C}}_r$, where the certified Teacher condition (S37) holds at z_k with $\delta_{\text{cert}} \ge \delta_{\text{total}}$. By the chain rule ($\tilde{h}_r \in C^1$, z absolutely continuous), for a.e. s, $\dot{\tilde{h}}_r(z) = \mathcal{L}_{f_r} \tilde{h}_r(z) + \mathcal{L}_{g_r} \tilde{h}_r(z) u + \mathcal{L}_{d_r} \tilde{h}_r(z) w + \nabla \tilde{h}_r(z)^T \Delta_\perp$, with $|\mathcal{L}_{d_r} \tilde{h}_r(z) w| \le \bar{d}_h \varepsilon_{\text{adv}}$ and $|\nabla \tilde{h}_r^T \Delta_\perp| \le L_h L_{\perp,P} \varepsilon_{\text{basis},P}$. Adding and subtracting (S37) evaluated at $z_k \in \mathcal{C}_{r,\text{act}}$, writing $\mathcal{L}_{g_r} \tilde{h}_r(z) u - \mathcal{L}_{g_r} \tilde{h}_r(z_k) \hat{\mathcal{T}}(z_k) = [\mathcal{L}_{g_r} \tilde{h}_r(z) - \mathcal{L}_{g_r} \tilde{h}_r(z_k)] u + \mathcal{L}_{g_r} \tilde{h}_r(z_k) [u - \hat{\mathcal{T}}(z_k)]$, and using the Lipschitz constants on $\mathcal{C}_{r,0}$, $\|u\|_2 \le u_{\max}$ and $\|z - z_k\|_2 \le M_z s$ (so that the three Lipschitz shifts sum to $L_{\text{hold}} s$) gives, for a.e. $s \in [0, T_s)$,

$$\dot{\tilde{h}}_r(z(t_k+s)) + \gamma_0 \tilde{h}_r(z(t_k+s)) \ge \delta_{\text{total}} - L_h L_{\perp,P} \varepsilon_{\text{basis},P} - \bar{d}_h \varepsilon_{\text{adv}} - L_{\text{hold}} s - \bar{g}_h \|u(t_k+s) - \hat{\mathcal{T}}(z_k)\|_2. \tag{S49}$$

Under (S9), the command error $e_u(t_k + s) := \|u(t_k + s) - \hat{\mathcal{T}}(z_k)\|_2$ against $\hat{\mathcal{T}}(z_k)$ at the current sample z_k satisfies:

- For $s \in [0, \tau_A)$: at $k = 0$, the standby initialization $u_{B,-1} := \Psi_B(z_0)$ gives $e_u(s) = \|\Psi_B(z_0) - \hat{\mathcal{T}}(z_0)\|_2 \le \varepsilon_{\text{CROWN}}(b_B) \le \varepsilon_{\text{CROWN}}(b_B) + \Delta u_{\text{lag}}$; for $k \ge 1$, $u(t_k + s) = u_{B,k-1} = \Psi_B(z_{k-1})$, so $e_u(t_k + s) \le \|\Psi_B(z_{k-1}) - \hat{\mathcal{T}}(z_{k-1})\|_2 + \|\hat{\mathcal{T}}(z_{k-1}) - \hat{\mathcal{T}}(z_k)\|_2 \le \varepsilon_{\text{CROWN}}(b_B) + \Delta u_{\text{lag}}$, where $\Delta u_{\text{lag}} = L_{\mathcal{T}} M_v T_s = L_{\mathcal{T},z} M_z T_s$.

- For $s \in [\tau_A, \tau_B)$: $u(t_k + s) = \Psi_A(Q_{b_A}(v_k))$ with $v_k = g_r^T P z_k$ the port vector (S6) ($z_k \in \tilde{\mathcal{C}}_r \subset \mathcal{C}_{r,0}$, so v_k lies in the LUT domain), so $e_u(t_k + s) \le \varepsilon_A(b_A)$ by (S29) (including the deadzone case $Q_{b_A}(v_k) = 0$); under the optional lower bound of (S30), moreover $\varepsilon_A(b_A) < \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}$.
- For $s \in [\tau_B, T_s)$: $u(t_k + s) = \Psi_B(z_k)$, so $e_u(t_k + s) \le \varepsilon_{\mathrm{CROWN}}(b_B)$.

Multiplying (S49) by the integrating factor $e^{\gamma_0 s} > 0$ and integrating from $s = 0$ to $s = T_s$ (the left side is $\frac{d}{ds}[e^{\gamma_0 s}\tilde{h}_r(z(t_k + s))]$ a.e., and $e^{\gamma_0 s}\tilde{h}_r(z(t_k + s))$ is absolutely continuous) yields

$$e^{\gamma_0 T_s}\tilde{h}_r(z(t_{k+1})) - \tilde{h}_r(z(t_k)) \ge \int_0^{T_s} e^{\gamma_0 s}\Big(\delta_{\mathrm{total}} - L_h L_{\perp,P}\varepsilon_{\mathrm{basis},P} - \overline{d}_h\varepsilon_{\mathrm{adv}} - L_{\mathrm{hold}}\, s - \overline{g}_h\, e_u(t_k + s)\Big) ds. \tag{S50}$$

Dividing (S50) by $\int_0^{T_s} e^{\gamma_0 s}\, ds = \frac{e^{\gamma_0 T_s} - 1}{\gamma_0} > 0$, using the exact sub-interval weights $w_1(\gamma_0), w_2(\gamma_0), w_3(\gamma_0)$ from (S31) with $w_1(\gamma_0) + w_2(\gamma_0) + w_3(\gamma_0) = 1$, and evaluating the exact integration-by-parts identity

$$\frac{\gamma_0}{e^{\gamma_0 T_s} - 1}\int_0^{T_s} s\, e^{\gamma_0 s}\, ds = \frac{(\gamma_0 T_s - 1)e^{\gamma_0 T_s} + 1}{\gamma_0(e^{\gamma_0 T_s} - 1)} = \varphi(\gamma_0 T_s)\, T_s,$$

we obtain

$$\frac{\gamma_0}{e^{\gamma_0 T_s} - 1}\int_0^{T_s} e^{\gamma_0 s}\, e_u(t_k + s)\, ds \le w_2(\gamma_0)\,\varepsilon_A(b_A) + \big(1 - w_2(\gamma_0)\big)\varepsilon_{\mathrm{CROWN}}(b_B) + w_1(\gamma_0)\,\Delta u_{\mathrm{lag}}.$$

Substituting δ_{total} from (S33) into (S50) makes the right-hand side of (S50) non-negative, proving $e^{\gamma_0 T_s}\tilde{h}_r(z(t_{k+1})) \ge \tilde{h}_r(z(t_k))$ and hence, by induction on $k \in \mathbb{N}_0$, $\tilde{h}_r(z(t_k)) \ge 0$ ($z(t_k) \in \tilde{\mathcal{C}}_r$) for all $k \in \mathbb{N}_0$ with $t_k < t^*$.

For any intermediate continuous time $t = t_k + \tau$ with $\tau \in (0, T_s)$, multiplying (S49) by $e^{\gamma_0 s}$, integrating from $s = 0$ to $s = \tau$, and dividing by $e^{\gamma_0 \tau}$ gives

$$\tilde{h}_r(z(t_k + \tau)) \ge e^{-\gamma_0\tau}\tilde{h}_r(z(t_k)) + \int_0^{\tau} e^{-\gamma_0(\tau - s)}\,\mathrm{RHS}(s)\, ds \ge -\int_0^{\tau}\big[\mathrm{RHS}(s)\big]_-\, ds,$$

where $[a]_- := \max(0, -a) \ge 0$ and we used $\tilde{h}_r(z(t_k)) \ge 0$ and $0 < e^{-\gamma_0(\tau - s)} \le 1$. Because $\delta_{\mathrm{total}} \ge L_h L_{\perp,P}\varepsilon_{\mathrm{basis},P} + \overline{d}_h\varepsilon_{\mathrm{adv}} + \overline{g}_h\varepsilon_{\mathrm{CROWN}}(b_B)$ (as $\varepsilon_A(b_A) \ge \varepsilon_{\mathrm{CROWN}}(b_B)$), the instantaneous deficit $[\mathrm{RHS}(s)]_-$ is bounded by $\overline{g}_h\Delta u_{\mathrm{lag}} + L_{\mathrm{hold}}s$ on $[0, \tau_A)$, by $\overline{g}_h(\varepsilon_A(b_A) - \varepsilon_{\mathrm{CROWN}}(b_B)) + L_{\mathrm{hold}}s$ on $[\tau_A, \tau_B)$, and by $L_{\mathrm{hold}}s$ on $[\tau_B, T_s)$. Integrating from $s = 0$ to $\tau < T_s$ yields $\int_0^{\tau}[\mathrm{RHS}(s)]_-\, ds \le \Delta h_{\mathrm{dip}}$. Consequently, $\tilde{h}_r(z(t)) \ge -\Delta h_{\mathrm{dip}}$ ($z(t) \in \mathcal{C}_r$) holds for all $t \in [0, t^*)$, and by the quadratic splitting identity (Lemma S2) and Step 1,

$$\begin{aligned} h(x(t)) &= h\big(\Phi z(t) + x_\perp(t)\big) = h(\Phi z(t)) - \|x_\perp(t)\|^2_{P_{\mathrm{full},\epsilon}} \\ &\ge h(\Phi z(t)) - \varepsilon^2_{\mathrm{basis},P} = \tilde{h}_r(z(t)) + \Delta h_{\mathrm{dip}} \ge 0 \quad \forall t \in [0, t^*), \end{aligned}$$

ensuring $x(t) \in \mathcal{C} \subset \mathcal{X}_{\mathrm{ODD}}$. Together with Step 0 ($t^* = \infty$) this proves conclusions (a)–(c) of Theorem S1. ∎

Step 3 (Proof of Corollary S1). Write $\lambda := \lambda_{K+1}$, $r_t := \|x_\perp(t)\|_{P_{\mathrm{full},\epsilon}}$, $\varepsilon_\sigma := \varepsilon_{\mathrm{basis},\sigma}(K,\alpha,T)$, $c_0 := \varepsilon_{\mathrm{basis},P}(T)$, so $\varepsilon_\sigma = c_0 + \Delta_\sigma$. All processes live on a filtered probability space satisfying the usual conditions; the strong solution exists up to its explosion time ζ.

(a) Disturbance bound. For any unit $a \in \mathbb{R}^{n_w}$, $\xi \mapsto a^T\psi(\xi)$ is 1-Lipschitz and integrable under $\mu_t, \mu_0 \in \mathcal{P}_2(\Xi)$, so Kantorovich–Rubinstein duality gives $a^T \int \psi\, d(\mu_t - \mu_0) \le W_1(\mu_t,\mu_0)$; taking the supremum over a and using $\int \psi\, d\mu_0 = 0$ and $W_1 \le W_2$ (Jensen), $\| \int \psi\, d\mu_t\|_2 \le W_2(\mu_t,\mu_0) \le r_{W_2} t$. Hence $\|w(t)\|_2 \le \varepsilon_{\mathrm{adv}} + r_{W_2} t \le \varepsilon_{\mathrm{adv}}(T)$ for $t \le T$, deterministically.

(b) Stopping times. Let $B_t := h(\Phi z_t) - \varepsilon_\sigma^2 = \tilde{h}_r(z_t) + \Delta h_{\mathrm{dip}}$, $\tau_{\mathrm{exit}} := \inf\{t \ge 0 : B_t \le 0\}$, $\tau_\perp := \inf\{t \ge 0 : r_t > \varepsilon_\sigma\}$, τ_{BM} as in (d), and $\tau_{\mathrm{stop}} := \tau_{\mathrm{exit}} \wedge \tau_{\mathrm{BM}} \wedge \tau_\perp$. For $t \le \tau_{\mathrm{exit}} \wedge \tau_\perp$ (by continuity of paths, at the stopping time as well), $B_t \ge 0$ and $r_t \le \varepsilon_\sigma$, so Lemma S2 gives $h(x_t) = h(\Phi z_t) - r_t^2 \ge B_t \ge 0$: $x_t \in \mathcal{C}$ and $z_t \in \mathcal{C}_{r,0}$. In particular $\tau_{\mathrm{exit}} \wedge \tau_\perp < \zeta$ (the compact set $\mathcal{C}$ must be left before explosion), every constant of Section S3 (with the replacements of the corollary) applies before τ_{stop}, and $\{\inf_{[0,T]} h(x_t) < 0\} \subseteq \{\tau_{\mathrm{exit}} \wedge \tau_\perp \le T\}$.

(c) Tail envelope via a generalized Itô formula. The tail obeys $dx_\perp = P_\perp[A_{\mathrm{full}}x + F_{\mathrm{nl}}(x) + Gu + Dw]dt + \Sigma_\perp(x)\, dW_t$. For the C^2 function $V := \frac{1}{2}r^2 = \frac{1}{2}x_\perp^T P_{\mathrm{full},\epsilon} x_\perp$, Itô's formula, Step 1, the bound $\|P_\perp D(w - w_{\mathrm{adv}})\|_{P_{\mathrm{full},\epsilon}} \le \|P_{\mathrm{full},\epsilon}^{1/2} P_\perp D\|_2 r_{W_2} T$ from (a), and $\frac{1}{2}\mathrm{tr}(\Sigma_\perp^T P_{\mathrm{full},\epsilon}\Sigma_\perp) = \frac{1}{2}\|P_{\mathrm{full},\epsilon}^{1/2} P_\perp \Sigma\|_F^2 \le \frac{1}{2}\overline{\sigma}_\perp^2$ give, for $t < \tau_{\mathrm{stop}} \wedge T$, $dV \le (-2\lambda V + M_\perp(T)\sqrt{2V} + \frac{1}{2}\overline{\sigma}_\perp^2)dt + dN_t$, $dN_t := x_\perp^T P_{\mathrm{full},\epsilon}\Sigma_\perp\, dW_t$, with $d\langle N\rangle_t = \|\Sigma_\perp^T P_{\mathrm{full},\epsilon} x_\perp\|_2^2 dt \le \|P_{\mathrm{full},\epsilon}^{1/2} P_\perp \Sigma\|_2^2\, r_t^2 dt \le \overline{\sigma}_\perp^2 r_t^2 dt$ ($\|\cdot\|_2 \le \|\cdot\|_F$). Let $g(V) := (\sqrt{2V} - c_0)_+^2$, so $U_t := g(V_t) = (r_t - c_0)_+^2$. For $\sqrt{2V} > c_0$, $g'(V) = 2(1 - c_0/r) \in [0,2)$ and $g''(V) = 2c_0/r^3 > 0$; $g' = g'' = 0$ for $\sqrt{2V} < c_0$; g' is continuous at $\sqrt{2V} = c_0$. Thus g is convex and $C^{1,1}$ on $[0,\infty)$ (its second derivative is bounded by $2/c_0^2$ where non-zero), and the Meyer–Itô formula (Revuz–Yor, Thm. VI.1.5), whose second-derivative measure $g''(V)dV$ has no atoms so that the local-time term reduces via the occupation-times formula, gives $dU = g'(V)dV + \frac{1}{2}g''(V)d\langle V\rangle$. On $\{r > c_0\}$, using $g', g'' \ge 0$, $M_\perp(T) = \lambda c_0$ and $d\langle V\rangle = d\langle N\rangle \le \overline{\sigma}_\perp^2 r^2 dt$:

$$\mathrm{drift}(U) \le \tfrac{2(r-c_0)}{r}\big(-\lambda r^2 + \lambda c_0 r + \tfrac{1}{2}\overline{\sigma}_\perp^2\big) + \tfrac{c_0}{r^3}\overline{\sigma}_\perp^2 r^2 = -2\lambda(r-c_0)^2 + \overline{\sigma}_\perp^2 = -2\lambda U + \overline{\sigma}_\perp^2,$$

and on $\{r \le c_0\}$ the drift is $0 \le -2\lambda U + \overline{\sigma}_\perp^2$. Hence $dU \le (-2\lambda U + \overline{\sigma}_\perp^2)dt + d\tilde{M}_t$ with $\tilde{M} := \int g'(V)dN$, $d\langle\tilde{M}\rangle \le \frac{4(r-c_0)^2}{r^2}\overline{\sigma}_\perp^2 r^2 dt = 4\overline{\sigma}_\perp^2 U\, dt$. This is the tail SDE with Hilbert–Schmidt norm. Applying the C^2 Itô formula to $e^{\eta U}$ with $\eta = \lambda/(2\overline{\sigma}_\perp^2)$,

$$\mathcal{L}e^{\eta U} \le \eta e^{\eta U}\big(-2\lambda U + \overline{\sigma}_\perp^2 + 2\eta\overline{\sigma}_\perp^2 U\big) = \eta e^{\eta U}\big(\overline{\sigma}_\perp^2 - \lambda U\big) = \frac{\lambda}{2}e^{\eta U}\big(1 - 2\eta U\big) \le \frac{\lambda}{2},$$

using $e^x(1-2x) \le e^x(1-x) \le 1$ for $x = \eta U \ge 0$. Let $\sigma := \tau_{\mathrm{stop}} \wedge T$. On $[0,\sigma]$, $U \le \Delta_\sigma^2$ and $r \le \varepsilon_\sigma$, so the stochastic integral in $e^{\eta U_{t\wedge\sigma}}$ has bounded integrand and is a true martingale; optional stopping gives $\mathbb{E}[e^{\eta U_\sigma}] \le e^{\eta U_0} + \frac{\lambda}{2}\mathbb{E}[\sigma] \le 1 + \frac{1}{2}\lambda T$, since $U_0 = 0$ by $r_0 \le M_\perp(T)/\lambda = c_0$. On $A_\perp := \{\tau_\perp \le T \wedge \tau_{\mathrm{BM}} \wedge \tau_{\mathrm{exit}}\}$, $\sigma = \tau_\perp$ and $U_\sigma = \Delta_\sigma^2$; Markov's inequality yields $\mathbb{P}(A_\perp) \le (1 + \frac{1}{2}\lambda T)e^{-\eta\Delta_\sigma^2} = \alpha/4$ by the definition of Δ_σ ($\mathbb{P}(A_\perp) = 0$ when $\overline{\sigma}_\perp = 0$, since then $U \equiv 0$).

(d) Brownian excursion tube. Let $t_k = \lfloor t/T_s \rfloor T_s$, and define the stopped projected Brownian increment

$$\xi_k(t) := \int_{t_k}^{t} \mathbb{I}_{\{\vartheta \le \tau_{\mathrm{exit}} \wedge \tau_\perp\}} \Sigma_r(x_\vartheta)\, dW_\vartheta$$

and $\tau_{\mathrm{BM}} := \inf\{t \in [0,T] : \|\xi_k(t)\|_2 > r_{\mathrm{BM}}(\alpha)\}$. Since $\|\xi_k(t)\|_2 > r_{\mathrm{BM}}(\alpha) \implies \max_{1\le j\le K} |\xi_{k,j}(t)| > r_{\mathrm{BM}}(\alpha)/\sqrt{K}$ and each coordinate $\xi_{k,j}(t)$ has quadratic variation $\langle \xi_{k,j}\rangle_t \le \overline{\sigma}_r^2 T_s$ on $[t_k, t_{k+1}]$ (the integrand is stopped while $x \in \mathcal{C}$, where $\|\Sigma_r\|_2 \le \overline{\sigma}_r$), the Dambis–Dubins–Schwarz exponential martingale inequality and union bound across KN_T events yield

$$\mathbb{P}(\tau_{\mathrm{BM}} \le T) \le 2KN_T \exp\!\left(-\frac{r_{\mathrm{BM}}(\alpha)^2}{2K\overline{\sigma}_r^2 T_s}\right) = 2KN_T \cdot \frac{\alpha}{8KN_T} = \frac{\alpha}{4}.$$

For $t < \tau_{\mathrm{stop}}$, $z_t - z_{t_k} = \int_{t_k}^t [f_r + g_r u + d_r w + \Delta_\perp]\, d\vartheta + \xi_k(t)$, and the drift has norm $\le M_z(T)$, so $\|z_t - z_{t_k}\|_2 \le M_z(T)(t - t_k) + r_{\mathrm{BM}}(\alpha)$ and $\|z_{t_k} - z_{t_{k-1}}\|_2 \le M_z(T)T_s + r_{\mathrm{BM}}(\alpha)$.

(e) Exponential barrier process. Let $\theta := \theta_{\mathrm{K}} = \ln(2/\alpha)/h_0$ and $Z_t := \exp(-\theta B_{t\wedge\tau_{\mathrm{stop}}})$. By the C^2 Itô formula applied to the smooth $\tilde{h}_r$ along $dz = [\cdot]dt + \Sigma_r dW$, $dB = (\mu_B + \frac{1}{2}\mathrm{tr}(\Sigma_r^T \nabla_z^2 \tilde{h}_r \Sigma_r))dt + \nabla_z \tilde{h}_r^T \Sigma_r\, dW$, and applied to $b \mapsto e^{-\theta b}$, for $t < \tau_{\mathrm{stop}}$, $\mathcal{L}Z_t = -\theta Z_t\big(\mu_B(t) + \frac{1}{2}\mathrm{tr}(\Sigma_r^T \nabla_z^2 \tilde{h}_r \Sigma_r) - \frac{1}{2}\theta \|\nabla_z \tilde{h}_r^T \Sigma_r\|_2^2\big)$. We split on the sample point $z_{t_k} \equiv z_k$, which is $\mathcal{F}_{t_k}$-measurable; since $B_{t_k} > 0$, $z_{t_k} \in \mathcal{C}_r = \mathcal{C}_{r,\mathrm{act}} \cup \mathcal{B}_{\mathrm{int}}$.

1. *Case A,* $z_{t_k} \in \mathcal{C}_{r,\mathrm{act}}$ ($\tilde{h}_r(z_{t_k}) \le h_{\mathrm{act}} = \overline{h}_r$). As in Step 2, but pointwise in t and without the factor $\gamma_0 \tilde{h}_r$ on the left: $\mu_B(t) \ge \delta_{\mathrm{cert}} - \gamma_0 \tilde{h}_r(z_{t_k}) - (L_{\mathcal{L}_{f_r}\tilde{h}_r} + L_{\mathcal{L}_{g_r}\tilde{h}_r} u_{\max})\|z_t - z_{t_k}\|_2 - \overline{g}_h \|u(t) - \hat{\mathcal{T}}(z_{t_k})\|_2 - \overline{d}_h \|w(t)\|_2 - L_h L_{\perp,P} \varepsilon_\sigma$. By (a), (d), the e_u bounds of Step 2 with the lag enlarged to $L_{\mathcal{T},z}(M_z(T)T_s + r_{\mathrm{BM}}) = \Delta u_{\mathrm{lag}} + L_{\mathcal{T},z} r_{\mathrm{BM}}$, and $\tilde{h}_r(z_{t_k}) \le \overline{h}_r$,

$$\mu_B(t) \ge \delta_{\mathrm{cert}} - \gamma_0 \overline{h}_r - \big[\delta_{\mathrm{total},\sigma} + \Delta\delta_{\mathrm{pw}}\big] - \overline{d}_h r_{W_2} t \ge \delta_{\mathrm{It\hat{o}}} + \tfrac{1}{2}\theta\sigma_h^2 + \big(\delta_{\mathrm{drift}} - \overline{d}_h r_{W_2} t\big),$$

   where we used $\delta_{\mathrm{total},\sigma} + \Delta\delta_{\mathrm{pw}} = L_h L_{\perp,P}\varepsilon_\sigma + \overline{g}_h \max(\varepsilon_A(b_A), \varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}) + \overline{d}_h \varepsilon_{\mathrm{adv}} + L_{\mathrm{hold}} T_s + (L_{\mathcal{L}_{f_r}\tilde{h}_r} + L_{\mathcal{L}_{g_r}\tilde{h}_r} u_{\max} + \gamma_0 L_h + \overline{g}_h L_{\mathcal{T},z}) r_{\mathrm{BM}}$, which dominates the subtracted terms (the $\gamma_0 L_h$ contributions are slack), and the hypothesis $\delta_{\mathrm{total},\sigma} + \delta_{\mathrm{stoch}} + \delta_{\mathrm{drift}} \le \delta_{\mathrm{cert}}$ with (S46). Since $\frac{1}{2}\mathrm{tr}(\cdot) \ge -\delta_{\mathrm{It\hat{o}}}$, $\|\nabla \tilde{h}_r^T \Sigma_r\|_2 \le \sigma_h$ and $t \le T \le \delta_{\mathrm{drift}}/(\overline{d}_h r_{W_2})$, the bracket in $\mathcal{L}Z_t$ is ≥ 0, so $\mathcal{L}Z_t \le 0$.

2. *Case B,* $z_{t_k} \in \mathcal{B}_{\mathrm{int}}$ ($\tilde{h}_r(z_{t_k}) > h_{\mathrm{act}}$). By (d) and L_h-Lipschitz continuity on $\mathcal{C}_{r,0}$, $\tilde{h}_r(z_t) > h_{\mathrm{act}} - L_h(M_z(T)T_s + r_{\mathrm{BM}}) = h_{\mathrm{act}}^-$, so $B_t > h_{\mathrm{act}}^- + \Delta h_{\mathrm{dip}}$ and $Z_t < e^{-\theta(h_{\mathrm{act}}^- + \Delta h_{\mathrm{dip}})}$; and $\mu_B(t) \ge -\underline{\mu}_{\mathrm{int}}^-$ because $(z_t, u(t), w(t), x_{\perp,t})$ is admissible in the supremum defining $\underline{\mu}_{\mathrm{int}}^-$. Hence $\mathcal{L}Z_t \le \theta Z_t(\underline{\mu}_{\mathrm{int}}^- + \delta_{\mathrm{It\hat{o}}} + \frac{1}{2}\theta\sigma_h^2) \le c_{\mathrm{int}}$.

(f) Kushner step. $Z_{t\wedge\tau_{\mathrm{stop}}} \in (0,1]$ (as $B \ge 0$ up to τ_{exit}), and its stochastic integral has integrand bounded by $\theta\sigma_h$, hence is a true martingale. By (e), $Y_t := Z_{t\wedge\tau_{\mathrm{stop}}} + c_{\mathrm{int}}(T - t\wedge\tau_{\mathrm{stop}})$, $t \in [0,T]$, is a non-negative continuous supermartingale, and Ville's (Doob's) maximal inequality gives $\mathbb{P}(\sup_{[0,T]} Y_t \ge 1) \le Y_0 = e^{-\theta\beta_0} + c_{\mathrm{int}} T$. On $A_{\mathrm{exit}} := \{\tau_{\mathrm{exit}} \le T \wedge \tau_{\mathrm{BM}} \wedge \tau_\perp\}$, $B_{\tau_{\mathrm{exit}}} = 0$, so $Y \ge Z = 1$ at τ_{exit}. The second bound in (S47) gives $1 - 2c_{\mathrm{int}}T/\alpha \ge e^{-\theta(\beta_0 - h_0)} > 0$, i.e. $e^{-\theta\beta_0} \le e^{-\theta h_0}(1 - 2c_{\mathrm{int}}T/\alpha) = \frac{\alpha}{2} - c_{\mathrm{int}}T$ (using $e^{-\theta h_0} = \alpha/2$); hence $\mathbb{P}(A_{\mathrm{exit}}) \le \alpha/2$.

(g) Union. If $\tau_{\mathrm{exit}} \wedge \tau_\perp \le T$, then either $\tau_{\mathrm{BM}} < \tau_{\mathrm{exit}} \wedge \tau_\perp$, so $\tau_{\mathrm{BM}} \le T$ (probability $\le \alpha/4$ by (d)), or $\tau_{\mathrm{stop}} = \tau_{\mathrm{exit}} \wedge \tau_\perp \le T$ and the event A_{exit} or $A_\perp$ occurs. Therefore, by (b),

$$\mathbb{P}\Big(\inf_{t\in[0,T]} h(x_t) < 0\Big) \le \mathbb{P}(\tau_{\mathrm{BM}} \le T) + \mathbb{P}(A_{\mathrm{exit}}) + \mathbb{P}(A_\perp) \le \frac{\alpha}{4} + \frac{\alpha}{2} + \frac{\alpha}{4} = \alpha.$$

Thus $\mathbb{P}\big(\inf_{t\in[0,T]} h(x(t)) \ge 0\big) \ge 1 - \alpha$. ∎

S5. Numerical Certification in Full

S5.1. Benchmark, implementation and protocol

The benchmark is the class-(H) telegraph reaction–diffusion–advection lattice

$$\begin{aligned}\dot{q} &= p,\\ \dot{p} &= -(\nu L_\Delta + \alpha_0 I)q - \gamma p - c\,W_f D_h W_f\, q\\ &\quad - \kappa(q - \sin q) + Bu + B_w w,\end{aligned}$$

on $N_m = 160$ nodes ($N = 320$ states) with Dirichlet data, $\nu = 0.05$, $\alpha_0 = 1$, $\gamma = 1$, $\kappa = 0.05$, two actuators ($m = 2$) and one disturbance channel. Here L_Δ is the discrete Laplacian, D_h is the skew central difference and the advection filter is

$$W_f = \Phi_f\,\mathrm{diag}(\underbrace{1,\dots,1}_{2},\, w,\dots,w)\,\Phi_f^\top,$$

where Φ_f holds the first $K_f = 4$ mass-normalised sine modes. The plain projector $\Pi_f = \Phi_f\Phi_f^\top$ is the special case $w = 1$. The cutoff-mode weight w is a disclosed benchmark parameter: it is a configuration literal, it is swept in Experiment 9, and the deployed value $w^\star = 2^{-11/2} \approx 0.0221$ is the one selected by that sweep. The ROM keeps $K_m = 2$ eigenmodes ($K = 4$), so $K_f > K_m$ and the advection couples the ROM to the tail.

The advection is not folded into the damped-wave operator. $A_{\text{full}} = (J_{\text{full}} - R_{\text{full}})P_{\text{full}}$ stays the pure damped wave, so $Q = \gamma P$ holds (relative error 7.039×10^{-15}), $\lambda_{K+1} = 0.5$ is unchanged, and (H2a) holds by eigenmode invariance (Lemma S7, residual 6.348×10^{-13}). For the same reason the block $P_0^{-1}\Phi^\top P_{\text{full}} A_{\text{full}} P_\perp$ vanishes to round-off (1.27×10^{-12}). The linear cross-coupling sits in the nonlinearity, $F_{\text{nl}}(x) = A_{\text{adv}}x + F_{\sin}(x)$ with

$$A_{\text{adv}} = \begin{bmatrix} 0 & 0\\ -c\,W_f D_h W_f & 0\end{bmatrix},$$

and its ROM–tail block $A_{\text{adv},r,\perp} := P_0^{-1}\Phi^\top P_{\text{full}} A_{\text{adv}} P_\perp$ of Lemma S5 has norm $\|A_{\text{adv},r,\perp}\|_2 = 0.106$ in R2 and 0.005301 in R1.

Two regimes are used. R2 has $c = 1$ ($\mathrm{Pe} = 20$) and is open-loop unstable. R1 has $c = 0.05$ ($\mathrm{Pe} = 1$). The sampling period of R2 is $T_s = 3 \times 10^{-4}$ with $\tau_A = 1.5 \times 10^{-5}$, $\tau_B = 3 \times 10^{-5}$. This is a trade-off forced by the active tail coupling: at the previous $T_s = 5 \times 10^{-4}$ the certificate refuses at $w^\star$, because $\delta_{\text{total}} = 0.01821 > \delta_{\text{cert}} = 0.01455$ and the authority condition (H6), (S45), fails (Table S5). R1 keeps $T_s = 0.002$. The short period is set by the hold term, not by the plant time scale: the slowest mode ($\omega_{\min} = 1.222$) is sampled 1.714×10^4 times per period $2\pi/\omega_{\min}$ in R2 (2571 in R1), and the hold term $\phi L_{\text{hold}} T_s$ of (S33), with $L_{\text{hold}} = 44.04$ and $\mathrm{Lip}(L_f h) = 1.77$, is a share 0.4872 of δ_{total} in R2 (0.5596 in R1, where $L_{\text{hold}} = 1.474$ and $\mathrm{Lip}(L_f h) = 0.7948$).

Every constant of Sections S1–S3 is computed at run time in float64 by a function that names the equation it implements. A pytest AST scan fails if a float literal appears in the numerical modules, and a perturbation test fails if any certificate quantity does not depend on the configuration inputs. The certificate raises (refuses) unless (H0)–(H7), the timing order and the codesign window all hold. This single flag is the Theorem S1 hypothesis row of Table S3. (H-dz) is checked separately and only for Proposition S3. The Path-A LUT is indexed in port space $v = g_r^\top P z$ by default.

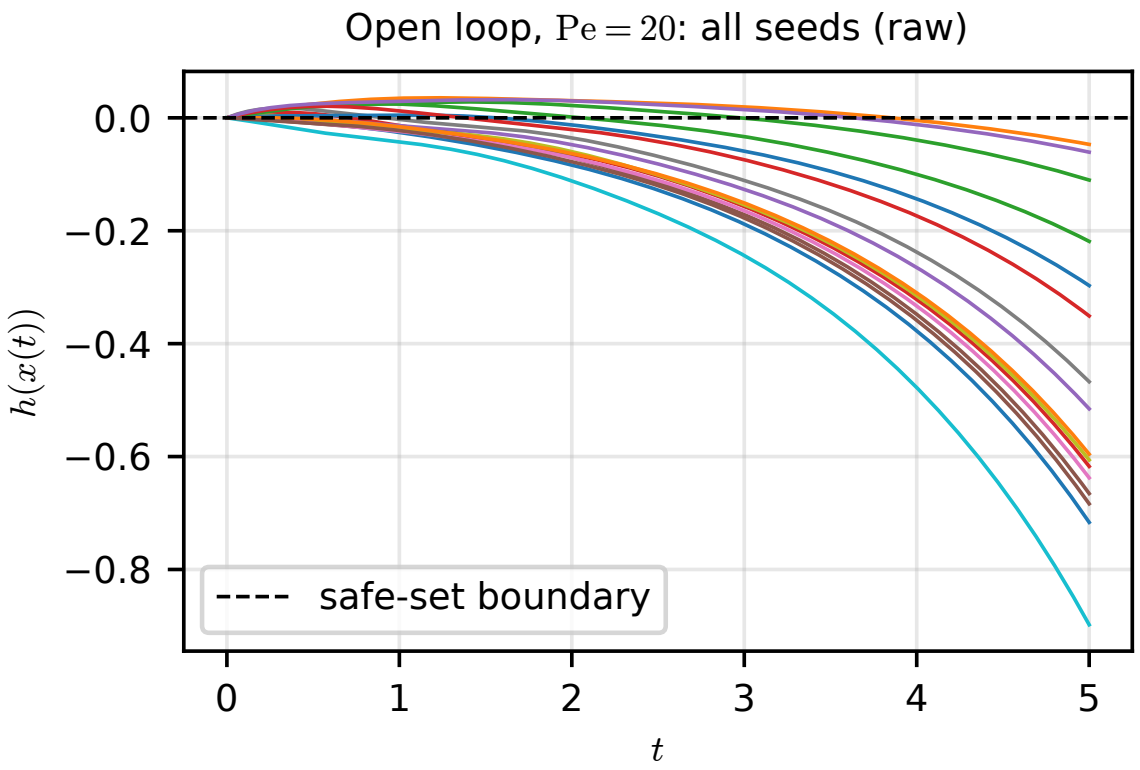


Figure S1: Experiment 0: raw open-loop $h(x(t))$ in R2 ($\mathrm{Pe} = 20$) for all seeds.

When $\mathcal{C}_{r,0} = \{\|z\|_P^2 \le E_{\mathrm{safe}}\}$ as in (S27), its range is $D_v = 2\sqrt{E_{\mathrm{safe}}}\max_j \sqrt{(g_r^\top P g_r)_{jj}} = 0.4472$ by (S7); the largest sampled value is 0.4472 (both to four significant digits). The modulus $\mathrm{Lip}(L_f h)$ that enters L_{hold} carries the factor of two of the gradient $\nabla_z(z^\top P z) = 2Pz$; it is cross-checked against finite-difference quotients of $L_f h$ at random pairs in $\mathcal{C}_{r,0}$, which must not exceed the certified modulus.

The full-order plant is integrated with an exponential integrator (ETD2RK in the lattice eigenbasis). The latch instants τ_A, τ_B are integration nodes, so the switching times are exact. The disturbance is the worst case of Remark S4 with $\epsilon_{\mathrm{adv}} = 0.002$. All seeds come from the configuration file. The suite runs 248 experiment assertions and 65 pytest tests on two CPU cores in 666.4 s, with a peak resident memory of 1193 MB (Table S21).

S5.2. Experiment 0: open-loop exit

With $\mathrm{Pe} = 20$ and $w = w^\star$ the linearisation has growth rate 0.2925 (Table S1). All 16 of 16 initial conditions satisfying (H7) leave $\mathcal{C}$ without control: the worst $\min_t h$ is -0.8979 and the latest exit is at $t = 3.856$ (Fig. S1, Table S4). The safety problem is therefore not trivial: invariance in the later experiments comes from the controller, not from passivity.

Table S1: Certificate constants (part 1 of 2), computed at run time in float64 from the lattice operators. K_f is the advection band, w the disclosed cutoff-mode weight (Table S6); $L_{\perp,P}(\mathrm{R2})/L_{\perp,P}(\mathrm{R1}) = 14.82$.

quantity	equation	R2 (Pe = 20)	R1 (Pe = 1)
N_m	–	160	160
N	–	320	320
m	–	2	2
ν	–	0.05	0.05
α_0 (plant)	–	1	1
γ	–	1	1
κ	–	0.05	0.05
c	–	1	0.05
K_m	–	2	2
K	–	4	4
full-weight advection modes	–	2	2
E_{safe}	(S5)	0.05	0.05
$\varepsilon_{\mathrm{adv}}$	(S12)	0.002	0.002
open-loop growth $\max \mathrm{Re}\,\lambda(A_{\mathrm{lin}})$	–	0.2925	-0.5
K_f	–	4	4
w (cutoff weight)	–	0.0221	0.0221
Pe	Remark S2	20	1
$\lambda_{\min}(P_{\mathrm{full},\epsilon})$	(S3)	0.6892	0.6892
$\omega_{\min}$	(S3)	1.222	1.222
$\mu_{2,P}(A_{r,0})$	(S26)	-0.5	-0.5
$\|Q - \gamma P\|/\|\gamma P\|$	(S26)	7.039×10^{-15}	7.039×10^{-15}
λ_{K+1}	(S11)	0.5	0.5
(H2a) residual $\|\Phi^\top P_{\mathrm{full},\epsilon} P_\perp\|$	Lemma S7	6.348×10^{-13}	6.348×10^{-13}
$\|A_{r,\perp}\|_2$ (eigenmode invariance)	Lemma S7	1.27×10^{-12}	1.27×10^{-12}
ρ_{ODD}	(S5)	0.2828	0.2828
D_{ODD}	(S27)	0.5656	0.5656
$\mathrm{Lip}(F_{\mathrm{nl}})$	Assumption S1	2.67	0.1354
$M_\perp$	(S12)	0.04916	0.003647
$\varepsilon_{\mathrm{basis},P}$	(S15)	0.09831	0.007293
$\|A_{\mathrm{adv},r,\perp}\|_2$	Lemma S5	0.106	0.005301
$L_{\perp,P}$ (advective, linear)	(S14)	0.04654	0.002327
$L_{\perp,P}$ (sin nonlinearity)	(S14)	8.719×10^{-4}	8.719×10^{-4}
$L_{\perp,P}$	(S14)	0.04741	0.003199
linear share of $L_{\perp,P}$	(S14)	0.9816	0.7274
L_h	(S28)	0.7865	0.7865
g_h	(S28)	0.4788	0.4788
d_h	(S28)	0.4472	0.4472
M_z	(S28)	4.324	0.7887
$\mathrm{Lip}(L_f h)$	(S28)	1.77	0.7948
$\mathrm{Lip}(L_g h)$	(S28)	2.394	2.394
$L_{\mathcal{T},z}$	(S19)	5.985	0.5985
$\eta_{\mathcal{T}}$	Lemma S3	4.975	0.4975
(H-sat) left side $L_{\mathcal{T},z}\sqrt{K} D_{\mathrm{ODD}}/2$	(S20)	3.386	0.3386
$u_{\max}$	(S20)	3.4	0.35
$\varepsilon_{\mathrm{CROWN}}$ (backward)	(S23)	0.004455	4.536×10^{-4}

Table S2: Certificate constants (part 2 of 2).

quantity	equation	R2 (Pe = 20)	R1 (Pe = 1)
E_q (explicit)	(S24)	0.005674	5.58×10^{-4}
$\varepsilon_{\mathrm{CROWN}}$	Prop. S1	0.004455	4.536×10^{-4}
D_v	(S7)	0.4472	0.4472
D_v (sampled sup)	(S7)	0.4472	0.4472
M_v	(S29)	5.176	0.9441
$\Delta_{q,A}$ (LUT step)	(S7)	8.735×10^{-4}	8.735×10^{-4}
ε_A	(S29)	0.007543	7.624×10^{-4}
Δu_{lag}	(S29)	0.007765	9.441×10^{-4}
window lower bound	(S30)	8	8
b_A	(S30)	9	9
window upper bound	(S30)	11	11
T_s	(S9)	3×10^{-4}	0.002
τ_A	(S9)	1.5×10^{-5}	1×10^{-4}
τ_B	(S9)	3×10^{-5}	2×10^{-4}
samples per period $2\pi/(\omega_{\min} T_s)$	(S9)	1.714×10^{4}	2571
b_B	Prop. S1	12	12
γ_0	(S37)	0.35	0.3
w_1	(S31)	0.05	0.04999
w_2	(S31)	0.05	0.04999
w_3	(S31)	0.9	0.9
$\phi(\gamma_0 T_s)$	(S33)	0.5	0.5
L_{hold}	(S32)	44.04	1.474
Δh_{dip}	(S32)	2.06×10^{-6}	3.008×10^{-6}
δ_{total}: tail term	(S33)	0.003666	1.835×10^{-5}
δ_{total}: quantisation term	(S33)	0.002207	2.246×10^{-4}
δ_{total}: lag term	(S33)	1.859×10^{-4}	2.26×10^{-5}
δ_{total}: adversary term	(S33)	8.944×10^{-4}	8.944×10^{-4}
δ_{total}: hold term	(S33)	0.006607	0.001474
δ_{total}	(S33)	0.01356	0.002634
tail share $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P} / \delta_{\mathrm{total}}$	(S33)	0.2704	0.006967
hold share of δ_{total}	(S33)	0.4872	0.5596
h_{act}	(S36)	0.02068	0.02559
$r_{\min}$	(S36)	0.1402	0.156
$\lambda_{\min}(M, P)$	Lemma S5	0.3731	0.9835
s_{nl}	Lemma S5	3.005×10^{-4}	3.005×10^{-4}
c_M	Lemma S5	0.3725	0.9829
δ_{cert}	Lemma S5	0.01456	0.03161
δ_{cert} branch	(S42)	1	1
margin $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$	(S45)	9.994×10^{-4}	0.02898
$\gamma_0 h_{\mathrm{act}}$ (Cor. S1)	Cor. S1	0.007237	0.007678
null-space rate	(S38)	1	1
μ_R	Prop. S3	0.5	0.5
L_{sec}	Prop. S3	2.392	0.1204
$\bar{\varepsilon}_q$	Prop. S3	0.01998	0.002342
r_{ss}	(S43)	∞	0.01162

Table S3: Hypothesis ledger of Theorem S1, hypotheses (H0)–(H7), and of Proposition S3, conditions (i) $\mu_R > L_{\text{sec}}$ and (H-dz); ✓ = satisfied.

hypothesis	content	R2	R1
(H0)	ODD containment and E_{safe}	✓	✓
(H1)	Assumption S1 constants on $\mathcal{X}_{\text{ODD}}$	✓	✓
(H2)	(H2a)–(H2c), $\gamma < 2\omega_{\min}$, $\epsilon = \gamma/2$	✓	✓
(H3)	Teacher: $H/2 \geq m$, $G_{\text{port}} \succ 0$, (H-sat)	✓	✓
(H4)	tail tolerance, h_{act} interval	✓	✓
(H5)	$b_A \geq 1$ satisfies (S8)	✓	✓
(H6)	authority (S45)	✓	✓
(H7)	initial conditions	✓	✓
(H-sat)	$L_{\mathcal{T},z}\sqrt{K}D_{\text{ODD}}/2 \leq u_{\max}$ (S20)	✓	✓
timing $0 < \tau_A < \tau_B < T_s$	latch order of (S9)	✓	✓
codesign window (S30)	b_A inside the window	✓	✓
Thm. S1 hypotheses	all of the above	✓	✓
$\mu_R > L_{\text{sec}}$	Prop. S3(i) contraction rate	×	✓
(H-dz)	$r_{\text{ss}} < r_{\min}$ (Prop. S3 only)	×	✓
Prop. S3 hypotheses	(i) $\mu_R > L_{\text{sec}}$, (H-dz) and (H3)	×	✓

Table S4: Experiment 0: open-loop ($u \equiv 0$) trajectories of the R2 benchmark (Pe $= 20$) from certified initial states.

seed	$h(x_0)$	$\min_t h(x(t))$	first exit time
0	2.06×10^{-6}	-0.7165	3×10^{-4}
1	2.06×10^{-6}	-0.04703	3.856
2	2.06×10^{-6}	-0.1102	2.983
3	2.06×10^{-6}	-0.6174	0.7137
4	2.06×10^{-6}	-0.5155	0.4689
5	2.06×10^{-6}	-0.6655	0.0657
6	2.06×10^{-6}	-0.6377	9×10^{-4}
7	2.06×10^{-6}	-0.4675	0.8838
8	2.06×10^{-6}	-0.606	0.4128
9	2.06×10^{-6}	-0.8979	3×10^{-4}
10	2.06×10^{-6}	-0.2972	1.566
11	2.06×10^{-6}	-0.5958	0.2889
12	2.06×10^{-6}	-0.2189	2.061
13	2.06×10^{-6}	-0.351	1.379
14	2.06×10^{-6}	-0.06081	3.636
15	2.06×10^{-6}	-0.6837	3×10^{-4}

S5.3. Experiment 1: constants, hypotheses and refusals

Tables S1 and S2 list every certificate constant for R2 and R1, and Table S3 lists the hypotheses. Proposition S2 holds to round-off: $\mu_{2,P}(A_{r,0}) = -0.5$.

The tail coupling is now active. In R2, $\|A_{\text{adv},r,\perp}\|_2 = 0.106$ and $L_{\perp,P} = 0.04741$. Of this, 0.04654 is the linear advective part and 8.719×10^{-4} the $\sin$ nonlinearity, so the advective part is a

fraction 0.9816 of the total. In R1, $L_{\perp,P} = 0.003199$, and the ratio between the regimes is 14.82. The tail term $L_h L_{\perp,P} \varepsilon_{\text{basis},P}$ is a share 0.2704 of $\delta_{\text{total}} = 0.01356$ in R2 (and a share 0.006967 of $\delta_{\text{total}} = 0.002634$ in R1). With $\varepsilon_{\text{basis},P} = 0.09831$, the R2 margin is $\delta_{\text{cert}} - \delta_{\text{total}} = 0.01456 - 0.01356 = 9.994 \times 10^{-4}$.

Lemma S5 gives $c_M = 0.3725$ from $\lambda_{\min}(M, P) = 0.3731$ and $s_{\text{nl}} = 3.005 \times 10^{-4}$ in R2, and $c_M = 0.9829$ from $\lambda_{\min}(M, P) = 0.9835$ and $s_{\text{nl}} = 3.005 \times 10^{-4}$ in R1. The codesign window is $[8, 11]$ and contains $b_A = 9$.

Table S5 shows the certificate refusing 6 inadmissible variants. The state-space LUT (the $m = K$ variant of Remark S3) has an empty window $[9, 5]$. The other refusals are $u_{\max}$ below the (H-sat) left side, unfiltered lattice advection, the plain projector Π_f, the previous sampling period and T_s, τ_A, τ_B scaled by 10. Proposition S3 needs (i) $\mu_R > L_{\text{sec}}$, (H-dz) and (H3) (Table S3). In R1 all three hold: $\mu_R = 0.5 > L_{\text{sec}} = 0.1204$ and $r_{\text{ss}} = 0.01162 < r_{\min} = 0.156$. In R2, $L_{\text{sec}} = 2.392$ exceeds $\mu_R = 0.5$, so condition (i) fails and Proposition S3 does not apply; this does not affect Theorem S1.

Table S5: Certificate refusals on non-admissible variants of R2. The LUT window is (S30) in the variant's own form (port space by default; the state-space row uses K, D_{ODD} and M_z).

configuration	refused	failing	δ_{total}	δ_{cert}	window $[\text{lo}, \text{hi}]$
state-space LUT ($m = K$ variant)	✓	(H5), codesign window (S30)	0.01377	0.01456	[9, 5]
$u_{\max}$ below the (H-sat) left side	✓	(H3)	0.01245	0.01483	[8, 11]
pure lattice advection (no spectral cutoff)	✓	(H4), (H6), (H7)	157.6	−3091	[3, 11]
plain projector Π_f ($w = 1$)	✓	(H4), (H6), (H7)	7.93	−8.456	[6, 11]
previous sampling period $T_s = 5 \times 10^{-4}$ at $w^\star$	✓	(H6)	0.01821	0.01455	[7, 11]
T_s, τ_A, τ_B ×10 (hold term)	✓	(H6)	0.07483	0.01438	[5, 11]

S5.4. Experiment 9: the cutoff-mode weight and the certifiable frontier

The weight w is swept over 17 geometric values from 1 down to 0.003906 at the R2 Péclet number (Table S6, Fig. S2). At every w the open loop leaves $\mathcal{C}$ on all seeds, so the plant is unsafe throughout the sweep. For $w \geq 0.08839$, the tail tolerance (H4) fails together with (H6) and (H7). For $0.03125 \leq w \leq 0.0625$, only the authority condition (H6) fails. The certificate accepts for $w \leq w^\star = 0.0221$. Below $w^\star$ the single-rate ablation (c) is no longer refused (first at $w = 0.01562$). The deployed $w^\star$ is therefore the largest grid value that is accepted with a positive authority margin $\delta_{\text{cert}} - \delta_{\text{total}}$ in (S45) while every ablation still breaks. $\|A_{\text{adv},r,\perp}\|_2$ is proportional to w, and the tail share of δ_{total} falls from 0.9968 at $w = 1$ to 0.2704 at $w^\star$.

For the plain projector ($w = 1$), the Péclet scan of Table S7 shows that the certifiable advection speeds ($c \leq 0.02$, $\text{Pe} \leq 0.4$) and the open-loop-unstable speeds ($c \geq 0.5$, $\text{Pe} \geq 10$) are disjoint. Under the Euclidean-ODD supremum that defines $M_\perp$ in (S12), full-strength cutoff-mode advection is not certifiable at any Péclet number where the open loop is unstable: no softening. The plain-projector result is the L^2-ODD conservatism of Remark S2. Keeping A_{adv} inside F_{nl} preserves (H2a) exactly (eigenmode invariance, Lemma S7) at the price of charging $A_{\text{adv},r,\perp}$ in $M_\perp$ and $L_{\perp,P}$. The weight w attenuates the cutoff modes, which is what an H^1-bounded ODD would do for a smooth field; certifying $w = 1$ at $\text{Pe} \geq 10$ would require that H^1 formulation, which is outside the present theorem.

Table S6: Disclosed sweep of the cutoff-mode advection weight w at the R2 Péclet number ($w = 1$: plain projector Π_f). The deployed $w^\star$ is the largest grid value accepted with a positive authority margin in (S45) and all ablations refused. Tail share: $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P} / \delta_{\mathrm{total}}$; margin: $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$; exits: open-loop exits of 16 seeds at $c = 1$; –: ablations not evaluated (the full certificate already refuses).

w	Thm. S1 hypotheses	failing	$\varepsilon_{\mathrm{basis},P}$	tail share	$\|A_{\mathrm{adv},r,\perp}\|_2$	margin	abl. refused	exits
1	×	(H4), (H6), (H7)	4.77	0.9968	4.798	−16.39	–	16/16
0.7071	×	(H4), (H6), (H7)	2.766	0.995	3.393	−6.088	–	16/16
0.5	×	(H4), (H6), (H7)	1.706	0.9911	2.399	−2.492	–	16/16
0.3536	×	(H4), (H6), (H7)	1.113	0.9831	1.696	−1.107	–	16/16
0.25	×	(H4), (H6), (H7)	0.7587	0.9677	1.2	−0.5211	–	16/16
0.1768	×	(H4), (H6), (H7)	0.5331	0.9388	0.8482	−0.2539	–	16/16
0.125	×	(H4), (H6), (H7)	0.3829	0.8879	0.5998	−0.1256	–	16/16
0.08839	×	(H4), (H6), (H7)	0.2801	0.8052	0.4241	−0.06177	–	16/16
0.0625	×	(H6)	0.2086	0.6865	0.2999	−0.02934	–	16/16
0.04419	×	(H6)	0.1585	0.5417	0.212	−0.01265	–	16/16
0.03125	×	(H6)	0.1232	0.395	0.1499	−0.003795	–	16/16
0.0221	✓	none	0.09831	0.2704	0.106	9.994×10^{-4}	✓	16/16
0.01562	✓	none	0.08073	0.1782	0.07497	0.003661	×	16/16
0.01105	✓	none	0.0683	0.1159	0.05301	0.00518	×	16/16
0.007812	✓	none	0.05952	0.0758	0.03748	0.006072	×	16/16
0.005524	✓	none	0.0533	0.05036	0.02651	0.006613	×	16/16
0.003906	✓	none	0.04891	0.03419	0.01874	0.006949	×	16/16

Table S7: Plain projector Π_f ($w = 1$): certificate acceptance versus open-loop instability over the advection speed c. The two sets are disjoint.

c	Pe	Thm. S1 hypotheses	failing	$\varepsilon_{\mathrm{basis},P}$	tail share $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P} / \delta_{\mathrm{total}}$	max Re eig$(A_{\mathrm{full}} + A_{\mathrm{adv}})$
0.005	0.1	✓	none	0.06197	0.05895	−0.5
0.01	0.2	✓	none	0.08563	0.1427	−0.5
0.02	0.4	✓	none	0.1329	0.3361	−0.5
0.03	0.6	×	(H6)	0.1803	0.5053	−0.5
0.05	1	×	(H4), (H6), (H7)	0.2749	0.7202	−0.5
0.1	2	×	(H4), (H6), (H7)	0.5114	0.9039	−0.5
0.2	4	×	(H4), (H6), (H7)	0.9846	0.9716	−0.5
0.3	6	×	(H4), (H6), (H7)	1.458	0.9859	−0.365
0.5	10	×	(H4), (H6), (H7)	2.404	0.9935	0.1025
1	20	×	(H4), (H6), (H7)	4.77	0.9968	0.8887

S5.5. Experiment 2: Proposition S1 (twin-CROWN)

For $b_B = 3, \ldots, 14$, with nearest and Adam-learned rounding, the twin-CROWN bound exceeds the largest Student–Teacher gap over 100016 samples (uniform samples of $\mathcal{Z}_{\mathrm{ODD}}$ plus all box vertices) by a factor of 1.887–2.844 (Fig. S3, Table S8). The bound is monotone non-increasing in b_B and halves with every additional bit of b_B. At $b_B = 12$, learned rounding lowers $\varepsilon_{\mathrm{CROWN}}$ from 0.004718 to 0.004455. The port box that indexes the Path-A table is certified as well.

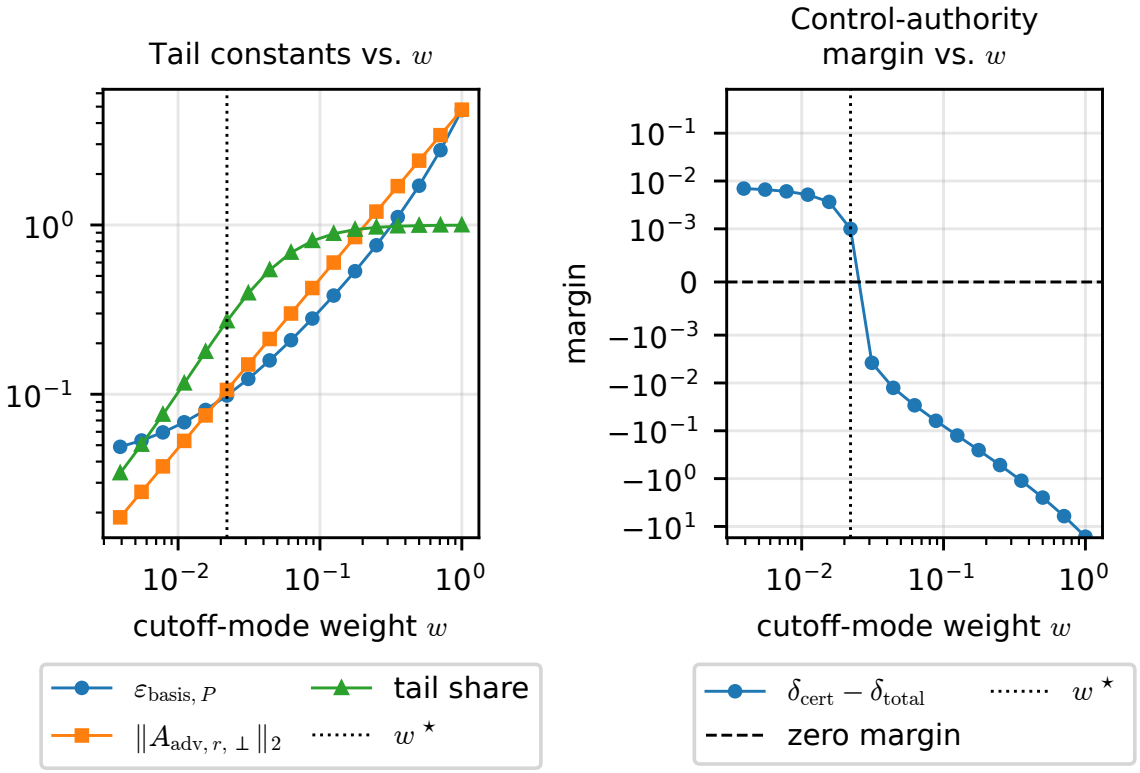


Figure S2: Experiment 9: tail constants (left) and the authority margin $\delta_{\text{cert}} - \delta_{\text{total}}$ of (S45) (right) versus the cutoff-mode weight w; the dotted line marks $w^\star$.

S5.6. Experiment 3: δ_{cert}, Lemma S5 and Lemma S4

In class (H), the certified bound of Lemma S5 is compared with the brute-force minimum of the exact bracket of (S37) over samples of $\mathcal{C}_{r,\text{act}}$. The bound is $0.01456 \le 0.01468$ in R2 and $0.03161 \le 0.03164$ in R1, so it is sound. In both regimes $\delta_{\text{total}} \le \delta_{\text{cert}}$ (Fig. S4).

On a class-(D) instance, the closed form (S39) equals the brute-force infimum of its affine bound in both branches and lies below the exact-bracket minimum (Table S9). This is a standalone check of the lemma, so δ_{total} is reported as N/A there. Using branch 1 unconditionally would overestimate the margin when $c < \gamma_0$.

S5.7. Experiment 4: Theorem S1 (positive case)

The full method runs with the worst-case adversary on 16 seeds per regime from two kinds of initial condition: the boundary start $\tilde{h}_r(z(0)) = 0$, with the tail at its largest admissible value (H7), and the interior start $h_r(z(0)) = 0.5\, h_{\text{act}}$. Conclusions (a)–(c) hold at every integrator node for both starts and both regimes (Fig. S5, Table S10); the largest tail ratio $\|x_\perp\|/\varepsilon_{\text{basis},P}$ (norm weighted by $P_{\text{full},\epsilon}$) is 1 in R2 and 1 in R1. For both starts the minimum of $h(x(t))$ over the run is attained at the initial instant ($t = 0$ for the boundary start, $t = 0$ for the interior start): $\min_t h$ is 2.06×10^{-6} in R2 and 3.008×10^{-6} in R1 from the boundary, 0.01034 and 0.0128 from the interior. The inter-sample dip

Table S8: Proposition S1: twin-CROWN bound vs. the sampled Student–Teacher gaps on 100016 points of $\mathcal{Z}_{\mathrm{ODD}}$ (100000 uniform samples plus all 16 box vertices).

b_B	rounding	$\varepsilon_{\mathrm{CROWN}}$	E_q (explicit)	max sampled gap	mean gap	bound / max gap
3	nearest	2.559	2.887	0.9001	0.3945	2.844
3	learned	2.326	3.189	0.8927	0.3574	2.606
4	nearest	1.329	1.481	0.5366	0.1701	2.476
4	learned	1.145	1.603	0.4509	0.1397	2.54
5	nearest	0.6217	0.7167	0.2263	0.07976	2.747
5	learned	0.5696	0.7491	0.2261	0.0715	2.519
6	nearest	0.319	0.3508	0.1449	0.03818	2.201
6	learned	0.2868	0.3616	0.1326	0.03538	2.162
7	nearest	0.143	0.1801	0.0664	0.01819	2.153
7	learned	0.1424	0.1838	0.0586	0.01816	2.43
8	nearest	0.07865	0.08686	0.03422	0.00951	2.298
8	learned	0.07095	0.09282	0.03621	0.00937	1.96
9	nearest	0.03591	0.04118	0.01733	0.004777	2.072
9	learned	0.03554	0.04131	0.01776	0.00476	2.001
10	nearest	0.01931	0.02345	0.01023	0.002453	1.888
10	learned	0.01781	0.02394	0.00809	0.002398	2.202
11	nearest	0.009407	0.01112	0.004111	0.001221	2.288
11	learned	0.008911	0.01101	0.003928	0.001202	2.268
12	nearest	0.004718	0.005478	0.002039	6.126×10^{-4}	2.314
12	learned	0.004455	0.005674	0.001966	6.014×10^{-4}	2.266
13	nearest	0.002385	0.002768	0.001043	3.084×10^{-4}	2.287
13	learned	0.002235	0.002777	0.001134	3.016×10^{-4}	1.97
14	nearest	0.001224	0.001345	5.346×10^{-4}	1.568×10^{-4}	2.29
14	learned	0.001131	0.001417	5.994×10^{-4}	1.509×10^{-4}	1.887

Table S9: Certified margin δ_{cert}: class-(H) bound of Lemma S5 ($c_M = \lambda_{\min}(M, P) - 2s_{\mathrm{nl}}$, two-branch infimum) and the class-(D) closed form (S39) vs. the brute-force infimum of the exact bracket over $\mathcal{C}_{r,\mathrm{act}}$. For class (D) the column c_M holds the closed-form slope c.

case	bound	c_M	$\lambda_{\min}(M, P)$	s_{nl}	γ_0	branch	δ_{cert} (bound)	brute-force inf	δ_{total}
R2	Lemma S5	0.3725	0.3731	3.005×10^{-4}	0.35	1	0.01456	0.01468	0.01356
R1	Lemma S5	0.9829	0.9835	3.005×10^{-4}	0.3	1	0.03161	0.03164	0.002634
class (D)	(S39)	2.59	–	–	0.3	1	0.0578	0.06345	N/A (standalone)
class (D)	(S39)	2.59	–	–	3	2	0.1032	0.1148	N/A (standalone)

floor $-\Delta h_{\mathrm{dip}}$ of conclusion (b) is therefore not exercised dynamically by these runs. Experiment 4b measures how much of each budget term the same trajectories use.

S5.8. Experiment 4b: budget utilisation along the certified trajectories

Experiment 4b re-runs the closed loops of Experiment 4 and evaluates, at each of the 1600032 (R2) and 240000 (R1) integrator nodes over both starts and all seeds, every term of (S49) next to the bound the proof uses for it (Table S11). Each fraction below is the realised term over its bound:

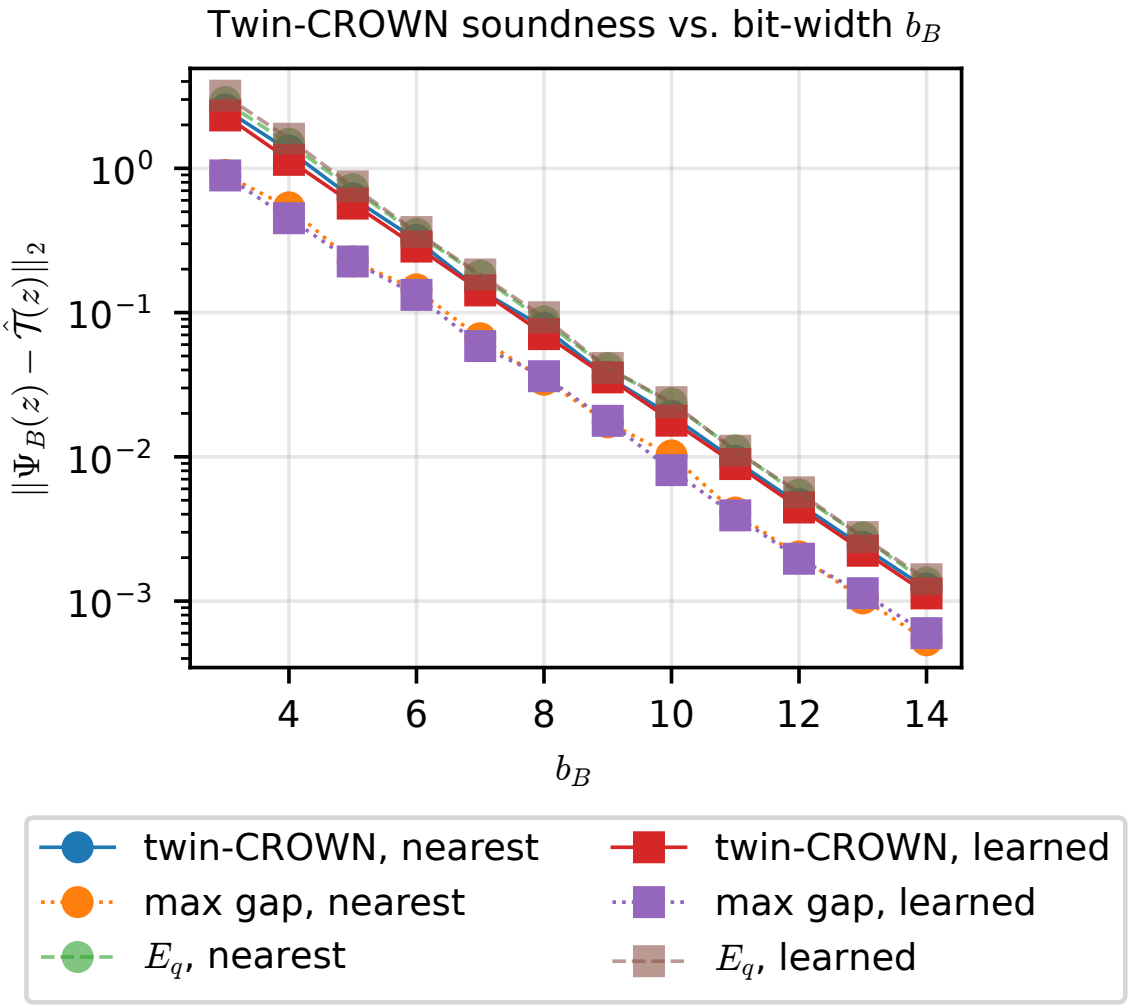


Figure S3: Experiment 2: certified $\varepsilon_{\mathrm{CROWN}}$, explicit E_q and the largest sampled gap versus b_B.

Table S10: Theorem S1: measured minima over all seeds and integrator nodes for the boundary start ($\tilde{h}_r(z(0)) = 0$) and the interior start ($h_r(z(0)) = 0.5\, h_{\mathrm{act}}$), worst-case adversary.

config	start	seeds	$\min_t h(x_t)$	time of min	$\min \tilde{h}_r$	max tail / $\varepsilon_{\mathrm{basis},P}$	margin $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$
R2	boundary	16	2.06×10^{-6}	0	-7.294×10^{-17}	1	9.994×10^{-4}
R2	interior	16	0.01034	0	6.709×10^{-4}	0.0583	9.994×10^{-4}
R1	boundary	16	3.008×10^{-6}	0	-7.774×10^{-17}	1	0.02898
R1	interior	16	0.0128	0	0.01274	0.04749	0.02898

the maximum over all nodes and, in parentheses, the median over the 102636 (R2) and 20238 (R1) nodes whose sample z_k lies in $\mathcal{C}_{r,\mathrm{act}}$, where the budget enters the proof. A fraction 0.9359 (R2) and

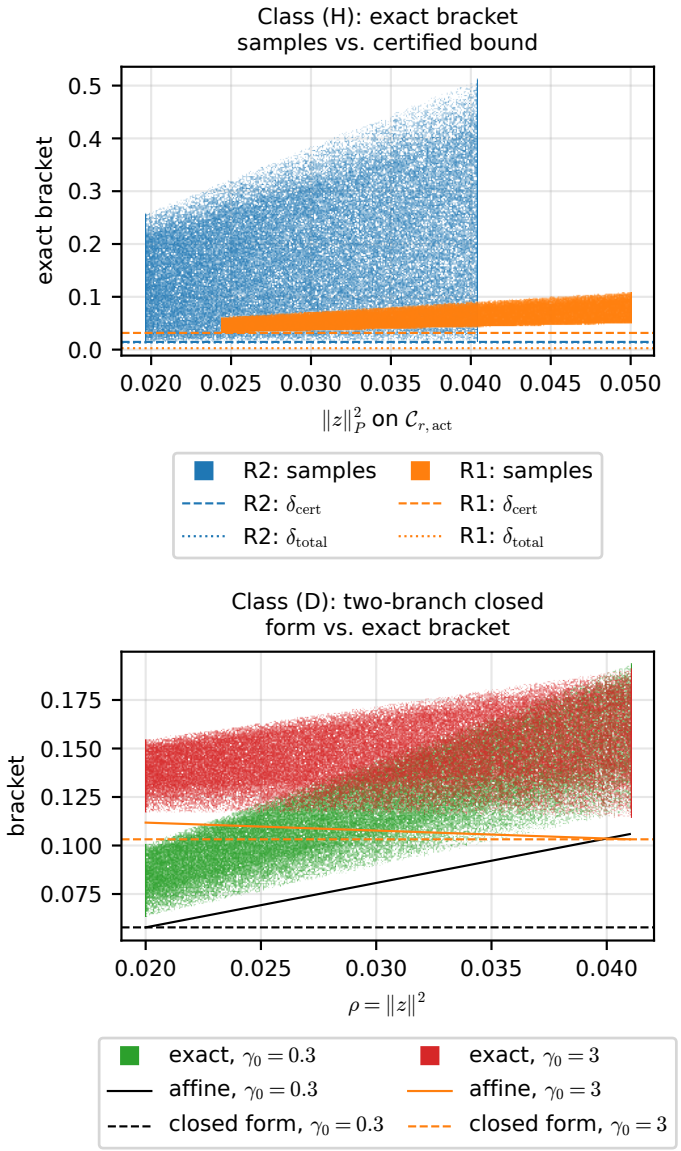


Figure S4: Experiment 3: exact bracket samples versus the certified δ_{cert} and δ_{total} (top); class-(D) closed form versus the exact bracket (bottom).

0.9157 (R1) of the nodes belong to periods with $z_k \in \mathcal{B}_{\mathrm{int}}$; there the proof uses only the Lipschitz branch, and its inequality holds at every such node.

- Adversary: 0.8772 (0.1454) in R2 and 0.9761 (0.2814) in R1. The worst-case w has norm $\varepsilon_{\mathrm{adv}}$ at every node, so the gap to one is the supremum of $\|d_r^\top \nabla \tilde{h}_r\|_2$ over $\mathcal{C}_{r,0}$ in $\overline{d}_h$.

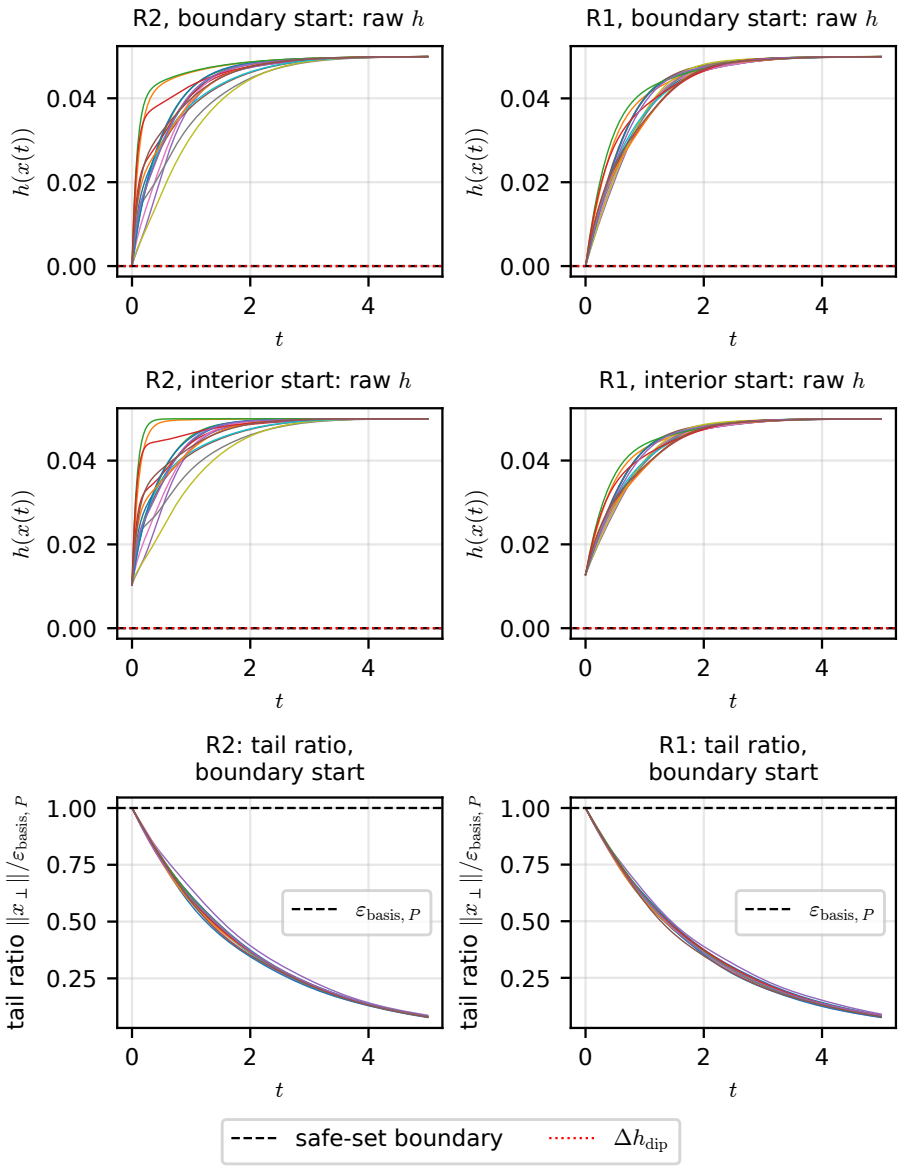


Figure S5: Experiment 4: raw closed-loop $h(x(t))$ for the boundary start (top row) and the interior start (middle row), and the tail ratio $\|x_\perp\|/\varepsilon_{\text{basis},P}$, with the norm weighted by $P_{\text{full},\epsilon}$, for the boundary start (bottom row); R2 left, R1 right.

- Quantization and lag: 0.3729 (0.01712) and 0.3672 (0.03416). The mismatch e_u reaches at most 0.5872 (R2) and 0.5705 (R1) of its phase bound, so $\varepsilon_{\text{CROWN}}$, ε_A and Δu_{lag} are the conservative constants of this term.

- Tail coupling: 0.3161 (0.003444) and 0.2606 (0.01455). Of the factors of $L_h L_{\perp,P} \varepsilon_{\text{basis},P}$, the two least used are the Lipschitz constant $L_{\perp,P}$ (median $\|\Delta_\perp\|_2/(L_{\perp,P}\|x_\perp\|) = 0.2554$ in R2 and 0.1842 in R1) and the alignment of $\nabla\tilde{h}_r$ with $\Delta_\perp$ (median cosine 0.2113 and 0.2817).

- Hold shift: 0.07439 (0.001438) and 0.09728 (0.0226), the smallest of the four fractions in both regimes. Two constants of L_{hold} are conservative here: $\|z - z_k\|_2$ reaches at most 0.3205 (R2) and 0.5853 (R1) of $M_z s$, and the hold shift at most 0.2323 (R2) and 0.1794 (R1) of $L_{\text{hold}}\|z - z_k\|_2/M_z$, its Lipschitz bound in z.

The realised rate $\dot{\tilde{h}}_r + \gamma_0 \tilde{h}_r$ minus the right-hand side of (S49) is positive at every active-shell node, so the soundness check passes. Its minimum is $0.6597\,\delta_{\text{total}}$ in R2 and $11.5\,\delta_{\text{total}}$ in R1, attained at a sampling instant, where the hold term vanishes; the Teacher margin at z_k (the left-hand side of (S37) minus δ_{total}) supplies a fraction 0.5084 (R2) and 0.9691 (R1) of it.

Table S11: Experiment 4b: realised fraction of each budget term of (S49) along the certified trajectories of Experiment 4 (both starts, all seeds, every integrator node, worst-case adversary); row (v): minimum and median of the realised rate minus the right-hand side of (S49), in units of δ_{total}. Columns $\mathcal{C}_{r,\text{act}}$: nodes of periods whose sample z_k lies in the active shell, where the budget enters the proof.

config	term	max, all nodes	median, all nodes	max, $\mathcal{C}_{r,\text{act}}$	median, $\mathcal{C}_{r,\text{act}}$
R2	(i) tail coupling	0.3161	2.063×10^{-4}	0.3161	0.003444
R2	(ii) adversary	0.8772	0.01896	0.8772	0.1454
R2	(iii) quantization/lag	0.3729	0.001841	0.3729	0.01712
R2	(iv) hold shift	0.07439	5.455×10^{-6}	0.07439	0.001438
R2	(iii) phase $[0, \tau_A)$	0.2057	7.39×10^{-4}	0.2057	0.008388
R2	(iii) phase $[\tau_A, \tau_B)$	0.3729	0.003766	0.3729	0.0348
R2	(iii) phase $[\tau_B, T_s)$	0.2551	0.002002	0.2551	0.01809
R2	(v) slack / δ_{total}: min, median	N/A	N/A	0.6597	2.807
R1	(i) tail coupling	0.2606	4.582×10^{-4}	0.2606	0.01455
R1	(ii) adversary	0.9761	0.04194	0.9761	0.2814
R1	(iii) quantization/lag	0.3672	0.005189	0.3672	0.03416
R1	(iv) hold shift	0.09728	8.842×10^{-4}	0.09728	0.0226
R1	(iii) phase $[0, \tau_A)$	0.2559	0.002249	0.2559	0.02988
R1	(iii) phase $[\tau_A, \tau_B)$	0.3672	0.009477	0.3672	0.05098
R1	(iii) phase $[\tau_B, T_s)$	0.2856	0.005166	0.2856	0.02683
R1	(v) slack / δ_{total}: min, median	N/A	N/A	11.5	17.56

S5.9. Experiment 5: Proposition S3 and ablation (d)

In R1, (H-dz) holds and the trajectories enter $\mathcal{B}_{\text{ss}}$. With the mid-tread deadzone, Path A is silent late in the run ($u_A \equiv 0$) for every $b_A \in \{3, 5, 7, 9\}$. Ablation (d) replaces it with a mid-rise quantiser and gives a persistent actuator limit cycle at every b_A, including the deployed $b_A = 9$ (Figs. S6 and S7, Table S12). The number of sign switches per unit time falls from 953.1 at $b_A = 3$ to 121.6 at $b_A = 9$, and the final lag error $\|u_A - \hat{\mathcal{T}}\|_2$ of the mid-rise quantiser at $b_A = 9$ is 3.507×10^{-4}. With $b_A = 1$, $r_{\text{ss}} = 0.3992 > r_{\min} = 0.156$, so (H-dz) fails and Proposition S3 is refused.

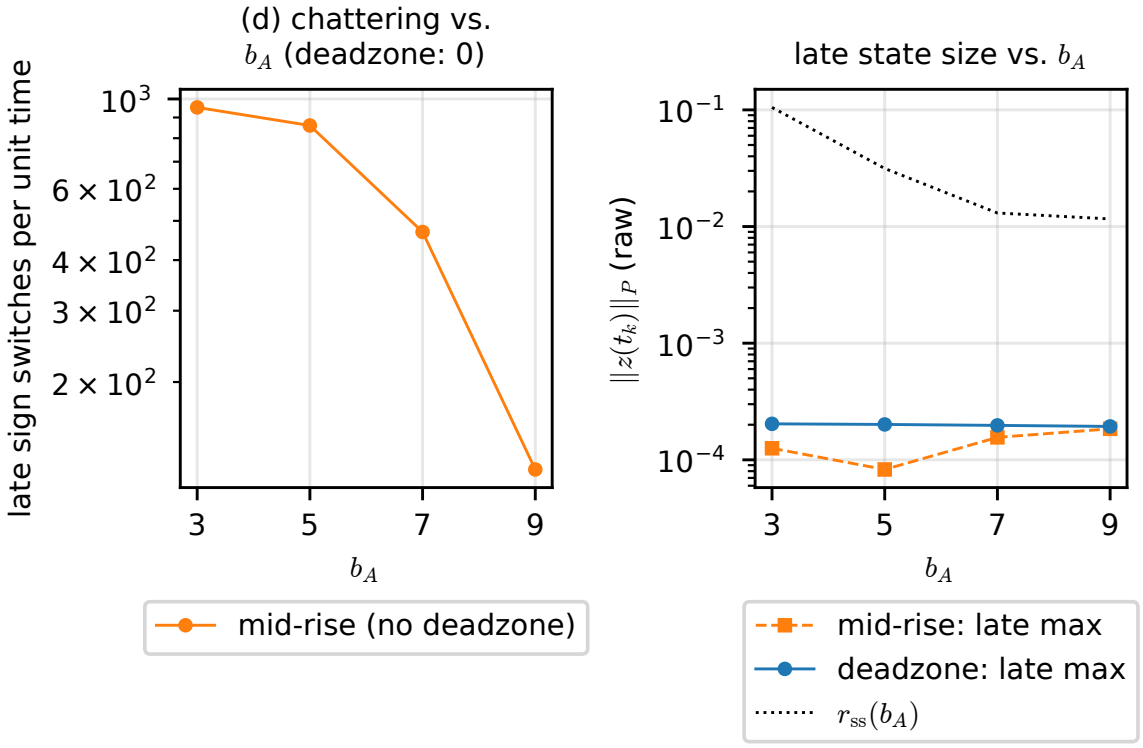


Figure S6: Experiment 5 / ablation (d): late sign switches of the mid-rise Path-A command (left) and the late-window state size (right) versus b_A.

R1 is open-loop stable, so the chattering appears in the actuator signal, not as a larger state amplitude.

S5.10. Experiment 6: ablations (a), (b), (c), (e)

Each ablation removes one component at the deployed $w^\star$. In every case the full certificate refuses and the full-order plant leaves $\mathcal{C}$ (Fig. S8, Table S13). For (a) and (b), a naive certificate with the corresponding term dropped would accept.

- (a) $K_m = 1$ without tail bookkeeping: 4 of 16 seeds exit.
- (b) $b_B = 2$ without the CROWN margin: 9 of 16 seeds exit.
- (c) Single-rate hold (Table S14). At the deployed T_s, δ_{total} rises from 0.01356 to $0.01702 > \delta_{\text{cert}}$, so (H6) fails. At $T_s = 0.2$, single rate gives 10 exits and multi-rate gives none.
- (e) $\epsilon_{\text{adv}} = 1$: 16 of 16 seeds exit. At the largest certifiable $\epsilon^\star_{\text{adv}} = 0.004223$ there are no exits.

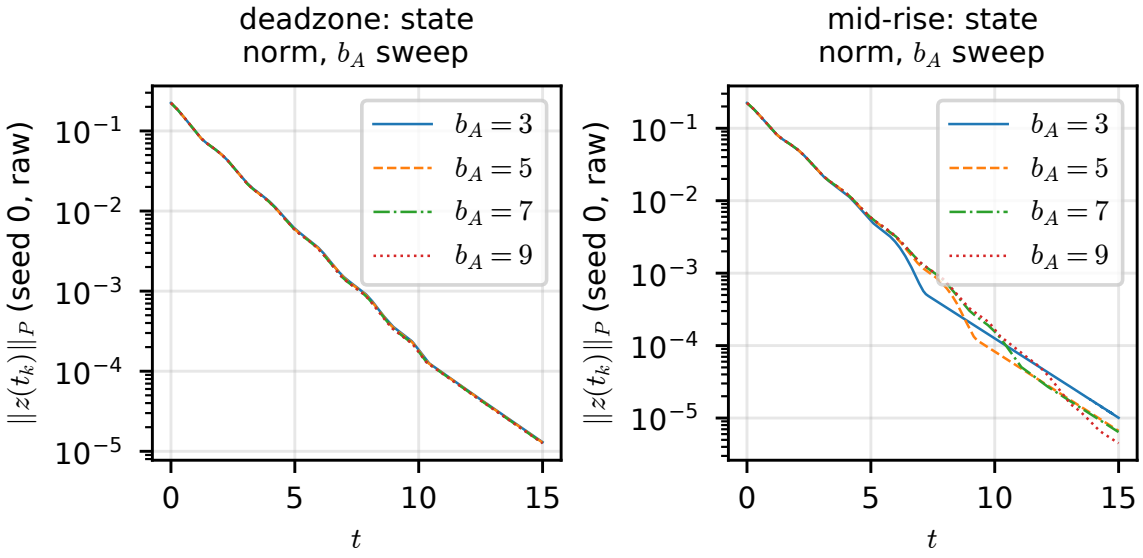


Figure S7: Experiment 5: $\|z(t_k)\|_P$ for the deadzone (left) and mid-rise (right) quantiser over the b_A sweep.

Table S12: Proposition S3 and ablation (d) on R1 (disturbance $w \equiv 0$): mid-tread LUT with deadzone vs. mid-rise LUT (no zero level) at the deployed $b_A = 9$ and over the sweep $b_A \in \{3, 5, 7, 9\}$; the last row is $b_A = 1$, which violates (H-dz). Final lag: min over seeds of $\|u_A(t_k) - \hat{\mathcal{T}}(z(t_k))\|_2$ at the last sample. For every swept b_A, $u_A \equiv 0$ in the late window with the deadzone and $u_A \neq 0$ at every late sample without it (asserted).

LUT	b_A	ε_A	r_{ss}	r_{min}	(H-dz)	late max $\|z\|_P$	final lag $\|u_A - \hat{\mathcal{T}}\|_2$	switches/time	min h
deadzone	3	0.02022	0.105	0.156	✓	2.035×10^{-4}	1.966×10^{-6}	0	3.939×10^{-6}
deadzone	5	0.005395	0.03145	0.156	✓	2.011×10^{-4}	1.914×10^{-6}	0	3.939×10^{-6}
deadzone	7	0.001689	0.01306	0.156	✓	1.974×10^{-4}	1.854×10^{-6}	0	3.939×10^{-6}
deadzone	9	7.624×10^{-4}	0.01162	0.156	✓	1.934×10^{-4}	1.833×10^{-6}	0	3.939×10^{-6}
mid-rise	3	0.02022	0.105	0.156	✓	1.257×10^{-4}	0.0197	953.1	3.939×10^{-6}
mid-rise	5	0.005395	0.03145	0.156	✓	8.268×10^{-5}	0.004851	859.9	3.939×10^{-6}
mid-rise	7	0.001689	0.01306	0.156	✓	1.559×10^{-4}	0.001286	469.4	3.939×10^{-6}
mid-rise	9	7.624×10^{-4}	0.01162	0.156	✓	1.845×10^{-4}	3.507×10^{-4}	121.6	3.939×10^{-6}
deadzone	1	0.07951	0.3992	0.156	×	2.067×10^{-4}	2.005×10^{-6}	0	3.939×10^{-6}

S5.11. Experiment 7: Corollary S1 and ablation (f)

In R1 the paths start on the level 0.9 of $\tilde{\mathcal{C}}_r$ ($h(x_0) = 0.005051$), with spatially smooth noise. The intensity is chosen by a grid search over $\sigma \in \{0.002, 0.003, 0.005, 0.007, 0.01, 0.014, 0.02\}$ (Table S15). $\sigma^\star = 0.007$ is the largest grid value for which the Corollary accepts at $T = 2$ and $\max_t |h(x_t) - h(x_0)| \geq 0.1\, h(x_0)$ holds on at least half of the paths; the Corollary refuses from $\sigma = 0.01$ on (failing condition: margin). On the screening paths at $\sigma^\star$ the criterion is driven mostly

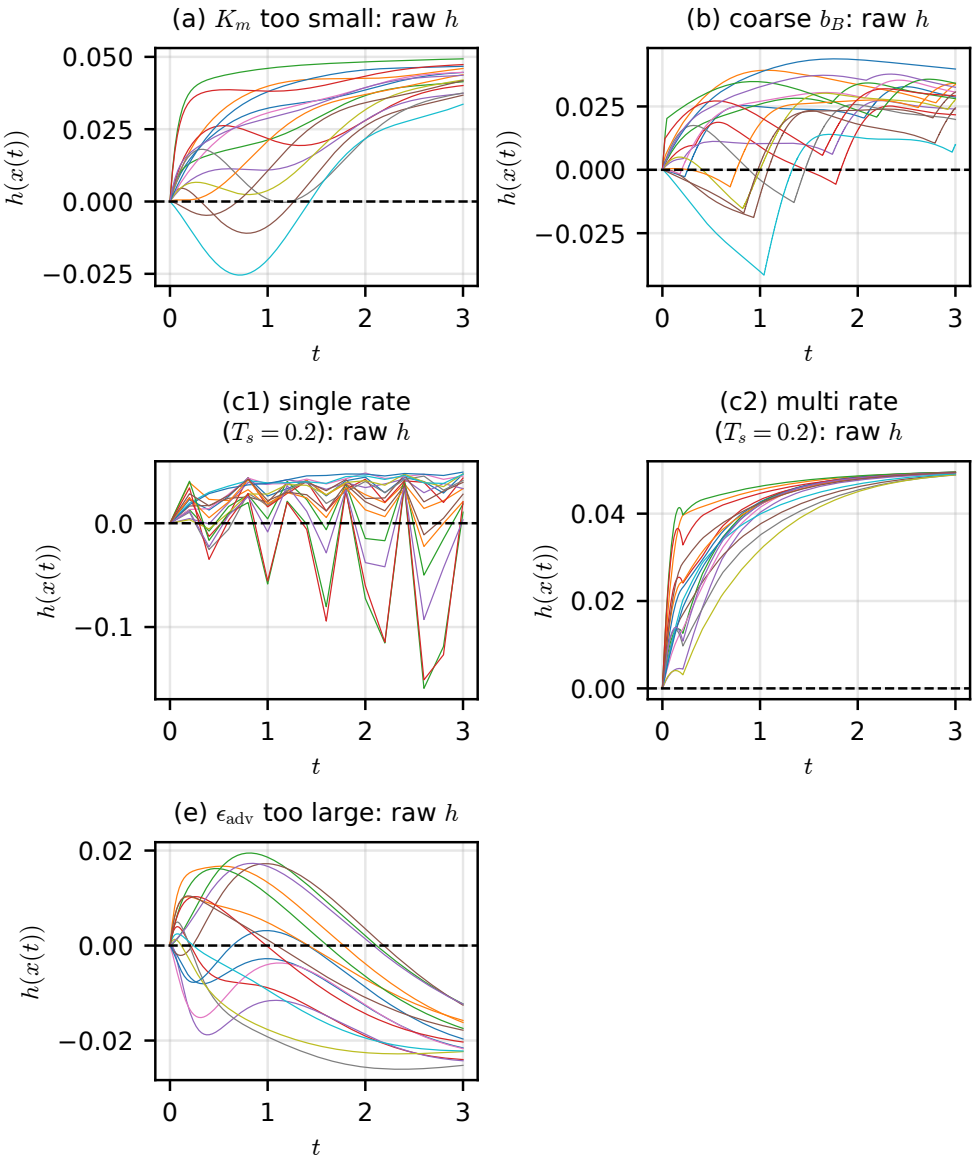


Figure S8: Experiment 6: raw $h(x(t))$ under each ablation; violations appear below $h = 0$.

by the closed-loop drift (median deviation $8.41\, h(x_0)$). The noise itself is not negligible: the median deviation from the noise-free trajectory is $0.174\, h(x_0)$ (Table S15).

Out of 2000 Monte Carlo paths, 0 exit. The exact Clopper–Pearson 95% upper limit is $0.001843 \leq \alpha = 0.05$ (Fig. S9, Table S16). The bound is very conservative: $T_{\text{TTL}} = 2022$ exceeds the horizon by a factor of 1011, and no exit is observed.

Ablation (f) uses spatially white small-scale noise. The full Corollary refuses it: the tail envelope $\Delta_\sigma = 1.1$ exceeds $\sqrt{E_{\text{safe}}} = 0.2236$. A cruder bound that ignores the tail diffusion ($\bar{\sigma}_\perp := 0$) accepts it, yet 2000 of 2000 paths exit (Clopper–Pearson lower limit $0.9982 > \alpha$). The tail-diffusion term is therefore load-bearing. The constants are split into Tables S17 and S18.

Table S13: Ablations at the deployed $w^\star$: every removed component yields a certificate refusal and a full-order safety violation under the worst-case adversary. Row (e): the largest certifiable adversary level is $\epsilon^\star_{\text{adv}} = 0.004223$.

ablation	refused	failing	δ_{total}	δ_{cert}	naive δ_{total}	naive δ_{cert}	exits (of 16)	$\min_t h$
(a) $K_m = 1$, no tail bookkeeping	✓	(H3), (H4), (H6), (H7)	7.593	−60.63	0.007956	0.0335	4	−0.02545
(b) $b_B = 2$, no CROWN margin	✓	(H6)	2.149	0.01456	0.01135	0.01456	9	−0.04155
(c) single rate $\tau_A = \tau_B = T_s$	✓	(H6)	0.01702	0.01456	–	–	10	−0.1591
(e) $\epsilon_{\text{adv}} = 1$	✓	(H6)	0.4617	0.01409	–	–	16	−0.02605

Table S14: Ablation (c): exponential weights w_1, w_2, Δh_{dip} and δ_{total} for multi-rate vs. single-rate hold.

schedule	τ_A	τ_B	w_1	w_2	Δh_{dip}	δ_{total}: lag term	δ_{total}	(H6)
multi rate	1.5×10^{-5}	3×10^{-5}	0.05	0.05	2.06×10^{-6}	1.859×10^{-4}	0.01356	✓
single rate	3×10^{-4}	3×10^{-4}	1	0	3.097×10^{-6}	0.003718	0.01702	×
single rate, $T_s = 0.2$	0.2	0.2	1	0	1.377	2.479	6.941	×
multi rate, $T_s = 0.2$	0.01	0.02	0.04835	0.04852	0.9057	0.1199	4.582	×

The Corollary is not exercised on R2. There the pointwise penalty $\gamma_0 h_{\text{act}} = 0.007237$ already exceeds the margin $\delta_{\text{cert}} - \delta_{\text{total}} = 9.994 \times 10^{-4}$, so its authority condition fails even at $\sigma = 0$ (asserted; Table S2). All stochastic runs use $r_{W_2} = 0$, i.e. no distributional drift of the adversary.

Table S15: Grid search over the profile-noise intensity σ (R1, $T = 2$, $\alpha = 0.05$). Fluct. share: fraction of 100 screening paths with $\max_t |h(x_t) - h(x_0)| \geq 0.1\, h(x_0)$; noise dev.: median of $\max_t |h(x_t) - h_{\text{det}}(t)|/h(x_0)$ (deviation from the noise-free run). $\sigma^\star = 0.007$.

σ	Corollary	failing condition	$\delta_{\text{total},\sigma} + \delta_{\text{stoch}} + \delta_{\text{drift}}$	$\delta_{\text{cert},\sigma}$	T_{TTL}	fluct. share	total dev.	noise dev.
0.002	✓	none	0.0161	0.03161	1015	1	8.399	0.04911
0.003	✓	none	0.0175	0.03161	1248	1	8.401	0.07428
0.005	✓	none	0.02084	0.03161	1667	1	8.405	0.1269
0.007	✓	none	0.02496	0.03161	2022	1	8.41	0.174
0.01	×	margin	0.03276	0.03161	2440	–	–	–
0.014	×	margin	0.04671	0.03161	2793	–	–	–
0.02	×	margin	0.07733	0.03161	2933	–	–	–

S5.12. Experiment 8: numerical method

At the convergence-test step, explicit Euler is unstable (amplification 3.73). The stiffness is moderate (spectral radius 72, stiffness ratio 43.83). Step halving gives error ratios of 4.002–4.045 for ETD2RK, i.e. order 2 (Fig. S10, Table S19). The simulator matches the closed-form augmented exponential of a latched linear system to 2.808×10^{-14}, whereas rounding the latch instants to a step

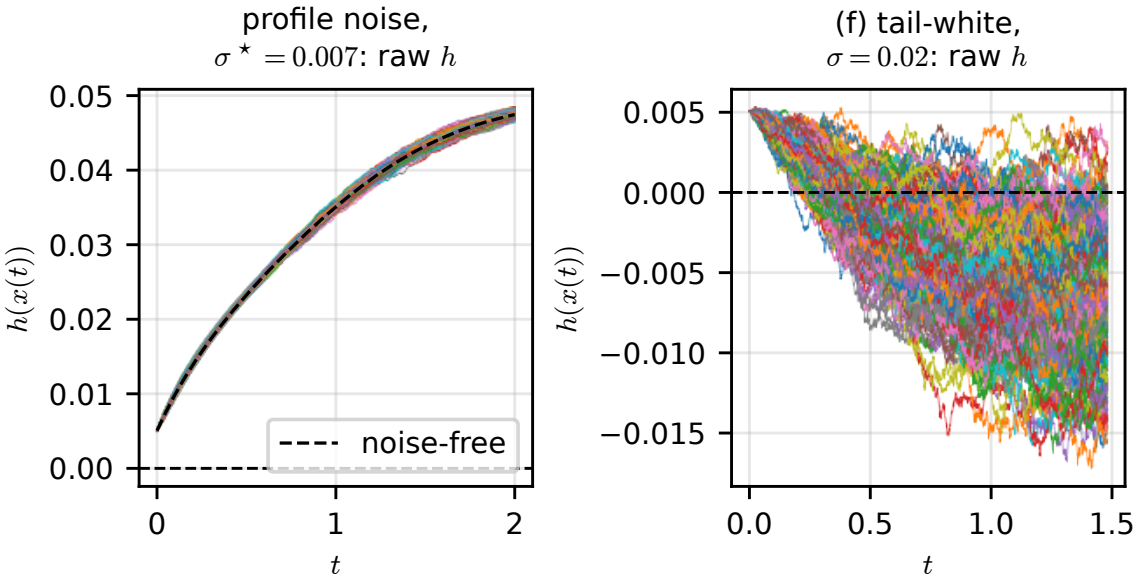


Figure S9: Experiment 7: raw stochastic $h(x_t)$ paths for profile noise at $\sigma^\star$ with the noise-free run (left) and for ablation (f) (right).

Table S16: Corollary S1 (R1, $T = 2$, $\alpha = 0.05$, $r_{W_2} = 0$, worst-case adversary, $z(0)$ on the level 0.9 of $\tilde{\mathcal{C}}_r$): Monte Carlo exit frequency vs. the bound α with the exact Clopper–Pearson interval; crude: bound with $\sigma_\perp := 0$. On R2 the Corollary refuses already at $\sigma = 0$: $\gamma_0 h_{\text{act}} = 0.007237$ exceeds the margin $\delta_{\text{cert}} - \delta_{\text{total}} = 9.994 \times 10^{-4}$.

noise	σ	$h(x_0)$	Corollary	crude	T_{TTL}	T_{TTL}/T	exits	CP 95% interval
profile, $\sigma^\star$	0.007	0.005051	✓	✓	2022	1011	0/2000	[0, 0.001843]
tail-white (f)	0.02	0.005051	×	✓	0	0	2000/2000	[0.9982, 1]

grid gives 0.01675. The sparse modal propagation equals a dense nodal ETD2RK to 1.928×10^{-14}, and two runs give identical SHA-256 hashes (Table S20).

Table S17: Corollary S1, diffusion-enlarged deterministic constants (full vs. crude bound); the tail envelope must stay below $\sqrt{E_{\text{safe}}} = 0.2236$.

bound	σ_r	$\sigma_\perp^2$	Δ_σ	$\varepsilon_{\text{basis},\sigma}$	$\delta_{\text{total},\sigma}$	$T_\perp$
profile, $\sigma^\star$, full	0.007	4.917×10^{-10}	9.703×10^{-5}	0.00739	0.002634	∞
profile, $\sigma^\star$, crude $\sigma_\perp := 0$	0.007	0	0	0.007293	0.002634	∞
tail-white (f), full	8.742×10^{-18}	0.0632	1.1	1.107	0.005408	0
tail-white (f), crude $\sigma_\perp := 0$	8.742×10^{-18}	0	0	0.007293	0.002634	∞

Table S18: Corollary S1, stochastic constants of (S46) and the Kushner bound (full vs. crude bound).

bound	r_{BM}	$\delta_{\mathrm{Itô}}$	σ_h	$\Delta\delta_{\mathrm{pw}}$	δ_{stoch}	c_{int}
profile, $\sigma^\star$, full	0.003237	4.9×10^{-5}	0.00313	0.008873	0.02074	5.146×10^{-6}
profile, $\sigma^\star$, crude $\sigma_\perp := 0$	0.003237	4.9×10^{-5}	0.00313	0.008873	0.02074	5.146×10^{-6}
tail-white (f), full	4.043×10^{-18}	8.786×10^{-35}	3.909×10^{-18}	0.001904	−0.1744	-1.837×10^{-7}
tail-white (f), crude $\sigma_\perp := 0$	4.043×10^{-18}	8.786×10^{-35}	3.909×10^{-18}	0.001896	0.009573	4.578×10^{-6}

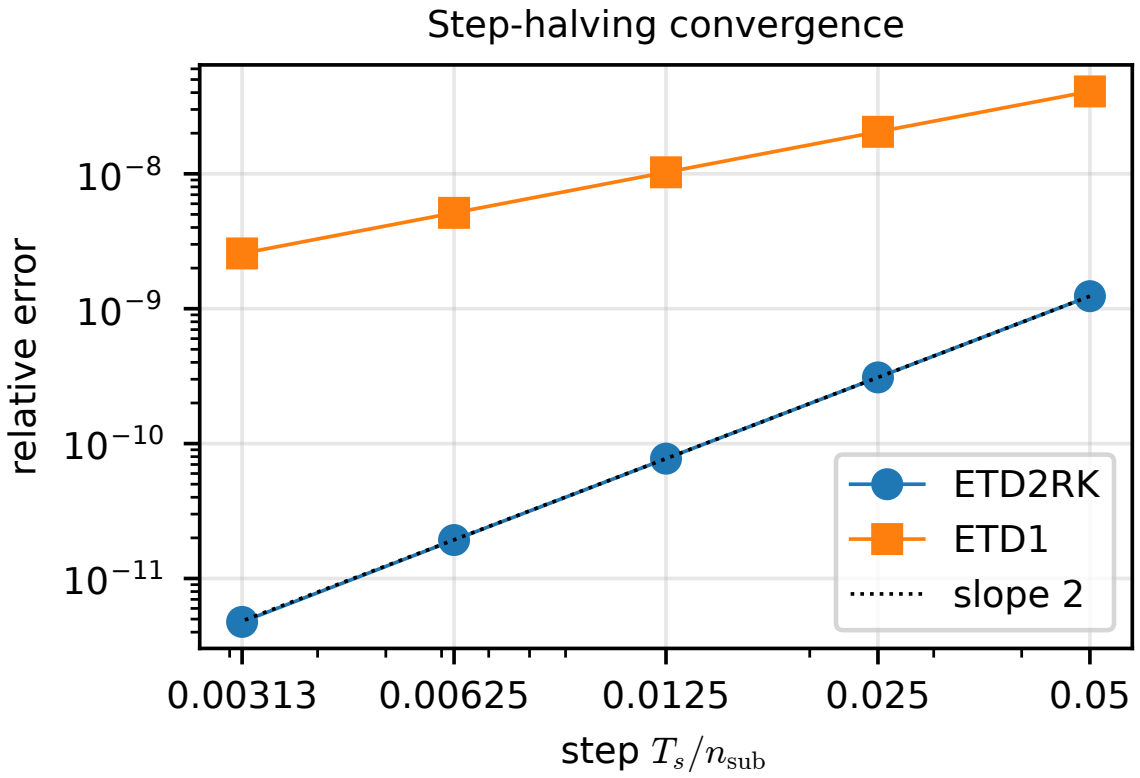


Figure S10: Experiment 8: step-halving convergence of ETD2RK and ETD1.

Table S19: Step-halving convergence of the exponential integrator (open-loop nonlinear lattice).

substeps per period	step	rel. error ETD2RK	ratio	rel. error ETD1
1	0.05	1.238×10^{-9}	–	4.081×10^{-8}
2	0.025	3.095×10^{-10}	4.002	2.048×10^{-8}
4	0.0125	7.731×10^{-11}	4.003	1.026×10^{-8}
8	0.00625	1.927×10^{-11}	4.011	5.135×10^{-9}
16	0.003125	4.765×10^{-12}	4.045	2.569×10^{-9}

Table S20: Numerical-integrity checks.

check	value
spectral radius $\max \lvert\lambda\rvert$	72
explicit Euler amplification at conv. step	3.73
stiffness ratio	43.83
latch: rel. error exact	2.808×10^{-14}
latch: rel. error grid-rounded	0.01675
modal vs nodal rel. error	1.928×10^{-14}
ETD2RK observed orders	2.001, 2.001, 2.004, 2.016
determinism SHA-256 (prefix)	`2bde49483745ec98`

Table S21: Measured wall-clock and peak resident memory per script (2 cores); experiment assertions passed: 248 of 248; pytest: 65 passed.

script	wall [s]	peak RSS [MB]
exp0 openloop	18.0	214
exp1 constants	4.3	112
exp2 crown	7.2	153
exp3 delta cert	5.9	206
exp4 theorem1	52.0	343
exp4b budget	213.4	1010
exp5 deadzone	53.3	209
exp6 ablations	66.4	297
exp7 stochastic	179.5	1050
exp8 numerics	9.1	220
exp9 wsweep	32.1	194
pytest	24.8	1193
total	666.4	1193

S6. Reproducibility

The code and data accompanying the paper, available at https://github.com/AnuKarthikaS/pgcbf, reproduce every table, figure, certificate constant, refusal check, and trajectory statistic in Section S5 and the main paper (Table S22).

Table S22: Repository layout and module responsibilities (Tables S1–S2: R2 = active, $c = 1$, $\mathrm{Pe} = 20$, $u_{\max} = 3.4$, $L_{\mathcal{T},z} = 5.985$, $\gamma_0 = 0.35$; R1 = passive, $c = 0.05$, $\mathrm{Pe} = 1$, $u_{\max} = 0.35$, $L_{\mathcal{T},z} = 0.5985$, $\gamma_0 = 0.3$).

Path	Contents
`config/default.json`	Configuration for the active regime R2 ($\mathrm{Pe} = 20$, $w^\star = 2^{-11/2}$)
`config/r1_passive.json`	Configuration for the passive regime R1 ($\mathrm{Pe} = 1$)
`src/pgcbf/lattice.py`	Full-order telegraph reaction–diffusion–advection lattice and operators
`src/pgcbf/rom.py`	Petrov–Galerkin projection, symplectic lifting, and eigenmode tables
`src/pgcbf/controller.py`	GroupSort-2 teacher, quantized student, and Path-A look-up table
`src/pgcbf/crown.py`	The twin-network CROWN pass and bound (Proposition S1)
`src/pgcbf/certificates.py`	Runtime evaluation of all constants and hypothesis checks (H0)–(H7), (H-dz)
`src/pgcbf/simulate.py`	Exact-latch modal ETD2RK integrator and worst-case disturbance law
`src/pgcbf/stochastic.py`	Euler–Maruyama stochastic integrator and exit bounds (Corollary S1)
`src/pgcbf/texnames.py`	Shared significant-digit formatter for CSV/LaTeX tables and prose
`src/pgcbf/plotting.py`	Matplotlib figure generation routines for Experiments 0–9
`experiments/`	Scripts `exp0_openloop.py` through `exp9_wsweep.py`
`tests/`	Unit, soundness, AST no-literal, perturbation, and linter test suites
`tools/`	Section and main-paper builders, red-team gates, PDF and page-sweep tools
`results/`	Generated JSON logs, CSV/LaTeX tables, PDF figures, and numeral ledger

Running `bash run_all.sh` from the repository root executes the certification and build pipeline:

1. It pins thread counts to 2 on CPU cores 0 and 1 (setting `OPENBLAS_NUM_THREADS=2`, `OMP_NUM_THREADS=2`, `MKL_NUM_THREADS=2`, and `PYTHONHASHSEED=0`) and clears prior generated outputs in `results/`.
2. It executes all eleven experiment scripts (`exp0_openloop` through `exp9_wsweep`, including `exp4b_budget`) and the `pytest` suite via `tools/timed_run.py`, recording wall-clock times, peak resident memory, and structured assertion logs in `results/logs/`.
3. It aggregates runtimes and assertion counts into `results/summary.md` via `tools/summarize.py`, builds Section S5 and `results/section_numbers.json` via `tools/build_section.py`, generates the main-paper figures via `tools/main_figures.py`, and resolves the main-paper template via `tools/build_main.py`.
4. It runs the LaTeX and figure linters, the automated red-team gates via `tools/redteam_gates.py`, and the standalone PDF compilation and page-sweep checks via `tools/standalone_pdf.py` and `tools/page_sweep.py`.

Across the complete execution of `run_all.sh`, all 248 of 248 experiment assertions pass and all 65 `pytest` unit and integration tests pass in 666.4 s of total wall-clock time on 2 CPU cores, with a peak resident memory of 1193 MB (Table S21). Independent executions of the closed-loop simulation in Experiment 8 produce the bit-for-bit identical trajectory SHA-256 hash `2bde49483745ec98a76b5fd814b08aad147b258c2ac39f417ca1133b03e93614`

(Table S20). As implemented in `src/pgcbf/certificates.py`, the certificate requires the codesign lower bound of (S30) that hypothesis (H5) of Theorem S1 marks optional.

S7. Correspondence Table

Table S23 maps each mathematical statement, key equation, table, and figure of the main paper to its counterpart, section, proof location, and page in this supplement, followed by the additional theoretical statements that appear only in this supplement.

Table S23: Correspondence between main-paper items and supplement items.

Main-paper item	Supplement item	Section (proof location)	Page (proof page)
Assumption 1 (plant, barrier, ODD)	Assumption S1	Section S1 (–)	p. 1 (–)
Assumption 2 (tail contraction, coupling)	Assumption S2	Section S2 (–)	p. 4 (–)
Section 2, reduction paragraph (quadratic split, inline)	Lemma S2	Section S2 (inline)	p. 5 (p. 5)
Proposition 1 (coupled twin-CROWN)	Proposition S1	Section S3 (Section S4.1)	p. 7 (p. 15)
Section 3, authority paragraph (P-weighted matrix measure, inline)	Proposition S2	Section S3 (Section S4.2)	p. 8 (p. 17)
Lemma 1 (class-(H) authority margin)	Lemma S5	Section S3 (Section S4.3)	p. 11 (p. 19)
Theorem 1 (sampled-data CBF safety)	Theorem S1	Section S3 (Section S4.4)	p. 12 (p. 21)
Section 3, adversary paragraph (worst-case w^*_{adv})	Remark S4	Section S3 (–)	p. 14 (–)
Section 4, stochastic-extension paragraph	Corollary S1	Section S3 (Section S4.4)	p. 14 (p. 21)
eq. (7) (budget)	Equations (S33), (S45)	Section S3 (Section S4.4)	pp. 10, 13 (p. 21)
Table 1 (budget by term, R1 and R2)	Tables S1, S2, S9	Section S5 (–)	pp. 28, 29, 34
Figure 1(a) (ablation panel)	Figure S8; Tables S13, S14	Section S5 (–)	pp. 41, 42, 42
Figure 1(b) (w-sweep)	Figure S2; Table S6	Section S5 (–)	pp. 33, 32
Additional theoretical items in this supplement			
– (ODD containment)	Lemma S1	Section S1 (inline)	p. 2 (p. 2)
– (dissipation tail rates)	Remark S1	Section S2 (–)	p. 5 (–)
– (sector-aligned LUT)	Lemma S3	Section S2 (Section S4.3)	p. 6 (p. 19)
– (class-(D) closed form)	Lemma S4	Section S3 (inline)	p. 11 (p. 11)
– (deadzone stability)	Proposition S3	Section S3 (Section S4.3)	p. 12 (p. 19)
– (spectral cutoff for RDA)	Remark S2	Section S3 (–)	p. 13 (–)
– (LUT indexing variants)	Remark S3	Section S3 (–)	p. 13 (–)
– (pencil contraction rate)	Lemma S6	Section S4.2 (inline)	p. 18 (p. 18)
– (characterization of (H2a))	Lemma S7	Section S4.2 (inline)	p. 18 (p. 18)
– (P-norm sector bound)	Lemma S8	Section S4.3 (inline)	p. 19 (p. 19)